



UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL
FACULDADE DE MEDICINA
PROGRAMA DE PÓS-GRADUAÇÃO EM PSIQUIATRIA E CIÊNCIAS DO
COMPORTAMENTO

JOÃO PEDRO GONÇALVES PACHECO

**THE DSM-5 LEVEL 1 CROSS-CUTTING SYMPTOM MEASURE AS A TOOL FOR
TRANSDIAGNOSTIC ASSESSMENT IN YOUTH:
Evidence from a Community Sample**

**ESCALA TRANSVERSAL DE SINTOMAS DE NÍVEL 1 DO DSM-5 COMO
FERRAMENTA PARA AVALIAÇÃO TRANSDIAGNÓSTICA EM JOVENS:
Evidências de uma Amostra Comunitária**

Porto Alegre
2026

JOÃO PEDRO GONÇALVES PACHECO

**THE DSM-5 LEVEL 1 CROSS-CUTTING SYMPTOM MEASURE AS A TOOL FOR
TRANSDIAGNOSTIC ASSESSMENT IN YOUTH:**

Evidence from a Community Sample

**ESCALA TRANSVERSAL DE SINTOMAS DE NÍVEL 1 DO DSM-5 COMO
FERRAMENTA PARA AVALIAÇÃO TRANSDIAGNÓSTICA EM JOVENS:**

Evidências de uma Amostra Comunitária

Tese apresentada ao Programa de Pós-Graduação em
Psiquiatria e Ciências do Comportamento da
Faculdade de Medicina da Universidade Federal do
Rio Grande do Sul como requisito parcial para a
obtenção do título de Doutor em Psiquiatria e
Ciências do Comportamento.

Orientador: Prof. Dr. Mauricio Hoffmann

Co-orientador: Prof. Dr. Christian Kieling

Porto Alegre

2026

CIP - Catalogação na Publicação

Pacheco, João Pedro Gonçalves

The DSM-5 Level 1 Cross-cutting Symptom Measure as
a Tool for Transdiagnostic Assessment in Youth:
Evidence from a Community Sample / João Pedro
Gonçalves Pacheco. -- 2026.

246 f.

Orientador: Maurício Scopel Hoffmann.

Coorientador: Christian Costa Kieling.

Tese (Doutorado) -- Universidade Federal do Rio
Grande do Sul, Faculdade de Medicina, Programa de
Pós-Graduação em Psiquiatria e Ciências do
Comportamento, Porto Alegre, BR-RS, 2026.

1. Epidemiologia psiquiátrica. 2. Infância e
adolescência. I. Hoffmann, Maurício Scopel, orient.
II. Kieling, Christian Costa, coorient. III. Título.

JOÃO PEDRO GONÇALVES PACHECO

**THE DSM-5 LEVEL 1 CROSS-CUTTING SYMPTOM MEASURE AS A TOOL FOR
TRANSDIAGNOSTIC ASSESSMENT IN YOUTH:**

Evidence from a Community Sample

**ESCALA TRANSVERSAL DE SINTOMAS DE NÍVEL 1 DO DSM-5 COMO
FERRAMENTA PARA AVALIAÇÃO TRANSDIAGNÓSTICA EM JOVENS:**

Evidências de uma Amostra Comunitária

Tese apresentada ao Programa de Pós-Graduação em Psiquiatria e Ciências do
Comportamento da Faculdade de Medicina da Universidade Federal do Rio Grande do Sul
como requisito parcial para a obtenção do título de Doutor em Psiquiatria e Ciências do
Comportamento.

Aprovado em: 31 de março de 2026

BANCA EXAMINADORA

Prof. Dr. Pedro V. da S. Magalhães
Universidade Federal Do Rio Grande do Sul

Prof.^a Dr.^a Leticia S. Czepielewski
Universidade Federal Do Rio Grande do Sul

Prof. Dr. Lucas Spanemberg
Pontifícia Universidade Católica do Rio Grande do Sul

Prof. Dr. Mauricio Scopel Hoffmann (orientador)
Universidade Federal Do Rio Grande do Sul

To Manoella, with all my love.

*And for every person whose life has been
touched by mental illness.*

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my advisor, Prof. Maurício, for his guidance, generosity, and support throughout this work, and to my co-advisor, Prof. Christian, for his valuable insights and thoughtful contributions during this process. I also thank the members of the examining committee for their time and for kindly agreeing to evaluate this thesis.

This research would not have been possible without the collaborators and participants of the 1993 Pelotas Birth Cohort study, to whom I extend my appreciation. I am also grateful to my co-authors for their collaboration in the studies derived from this work, as well as to Claudia Buchweitz for insightful conversations along the way. I thank the Graduate Program in Psychiatry and Claudia Grabinski for their essential institutional support.

My professional development has been shaped by many colleagues and mentors. I am thankful to my friends and colleagues at the Psychiatry Residency Program and the Department of Neuropsychiatry at the Federal University of Santa Maria for their support throughout my training. I also acknowledge my previous supervisors and co-supervisors, Prof. Diana Frauches, Prof. Gustavo Pinasco, Prof. Marcos Meirelles, Prof. Chris Connolly, and Prof. Vitor Calegaro, for their important role in my academic development. I extend my gratitude to André for his perspective and support during this journey. I am also grateful to my patients, whose experiences continually remind me of the complexity of mental health beyond theoretical models.

Finally, I am grateful to my family for their constant encouragement: my mother, Luciana, and Nilton; my father, João Luiz, and Karina; my siblings, Isabella and João Carlos; and my grandparents, Lucia, Carlos, Clérída, and José. I am also thankful to Efígenia, for her support over the years. I am deeply grateful to Manoella for her companionship throughout this journey, and to her family for their support. I also thank my pets for their companionship during the many hours spent developing ideas, coding, writing, submitting papers, and preparing presentations. To all my friends, thank you for your constant encouragement along this journey.

*Some people claim that an epidemiologist sees
the entire world in a 2x2 table. (...)*

*Regrettably, neither life nor tests are so simple;
grey zones abound.*

- David A Grimes, Kenneth F Schulz

*We shall not cease from exploration
And the end of all our exploring
Will be to arrive where we started
And know the place for the first time.*

- T. S. Eliot

RESUMO

Os transtornos mentais frequentemente se iniciam na adolescência e no início da vida adulta e são caracterizados por sobreposição de sintomas, comorbidade e progressão gradual. Essas características desafiam estratégias tradicionais de rastreamento baseadas em instrumentos específicos para cada transtorno, sobretudo em contextos com restrições de recursos, nos quais são necessárias abordagens eficientes de detecção precoce. Nesse cenário, instrumentos multidimensionais e transdiagnósticos podem representar uma alternativa pragmática para identificar sofrimento psíquico e organizar o cuidado em jovens. Esta tese avaliou o desempenho da Escala Transversal de Sintomas de Nível 1 do DSM-5 (*DSM-5 Level 1 Cross-Cutting Symptom Measure*; DSM-XC), proposta pela American Psychiatric Association, como instrumento de rastreamento dimensional em uma amostra populacional de adultos jovens da Coorte de Nascimentos de Pelotas de 1993. Desenvolvido com base no modelo de revisão de sistemas da medicina, o instrumento examina domínios psicopatológicos relevantes no contexto assistencial. Contudo, a capacidade de seus domínios em informar a probabilidade de diagnósticos específicos, tanto na triagem inicial quanto entre indivíduos com transtornos mentais, não havia sido previamente avaliada, e os pontos de corte propostos não haviam sido empiricamente testados. A investigação foi organizada em três estudos. O primeiro examinou se os domínios da DSM-XC funcionam como indicadores específicos de transtornos e avaliou o impacto do enquadramento adverbial na acurácia diagnóstica. Os resultados indicaram que os domínios operam como indicadores transdiagnósticos, modificando simultaneamente a probabilidade de múltiplos transtornos internalizantes. A formulação baseada em frequência apresentou maior especificidade e maiores razões de verossimilhança positivas do que a formulação baseada em intensidade. O segundo estudo avaliou a capacidade dos domínios da DSM-XC em detectar comorbidade psiquiátrica. Embora os pontos de corte originais tenham se mostrado úteis para excluir diagnósticos adicionais, sua capacidade de confirmação foi limitada. A elevação dos pontos de corte produziu melhora modesta na confirmação de depressão, ansiedade e transtornos por uso de substâncias, com pequenas mudanças nas probabilidades pós-teste. Escores elevados em domínios específicos não confirmam comorbidades, mas sinalizam a necessidade de investigação clínica sistemática. Em contraste, o escore total demonstrou utilidade como indicador de elevada carga sintomática e potencial multimorbidade. O terceiro estudo examinou se perfis sintomáticos derivados da DSM-XC eram consistentes com um modelo transdiagnóstico de estadiamento clínico na comunidade. Padrões em transtornos do humor, ansiedade e estresse pós-traumático convergiram ao longo de *continuum* de sintomatologia, e, correspondendo a gradientes crescentes de impacto em desfechos de saúde e funcionamento. Estágios iniciais associaram-se a riscos cardiometabólicos e comportamentais, enquanto estágios persistentes ou recorrentes concentraram maiores níveis de suicidalidade e desvantagem socioeconômica. Em conjunto, os achados sustentam a DSM-XC como instrumento de rastreio dimensional capaz de captar gradientes clinicamente relevantes de sofrimento psíquico em jovens e orientar a investigação diagnóstica em nível populacional. Estudos futuros devem avaliar se instrumentos breves e multidimensionais podem apoiar a alocação de cuidados informada por estágios em contextos clínicos e de saúde pública.

Palavras-chave: Epidemiologia psiquiátrica; Infância e adolescência; Acurácia diagnóstica; Estadiamento; Triagem.

ABSTRACT

Mental disorders frequently emerge during adolescence and early adulthood and are characterized by symptom overlap, comorbidity, and gradual progression. These features challenge traditional screening strategies based on disorder-specific instruments, particularly in resource-constrained settings where efficient early detection approaches are required. In this context, multidimensional and transdiagnostic instruments may represent a pragmatic alternative for identifying psychological distress and organizing care in young people. This thesis evaluated the performance of the DSM-5 Level 1 Cross-Cutting Symptom Measure (DSM-XC), proposed by the American Psychiatric Association, as a dimensional screening instrument in a population-based sample of young adults from the 1993 Pelotas Birth Cohort. Developed based on the medical review of systems model, the instrument examines psychopathological domains relevant to clinical care. However, the extent to which its domains inform the probability of specific diagnoses, both in initial screening and among individuals with established mental disorders, had not been previously evaluated, and the proposed cut-off points had not been empirically tested. The investigation was organized into three studies. The first examined whether DSM-XC domains function as disorder-specific indicators and assessed the impact of adverbial symptom framing on diagnostic accuracy. Results indicated that the domains operate as transdiagnostic indicators, simultaneously modifying the probability of multiple internalizing disorders. Frequency-based item formulation showed greater specificity and higher positive likelihood ratios than intensity-based formulation. The second study evaluated the ability of DSM-XC domains to detect psychiatric comorbidity. Although the original cut-off points were useful for ruling out additional diagnoses, their rule-in capacity was limited. Increasing cut-off thresholds resulted in modest improvement in confirming depression, anxiety, and substance use disorders, with only small changes in post-test probabilities. Elevated scores in specific domains do not confirm comorbidities but signal the need for systematic clinical investigation. In contrast, the total DSM-XC score demonstrated utility as an indicator of elevated symptom burden and potential multimorbidity. The third study examined whether DSM-XC-derived symptom profiles were consistent with a transdiagnostic clinical staging model in the community. Patterns across mood, anxiety, and post-traumatic stress disorders converged along a symptom continuum corresponding to increasing gradients of impact on health and functioning outcomes. Early stages were associated with cardiometabolic and behavioral risks, whereas persistent or recurrent stages concentrated higher levels of suicidality and socioeconomic disadvantage. Taken together, these findings support the DSM-XC as a dimensional screening instrument capable of capturing clinically meaningful gradients of psychological distress in youth and guiding diagnostic investigation at the population level. Future studies should assess whether brief, multidimensional instruments can support stage-informed care allocation in clinical and public health settings.

Keywords: Psychiatric epidemiology, Childhood and adolescence, Diagnostic accuracy, Staging, Screening.

ABBREVIATIONS AND ACRONYMS

AUD	Alcohol Use Disorder
BD	Bipolar Disorder
CMMHSI	Common Measures in Mental Health Science Initiative
DSM	Diagnostic and Statistical Manual of Mental Disorders
DSM-XC	DSM-5 Level 1 Cross-Cutting Symptom Measure
GAD	Generalized Anxiety Disorder
HiTOP	Hierarchical Taxonomy of Psychopathology
LMICs	Low- and Middle-Income Countries
MDD	Major Depressive Disorder
MINI	Mini International Neuropsychiatric Interview
NPV	Negative Predictive Value
PHQ-9	Patient Health Questionnaire-9
PPV	Positive Predictive Value
PTSD	Post-Traumatic Stress Disorder
ROC	Receiver Operating Characteristic
SAD	Social Anxiety Disorder
USPSTF	U.S. Preventive Services Task Force
YLDs	Years Lived with Disability

TABLE OF CONTENTS

1 INTRODUCTION	12
1.1 Gap 1: Assumption of disorder specificity	16
1.2 Gap 2: Influence of symptom framing	17
1.3 Gap 3: Detection of psychiatric comorbidity	18
1.4 Gap 4: Internal coherence of clinical staging	19
1.5 Rationale	20
2 OBJECTIVES	22
2.1 General objective	22
2.2 Specific objectives	22
3 ARTICLES	23
3.1 Article 1	24
4 FINAL REMARKS	139
4.1 Synthesis of main findings	139
4.2 Implications for clinical assessment and public health	141
4.3 Methodological considerations and limitations	142
4.4 Future research directions	143
4.5 Conclusion	144
5 THESIS REFERENCES	146
6 APPENDIX	152

1 INTRODUCTION

Mental disorders constitute one of the leading contributors to health-related burden during adolescence and young adulthood. Epidemiological evidence indicates that more than three-quarters of mental disorders diagnosed in adulthood have their onset before the age of 25 years (Solmi et al. 2022); moreover, across the life course, approximately 25% of all years lived with disability (YLDs) attributable to mental disorders accumulate before this age (Kieling et al. 2024). As a result, adolescence and young adulthood represent a critical developmental period during which preventive and early-intervention strategies may substantially influence long-term clinical outcomes (Davey and McGorry 2019; McGorry et al. 2024).

This developmental window is particularly relevant for low- and middle-income countries (LMICs), where the majority of the global youth population resides (Kieling et al. 2022), and mental health care systems frequently operate under substantial resource constraints that require contextually adapted and scalable approaches (Patel et al. 2013; Rose-Clarke et al. 2024). Extending detection efforts to community-based samples in LMICs therefore requires the development and evaluation of brief, freely available assessment tools capable of supporting timely identification without imposing substantial financial or administrative burden on already resource-constrained systems (Farber et al. 2023; Kohrt et al. 2018).

Accumulating evidence in mental health research indicates that psychiatric disorders are highly interconnected, with direct implications for screening and early-intervention strategies. Converging findings from genetic architecture (Bourque et al. 2024; Jami et al. 2022), neural correlates (Vanes and Dolan 2021; Xie et al. 2023), and latent variable models (Eaton et al. 2023), among other lines of inquiry, suggest that the boundaries between disorders are less clear-cut than traditionally assumed. This reconceptualization has been captured by the notion of transdiagnostic psychiatry, which emphasizes dimensions, processes, and mechanisms that cut across or move beyond traditional diagnostic boundaries (Fusar-Poli et al. 2019). In this framework, transdiagnostic approaches may target characteristics shared across disorders (e.g., cognitive patterns) or those extending beyond diagnostic categories altogether (e.g., functional impairment). In the present thesis, the term “transdiagnostic” is used in an operational sense, corresponding to an across-diagnoses approach. That is, rather than proposing novel constructs beyond existing diagnostic systems or analysing overarching characteristics of mental disorders (e.g., suffering), this work

examines whether a brief symptom checklist nested into domains function as non-specific indicators that cut across multiple disorders. Accordingly, the focus is on evaluating the extent to which these measures present shared diagnostic accuracy and clinical utility across established diagnostic categories. In parallel, substantial evidence indicates that clinical presentations are dimensional in nature (Fried et al. 2022; Haslam et al. 2012; Hyman 2021), spanning a continuum from lower to higher levels of severity without a sharp boundary between health and illness. This dimensional view contrasts with earlier categorical conceptions of mental disorder rooted in Kraepelinian diagnostic dichotomies.

In this context, there has been growing interest in the development of assessment approaches aligned with transdiagnostic and dimensional models of psychopathology (Eaton et al. 2023; Forbes et al. 2016; Kotov and et al. 2017; Narrow et al. 2013; Shah et al. 2020), and several dimensional, measurement-based frameworks have been proposed in recent years. Notable examples include the DSM's dimensional symptom measures (American Psychiatric Association 2022; Oquendo et al. 2026), self-report measures derived from the Hierarchical Taxonomy of Psychopathology (HiTOP) (Simms et al. 2023), and service-oriented personalized care models embedded within emerging youth mental health systems. For instance, the Australian "Right Care, First Time" framework (Hickie et al. 2019) operationalizes several transdiagnostic and dimensional principles within an applied model of youth mental health service delivery.

The DSM-5 Level 1 Cross-Cutting Symptom Measure (DSM-XC) represents a prominent illustration of these transdiagnostic, dimensional assessment approaches and is supported by the Common Measures in Mental Health Science Initiative (CMMHSI) (Farber et al. 2023). As part of the *Diagnostic and Statistical Manual of Mental Disorders*, one of the most widely adopted classificatory systems in psychiatric diagnosis and epidemiology, assessment measures introduced within this manual are likely to influence clinical assessment practices, service organization, and population-level surveillance. Unlike traditional screening tools, the DSM-XC is designed to function as a mental health "review of systems", enabling systematic assessment across 13 psychopathological domains, including depression, anxiety, substance use, and psychosis (American Psychiatric Association 2013). Of particular relevance to LMICs, the DSM-XC was selected within the CMMHSI in part due to practical implementation considerations, including free availability, brevity, feasibility of administration and interpretation, prior use in research settings, and potential relevance across diverse mental health contexts. These characteristics position the DSM-XC as a potentially scalable standard measure, particularly in settings where cost and respondent burden may

limit the uptake of other tools. Given its intended role as a standardized multidimensional screener within routine care and research environments, rigorous empirical evaluation of its measurement properties is essential to ensure that its implementation meaningfully improves early identification and clinical decision-making in youth mental health contexts.

The DSM-XC was conceptualized and developed by the DSM-5 Task Force and Work Groups (Clarke and Kuhl 2014) to address the limitations of strict categorical diagnostic systems by capturing cross-cutting symptomatology commonly observed in clinical practice. It was designed to be administered prior to clinical evaluation to inform decision-making, rather than acting as a standalone diagnostic screener. Many of its Level 1 items were derived from established instruments, such as the Patient Health Questionnaire-2 (PHQ-2) for depression. Early feasibility and clinical utility were established during the large-scale DSM-5 field trials, which utilized web-based electronic data capture across diverse adult and pediatric academic clinical settings in the United States and Canada (Clarke et al. 2013). During these trials, both patients and clinicians reported that the measure was simple, easy to use, and clinically meaningful (Mościcki et al. 2013). Initial psychometric evaluations within these trials demonstrated mostly good-to-excellent test-retest reliability coefficients among 2,246 adult patients presenting to these academic clinics with various existing mental health diagnoses (Narrow et al. 2013). However, the measure was published in the manual and recommended for routine practice based primarily on these reliability metrics, without any formal validation studies having been carried out to estimate internal consistency, dimensionality, predictive value, or criterion validity against gold-standard diagnostic interviews.

Despite its clinical availability, subsequent independent efforts to validate the DSM-XC have largely relied on convenience samples and comparisons with other self-report scales rather than clinical diagnoses. For example, investigations into its construct validity among a convenience sample of non-treatment-seeking college students supported both its convergent and divergent validity (Bravo et al. 2018); however, this was demonstrated solely through correlations with other self-report measures, such as comparing the DSM-XC Depression domain with the 21-item Depression Anxiety Stress Scales (DASS-21) and the Substance Use domain with the 10-item Alcohol Use Disorder Identification Test (AUDIT). Similarly, the largest factor analytic study to date (Gibbons et al. 2022) sought to clarify the instrument's underlying dimensionality using an online convenience sample of 3,533 adults (mean age 46.5 ± 14.8 years) recruited during the COVID-19 pandemic. While this sample was not fully representative of the general population, comprising predominantly white

(90.2%), female (83.2%), and highly educated (52.3% with advanced degrees) individuals, the confirmatory factor analysis indicated that a six-factor model (mood, worry, activation, somatic symptoms, thought content, and substance use) and a bifactor solution (a general psychopathology factor alongside mood and anxiety residuals) fit the data well.

Studies attempting to assess the DSM-XC's actual screening capabilities have similarly suffered from significant methodological and sampling limitations. The measure has demonstrated varying degrees of sensitivity and specificity, generally showing acceptable sensitivity but poor specificity, when applied to a highly morbid clinical population of inpatients and outpatients with substance use disorders at a correctional facility (Bastiaens and Galus 2018). Other studies evaluating its screening utility have utilized internet convenience samples (Doss and Lowmaster 2022) or screened healthy research volunteers via online portals (Mahoney et al. 2020), comparing DSM-XC flags against clinical triage decisions rather than comprehensive diagnostic evaluations.

While these previous studies suggest the DSM-XC can capture relevant psychopathological variance, their reliance on scale-to-scale correlations, convenience samples, or basic triage decisions fails to provide a robust diagnostic validation. Consequently, there remains a critical lack of large-scale, population-based studies that rigorously validate the DSM-XC against gold-standard structured clinical interviews across multiple diagnoses to determine its true practical utility in the community.

Despite its increasing adoption in clinical and research settings, comprehensive cultural adaptations and translations of the DSM-XC remain limited. While a few studies have utilized translated versions, such as Turkish (Çökmüş et al. 2017), Hindi (Goel and Kataria 2018), and Urdu (Ishfaq and Kamal 2019), the potential biases introduced during the translation process and the cross-cultural equivalence of the measure's dimensional constructs are often underexplored. In Brazil, rigorous evaluations of the instrument's semantic and cultural adaptation have been similarly scarce, raising questions about how well the measure's items capture the nuanced expressions of psychological distress in the local context. For the present work, a Brazilian Portuguese adaptation of the DSM-XC was employed. As detailed in the subsequent chapters, this version included specific modifications, such as the separation of double-barreled questions into distinct intensity and frequency scales, to better suit the cognitive demands of the community-based youth sample (the full Portuguese instrument used is provided in the Appendix).

Despite its promise, several methodological issues remain insufficiently examined in the application of the DSM-XC specifically and of brief mental health screening tools for

youth more broadly. In particular, key assumptions underlying current screening paradigms, including the disorder-specific functioning of symptom assessment measures, the impact of adverbial framing on symptom reporting (e.g., whether symptoms are rated based on frequency, such as “how often”, or intensity, such as “how much”), the capacity of measures to detect psychiatric comorbidity, and their potential role in operationalizing transdiagnostic clinical staging, have rarely been evaluated within population-based samples. These limitations may constrain the utility of multidimensional tools for timely identification and highlight the need for empirical investigation into how brief cross-cutting measures perform beyond single-disorder detection.

This thesis examines these challenges from the perspective of the performance of the DSM-XC, focusing on whether this tool operates as intended (i.e., multidimensionally) across key measurement properties relevant to screening. The empirical work is presented in three papers: the first addresses assumptions related to disorder-specific functioning and adverbial framing, the second examines its capacity to detect psychiatric comorbidity, and the third explores its potential role in informing transdiagnostic clinical staging. These aspects are described in greater detail in the sections below.

1.1 Gap 1: Assumption of disorder specificity

Despite growing interest in transdiagnostic and dimensional approaches, many brief screening tools are developed and interpreted based on the assumption of disorder-specific functioning, a premise that remains largely unexamined empirically. Recognizing the importance of early identification, major expert panels such as the U.S. Preventive Services Task Force (USPSTF) recommend systematic screening for conditions like anxiety, depression, and suicide risk in primary care settings among adolescents and young adults (US Preventive Services Task Force et al. 2023b; US Preventive Services Task Force et al. 2023a). However, these guidelines focus heavily on discrete diagnostic categories, with separate tools employed to screen for different disorders. Widely used assessment measures such as the Patient Health Questionnaire-9 (PHQ-9) (Kroenke et al. 2001) for depression and the Generalized Anxiety Disorder-7 (GAD-7) (Spitzer et al. 2006) for anxiety further exemplify this approach, demonstrating robust diagnostic accuracy for their respective target conditions (Negeri et al. 2021; Plummer et al. 2016), but are rarely investigated in detection of closely related internalizing disorders. In fact, empirical evidence shows that these tools are not specific for detecting their intended diagnostic targets (Jobim et al. 2025).

This practice rests on the assumption that a single assessment measure should map exclusively to a single mental health condition. While operationally convenient, this assumption contrasts with the typical presentation of mental health symptoms in clinical practice, where comorbidity and shared etiological factors are the norm rather than the exception (Krueger and Eaton 2015; McGrath et al. 2020; Plana-Ripoll et al. 2019). Furthermore, recent empirical work highlights that symptoms such as insomnia and irritability often overlap across diagnostic criteria (Forbes et al. 2024), challenging the current practice of assessing disorders in isolation. Despite the prevalence of disorder-specific screening recommendations, diagnostic accuracy research has rarely evaluated multiple mental health conditions simultaneously to test the potential limitations of this approach (Caspi et al. 2025; Jobim et al. 2025; Kohrt et al. 2025).

Although the DSM-XC utilizes a dimensional assessment structure to capture symptom severity, it retains a pragmatic link to traditional diagnostic categories through a two-tiered process. A positive result on the Level 1 screener, defined by specific thresholds for each domain, acts as a trigger for a more detailed Level 2 domain-specific assessment (American Psychiatric Association 2013).

1.2 Gap 2: Influence of symptom framing

Despite the potential of the DSM-XC as a comprehensive screening tool, critical methodological issues remain regarding its original operationalization. The assessment measure employs a double-barreled question structure, asking respondents “how much (or how often)” they have been bothered by a symptom over the past two weeks. This combined phrasing forces patients to simultaneously evaluate and integrate two distinct aspects of their experience, severity (intensity) and duration or recurrence (frequency), into a single response. Evidence from pilot testing in the Pelotas Birth Cohort has indicated that this ambiguity creates cognitive demand and confusion, as participants struggle to interpret the combined prompt, potentially compromising the clarity and accuracy of the assessment.

The impact of such framing choices is not merely semantic; methodological research suggests that asking about intensity versus frequency can elicit significantly different response patterns and psychometric properties (Gong et al. 2013; Levin et al. 1998). For instance, frequency-based ratings have been associated with higher reliability, criterion validity (Krishnakumar et al. 2021), and stability over time (Krabbe and Forkmann 2014) when compared to intensity ratings, which may be more susceptible to subjective interpretation. Furthermore, the lack of standardization in adverbial framing contributes to substantial

heterogeneity in mental health assessment, undermining the comparability of data across studies and clinical settings (Newson et al. 2020).

Despite these methodological concerns, the relative diagnostic accuracy of these distinct framings has not been systematically evaluated within the context of the DSM-XC. It remains unclear whether separating these dimensions, specifically, asking “how often” versus “how much”, would improve the measure's ability to detect mental disorders or rule them out. Determining which framing provides superior screening utility is therefore essential, not only for optimizing the diagnostic performance of the DSM-XC but also for establishing clearer standards for the future development of psychiatric assessment measures.

1.3 Gap 3: Detection of psychiatric comorbidity

Beyond understanding the impact of adverbial framing, the capacity of screening tools to accurately detect psychiatric comorbidity remains a significant clinical gap. Psychiatric comorbidity is consistently associated with poorer long-term outcomes, including elevated risk of chronic physical illness (Scott et al. 2016), higher healthcare costs (Christensen et al. 2023), and reduced life expectancy (Plana-Ripoll et al. 2020). Moreover, compared with individuals whose disorders first emerge later in life, those with early-onset conditions demonstrate a more rapid accumulation of additional psychiatric diagnoses over time, (McGrath et al. 2020) underscoring the importance of early, comprehensive assessment strategies.

Despite these consequences, comorbidity is frequently under-recognized in routine clinical settings, particularly when clinicians rely on unstructured interviews rather than structured assessments (Zimmerman and Mattia 1999). In the context of time-constrained practice, clinicians may prioritize the first diagnosed condition or the “chief complaint”, often limiting further investigation into additional disorders. While structured interviews could mitigate this oversight, they are typically too time-consuming for routine use in clinical settings. Consequently, brief screening tools are essential to support the systematic early detection of multimorbidity and ensure that complex clinical presentations are not overlooked (Arrarás et al. 2023). However, empirical work on screening tools designed specifically to detect comorbidity is limited. Most existing research is derived from the Psychiatric Diagnostic Screening Questionnaire (PDSQ), which was validated primarily in adults and may not generalize well to youth or early-intervention contexts (Zimmerman and Chelminski 2006; Zimmerman and Mattia 2001). Although the DSM-XC was explicitly developed to assist clinicians in identifying comorbidities that could impact treatment and prognosis

(American Psychiatric Association 2013), its actual diagnostic performance in detecting secondary disorders among individuals who already have a primary diagnosis has not been empirically tested. Consequently, there is a significant gap in understanding how brief screeners perform in the context of multimorbidity. Establishing the ability of the DSM-XC to effectively rule out or confirm additional diagnoses is therefore important to support its use as a comprehensive tool for comorbidity assessment.

1.4 Gap 4: Internal coherence of clinical staging

Finally, the dimensional structure of the DSM-XC offers a standardized metric to evaluate the coherence of transdiagnostic staging frameworks. Clinical staging conceptualizes mental disorders not as static categories, but as dynamic processes progressing along a continuum of severity, ranging from asymptomatic states (Stage 0) and non-specific distress (Stage 1a) to subthreshold phenotypes (Stage 1b) and established or persistent illness (Stages 2 to 4). The utility of this approach has increasingly extended beyond conceptual models into service delivery, informing the redesign of youth mental health systems. A prominent example is the “Right Care, First Time” model (Hickie et al. 2019), which explicitly integrates clinical staging with multidimensional assessment to guide early intervention.

However, such staging models have been primarily developed and implemented in help-seeking cohorts within specialized services in high-income countries. Although these clinical settings are appropriate for evaluating service-level applications of staging, they tend to include individuals who are already experiencing more severe or impairing forms of illness. In community populations, particularly in low- and middle-income countries, mental health symptoms often emerge outside formal care pathways, and individuals at earlier stages are less likely to be identified through service-based sampling. As a result, whether staging frameworks retain their construct validity (Clarke et al. 2025) when applied beyond clinical samples remains uncertain.

This limitation is compounded by the absence of a standardized approach for operationalizing clinical stages in population-based settings. A brief, targeted literature review conducted for this thesis identified substantial heterogeneity in how stages are defined across disorders, instruments, and thresholds, with most community-based applications focusing on specific transitions between stages or on longitudinal progression across symptom domains. Only a small number of empirical studies (Chen et al. 2025; Ratheesh et al. 2025; Ratheesh et al. 2023) have attempted to operationalize transdiagnostic staging models in longitudinal community cohorts.

In the absence of a gold-standard framework for defining stages across mood, anxiety, and stress-related disorders in community samples, the staging model employed in the third study of this thesis was developed following a review of the available literature and recent advances in public mental health staging (Buchweitz et al. 2026; Chen et al. 2025; Dwyer et al. 2025; Haas et al. 2025; Scott et al. 2024). Based on this review, an expert panel composed of four psychiatrists, including two with prior research experience in clinical staging models, reached consensus on operational criteria integrating symptom presence, diagnostic status, impairment, and chronicity for each disorder category.

In this context, stage-specific symptom profiles were examined among individuals classified within the same clinical stage, irrespective of their primary diagnosis. These profiles were defined as the pattern of cross-cutting symptoms observed via the DSM-XC within each stage. Specifically, this thesis employs the DSM-XC to examine whether these patterns remain consistent across different diagnostic categories within the same stage (as defined by structured clinical interviews), thereby testing the internal coherence of the transdiagnostic model.

Furthermore, this framework was examined to determine whether successive stages correspond to objective gradients in external indicators of clinical burden. Specifically, transdiagnostic stages were compared with respect to the prevalence of cardiometabolic risk, suicidal ideation, and NEET status (not in education, employment, or training), in order to assess whether stage progression is associated with systematic differences in physical health vulnerability and socio-occupational disengagement.

1.5 Rationale

Four critical knowledge gaps drive the present thesis, highlighting the need for a detailed examination of the properties and functions of brief screeners like the DSM-XC. To address these gaps and target key limitations of current screening paradigms, the present analysis adopts a clinimetric (Fava et al. 2025) rather than a strictly psychometric approach. This framework is particularly appropriate because the DSM-XC is already published and indicated for routine clinical use from the DSM-5 onwards. While traditional psychometric evaluations emphasize the homogeneity of scale components and unidimensionality, a clinimetric approach is designed to accommodate the heterogeneous features of clinical variables. Consequently, rather than focusing on internal consistency or latent factor models, this thesis evaluates the DSM-XC based on its discrimination properties—specifically, its capacity to accurately differentiate among distinct clinical subgroups. By examining the

instrument's ability to provide a broad rating of clinical phenomena, this research investigates its practical utility for determining diagnostic probability, detecting psychiatric comorbidity, and characterizing progressive illness staging within a population-based setting, ultimately aiming to support clinical judgment and decision-making in epidemiology and public health.

First, while guidelines prioritize disorder-specific screening, it remains uncertain whether short measures operate in a truly disorder-specific manner or if they function transdiagnostically, influencing the likelihood of multiple disorders simultaneously.

Second, the specific diagnostic impact of assessing symptom frequency (“how often”) versus intensity (“how much”) remains untested. To address these ambiguities, the first study assesses the screening properties of the DSM-XC domains for eight distinct psychiatric disorders, directly comparing the diagnostic accuracy of intensity-based versus frequency-based scoring.

Third, despite the clinical importance of identifying multimorbidity, the accuracy of cross-cutting measures for specifically detecting comorbid psychiatric disorders among youth has not been evaluated against optimized thresholds. To bridge this gap, the second study evaluates the capacity of the DSM-XC to identify psychiatric comorbidity among youth with a primary diagnosis, testing original DSM-5 thresholds against optimized cut-offs derived to enhance detection. In this study, we adopted a disorder-specific approach, examining seven psychiatric disorders that had direct correspondent domains within the DSM-XC available in the cohort’s assessment battery.

Finally, while clinical staging offers a promising framework for tracking illness progression, the potential of dimensional screeners to characterize successive transdiagnostic stages of illness severity in the community has not been established. Consequently, the third study assesses a transdiagnostic clinical staging framework encompassing mood, anxiety, and post-traumatic stress disorders, verifying if the DSM-XC captures coherent cross-cutting symptom profiles across progressive stages and quantifying their association with objective indicators of health, social, and functional burden. This study focused exclusively on these disorder categories because they represent the psychiatric conditions with the highest health-related burden among youth.

Collectively, these studies examine how the DSM-XC performs in detecting mental disorders and for characterizing gradients of symptom expression across successive clinical stages among youth in a population-based setting.

2 OBJECTIVES

2.1 General objective

To evaluate the clinical utility of the DSM-5 Level 1 Cross-Cutting Symptom Measure (DSM-XC) as a first-step mental disorder detection tool, comorbidity assessment, and characterization of transdiagnostic mental health stages in a community-based sample of youth.

2.2 Specific objectives

1. Article 1. To assess the screening properties of the DSM-XC domains in detecting eight distinct psychiatric disorders within the community.
2. Article 1. To evaluate the impact of adverbial phrasing by comparing the diagnostic accuracy of intensity-based versus frequency-based scoring.
3. Article 2. To evaluate the diagnostic performance of the DSM-XC in identifying comorbid psychiatric conditions among youth already presenting with a primary diagnosis, and
4. Article 2. To test the utility of original DSM-5 recommended thresholds versus optimized cut-offs derived to enhance comorbidity detection.
5. Article 3. To assess whether a transdiagnostic staging framework, encompassing mood, anxiety, and post-traumatic stress disorders, captures coherent gradients of cross-cutting symptoms.
6. Article 3. To examine the association of transdiagnostic stages and indicators of health, social, and functional burden.

3 ARTICLES

Article 1.

How much or how often? Examining the screening properties of the DSM Cross-Cutting Symptom Measure in a youth population-based sample.

Psychological Medicine. Cambridge University Press.

Received: 26 September 2023; Accepted: 15 March 2024.

<https://doi.org/10.1017/S0033291724000849>

Article 2.

Diagnostic Performance of the DSM-5 Level 1 Cross-cutting Symptom Measure to Detect Psychiatric Comorbidity Among Youth: a Community-based Study.

Manuscript submitted to *Early Intervention in Psychiatry*.

Article 3.

Assessing Mental Health Transdiagnostic Staging in the Community: Evidence from a Brazilian Youth Population-Based Sample

Manuscript submitted to *The Lancet Child and Adolescent Health*.

3.1 Article 1

How much or how often? Examining the screening properties of the DSM Cross-Cutting Symptom Measure in a youth population-based sample

João Pedro Gonçalves Pacheco, M.D., M.Sc.^{1,2,3}, Christian Kieling, M.D., Ph.D.^{2,4,5}, Pedro H. Manfro, M.D., Ph.D.², Ana M. B. Menezes, M.D., Ph.D.⁶, Helen Gonçalves, Ph.D.⁶, Isabel O. Oliveira, Ph.D.⁶, Fernando C. Wehrmeister, Ph.D.^{6,7}, Luis Augusto Rohde, M.D., Ph.D.^{5,8,9}, Mauricio Scopel Hoffmann, M.D., Ph.D.^{1,2,3,10}

¹ Department of Neuropsychiatry, Universidade Federal de Santa Maria, Avenida Roraima 1000, Santa Maria, 97105-900, Brazil.

² Graduate Program in Psychiatry and Behavioral Sciences, Universidade Federal do Rio Grande do Sul, rua Ramiro Barcelos 2350, Porto Alegre, 90035-003, Brazil.

³ Mental Health Epidemiology Group, Santa Maria, Brazil

⁴ Department of Psychiatry and Legal Medicine, Universidade Federal do Rio Grande do Sul (UFRGS), Porto Alegre, RS, Brazil

⁵ ADHD Outpatient Program & Developmental Psychiatry Program, Hospital de Clínicas de Porto Alegre, Federal University of Rio Grande do Sul, Brazil

⁶ Postgraduate Program in Epidemiology, Universidade Federal de Pelotas, Pelotas, RS, Brazil

⁷ Institute for Global Public Health, Rady Faculty of Health Sciences, University of Manitoba, Canada

⁸ National Institute of Developmental Psychiatry, Brazil

⁹ UniEduK, Brazil

¹⁰ Care Policy and Evaluation Centre, London School of Economics and Political Science, Houghton Street, London WC2A 2AE United Kingdom.

Conflict of interest statement: CK is the founder of Wida, a digital mental health platform. LAR has received grant or research support from, served as a consultant to, and served on the speakers' bureau of Abbott, Aché, Bial, Medice, Novartis/Sandoz, Pfizer/Upjohn, and Shire/Takeda in the last three years. The ADHD and Juvenile Bipolar Disorder Outpatient Programs chaired by Dr Rohde have received unrestricted educational and research support from the following pharmaceutical companies in the last three years: Novartis/Sandoz and Shire/Takeda. He has received authorship royalties from Oxford Press and ArtMed.

Acknowledgments: This work was supported by the Brazilian public funding agency Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) grant 466826/2014-1. CK is a CNPq researcher and a UK Academy of Medical Sciences Newton Advanced Fellow. MSH is a postdoctoral research fellow at UFRGS, supported by the United States National Institutes of Health grant R01MH120482-01.

Funding Statement: This work was supported by the Brazilian public funding agency Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) grant 466826/2014-1.

Ethical standards: The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008.

Correspondence: João Pedro Gonçalves Pacheco, Department of Neuropsychiatry, Universidade Federal de Santa Maria, Avenida Roraima 1000, Santa Maria, 97105-900, Brazil. joao.pacheco@ufsm.br

Abstract

Background. The DSM Level 1 Cross-Cutting Symptom Measure (DSM-XC) allows for assessing multiple psychopathological domains. However, its capability to screen for mental disorders in a population-based sample and the impact of adverbial framings (intensity and frequency) on its performance are unknown.

Methods. The study was based on cross-sectional data from the 1993 Pelotas birth cohort in Brazil. Participants with completed DSM-XC and structured diagnostic interviews ($n=3,578$, aged 22, 53.6% females) were included. Sensitivity, specificity, positive (LR+) and negative (LR-) likelihood ratios for each of the 13 DSM-XC domains were estimated for detecting 5 internalizing disorders (bipolar, generalized anxiety, major depressive, post-traumatic stress, and social anxiety disorders) and 3 externalizing disorders (antisocial personality, attention-deficit/hyperactivity, and alcohol use disorders). Sensitivities and specificities >0.75 , $LR+>2$ and $LR-<0.5$ were considered meaningful. Values were calculated for the DSM-XC's original scoring and for adverbial framings.

Results. Several DSM-XC domains demonstrated meaningful screening properties. The anxiety domain exhibited acceptable sensitivity and LR- values for all internalizing disorders. The suicidal ideation, psychosis, memory, repetitive thoughts and behaviours, and dissociation domains displayed acceptable specificity for all disorders. Domains also yielded small but meaningful LR+ values for internalizing disorders. However, LR+ and LR- values were not generally meaningful for externalizing disorders. Frequency-framed questions improved screening properties.

Conclusions. The DSM-XC domains showed transdiagnostic screening properties, providing small but meaningful changes in the likelihood of internalizing disorders in the community, which can be improved by asking frequency of symptoms compared to intensity. The DSM-XC is currently lacking meaningful domains for externalizing disorders.

Keywords: psychiatric, diagnosis, transdiagnostic, questionnaires, assessment, cross-disorder

Introduction

There is a growing clinical and academic interest in investigating symptoms that cut across multiple mental disorders (Eaton et al., 2023; Forbes et al., 2023; Gibbons, Farmer, Shaw, & Chung, 2023), such as anxiety, insomnia, depressed mood, hallucinations, among others. Both authoritative texts (American Psychiatric Association, 2013) and empirical studies (Forbes, Tackett, Markon, & Krueger, 2016; Waszczuk et al., 2017) have encouraged dimensional approaches to psychopathology assessment, with the aim of elucidating shared features among different disorders. An emerging tool is The Diagnostic and Statistical Manual of Mental Disorders-5 (DSM-5) Level 1 Cross-Cutting Symptom Measure (DSM-XC), which consists of 23 self-reported questions that explore 13 different mental health domains. However, the current body of evidence supporting the utility of cross-cutting symptom assessments, including the DSM-XC, remains limited. This gap presents an unexplored opportunity not only for understanding the connections between different diagnostic categories but also for optimizing community-based mental disorder screening.

The U.S. Preventive Services Task Force (USPSTF) supports the use of disorder-specific scales in screening recommendations for anxiety (US Preventive Services Task Force et al., 2023a), depression (US Preventive Services Task Force et al., 2023b), and unhealthy drug and alcohol use among adults (US Preventive Services Task Force et al., 2020; US Preventive Services Task Force et al., 2018). Notable scales presenting good diagnostic accuracy for these purposes (Mulvaney-Day et al., 2018) include the Patient Health Questionnaire-9 (PHQ-9, sensitivity=0.85, specificity=0.85) and Center for Epidemiological Studies Depression (CES-D, sensitivity=0.83, specificity=0.78) for major depression (Negeri et al., 2021; Vilagut, Forero, Barbaglia, & Alonso, 2016), the General Anxiety Disorder-7 (GAD-7) for generalized anxiety disorder (sensitivity=0.83, specificity=0.84) (Plummer, Manea, Trepel, & McMillan, 2016), the National Institute on Alcohol Abuse and Alcoholism (NIAAA) for current alcohol use disorder (sensitivity=0.88, specificity=0.67) (Smith, Schmidt, Allensworth-Davies, & Saitz, 2009), The Alcohol, Smoking and Substance Involvement Screening Test (ASSIST) for tobacco, alcohol, and cannabis use disorders (sensitivities=0.95-1.00; specificities=0.79-0.93) (Gryczynski et al., 2015), and the Tobacco, Alcohol, Prescription Medication and Other Substance Use (TAPS, sensitivities=0.70-0.74, specificities 0.85-0.95 for tobacco, alcohol and marijuana) (McNeely et al., 2016), among others. Nevertheless, current recommendations lack the inclusion of multi-disorder tools. Only a handful of such screening instruments are available, such as the PHQ-4, The Hospital

Anxiety and Depression Scale (HADS), and the Mental Health Index (MHI-5). The PHQ-4 (Löwe et al., 2010) is an ultra-short assessment that combines the PHQ-2 (sensitivity = 0.89, specificity = 0.75) (Mitchell, Yadegarfar, Gill, & Stubbs, 2016) and GAD-2 (sensitivity = 0.80, specificity = 0.81) (Plummer et al., 2016), facilitating simultaneous screening for depression and anxiety. The HADS is a lengthier but commonly used tool for identifying depression and anxiety in medically ill patients, with sensitivities and specificities for depression and anxiety subscales of approximately 0.80 (Bjelland, Dahl, Haug, & Neckelmann, 2002). However, the extent of diagnostic insight provided by comprehensive assessments remains largely unexplored for psychiatric disorders.

The symptoms included in the DSM-XC exhibit similarities to the PHQ-2 (Löwe, Kroenke, & Gräfe, 2005; Mitchell et al., 2016) and the GAD-2 (Plummer et al., 2016). However, the utility of these scales has been ascertained only for the specific diagnoses they were designed to screen for. Given the recognized high comorbidity among mental disorders (Plana-Ripoll et al., 2019) and the occurrence of overlapping symptoms across diagnoses (Forbes et al., 2023), it is pertinent to consider expanding the use of these brief screeners to identify multiple disorders. In addition, a comprehensive examination of the screening properties of the 13 DSM-XC domains can help pinpoint which cross-cutting symptoms are more effective in either confirming or ruling out specific diagnoses.

Previous studies suggested the DSM-XC (Bastiaens & Galus, 2018; Doss & Lowmaster, 2022; Mahoney et al., 2020) could be used for screening, but they lacked the use of diagnostic interviews and assessed convenience samples with limited generalizability and ecological validity. Additionally, the DSM-XC merges questioning about the intensity (“how much”) and frequency (“how often”) of symptoms, which could impact its psychometric and screening properties (Gong et al., 2013; Krabbe & Forkmann, 2014; Krishnakumar et al., 2021; Levin, Schneider, & Gaeth, 1998), but empirical studies on the impact on screening utility of different framings is not known.

Our aim was to assess the screening properties (sensitivity, specificity, and likelihood ratios [LRs]) of the DSM-XC’s domains to detect 8 mental disorders, representative of both internalizing and externalizing spectra, in the community. We will address this by assessing the instruments’ original operationalization merging intensity and frequency of symptoms and conduct an analysis for both framings separately.

Methods

Assessments

Participants

We analysed data from the latest assessment of 22 years old of an on-going longitudinal population-based study of the city of Pelotas, Brazil, that recruited all children born alive in 1993 (Victora et al., 2008), with a retention rate of 76.3%. Detailed information on the cohort is described elsewhere (Gonçalves et al., 2018; Victora et al., 2006). Only individuals with completed DSM-XC and MINI data were included in the present study. Written informed consent was obtained from all participants. The study was approved by the Institutional Review Board of the School of Medicine, Universidade Federal de Pelotas (approval number 1.250.366).

The DSM-5 Level 1 Cross-Cutting Symptom Measure

The DSM-XC is a self-reported measure that examines different mental health domains to assist clinicians in identifying comorbidity that could impact a patient's treatment and prognosis (American Psychiatric Association, 2013). The cross-cutting measures for adults involve two levels of inquiry. At Level 1, the questionnaire consists of 23 items distributed in 13 domains: depression, anger, mania, anxiety, somatic symptoms, suicidal ideation, psychosis, sleep problems, memory, repetitive thoughts and behaviours, dissociation, personality functioning, and substance use. Items ask, “how much (or how often) have you been bothered by” a given problem during the previous two weeks and provide a 5-point Likert scale ranging from 0 = “None (not at all)” to 4 = “Severe (nearly every day)”. A rating of mild (i.e., 2) or greater on any item within a domain, except for substance use, suicidal ideation, and psychosis is considered a positive screen. For substance use, suicidal ideation, and psychosis, a rating of slight (i.e., 1) or greater on any item within the domain is considered a positive domain. The threshold scores in any of its domains indicate the need for a thorough clinical investigation with the respective Level 2 Cross-Cutting Symptom Measure. Level 2 questions offer a more detailed assessment of specific domains. Except for suicidal ideation, psychosis, memory problems, dissociation and personality functioning, each domain in the adult version of the DSM-XC corresponds to a Level 2 measure, which contains more questions about each screened disorder. Additional details on Level 1 and Level 2 measures can be found online at www.psychiatry.org/dsm5.

In the original DSM-XC, there are two simultaneous framings when asking about symptoms: intensity (i.e., how much) and frequency (i.e., how often), requiring the respondent to choose a single answer. After a pilot test in Brazil, we identified that some participants had difficulty understanding the double-barrelled questions, which led to the decision to split each question into two, and answers were recorded on two separate 5-point scales: one for intensity and another for frequency. To compute the score as intended by the original DSM-XC “OR” rule, the highest score in each question was considered when analysed jointly. For example, an answer of 0 (none) for how much and 2 (several days) for how often, or vice versa, was coded as 2. An additional adaptation was made to the instrument by dividing the last item (item 23) into two (items 23 and 24, as shown in Supplementary Table 1) to allow for the exploration of substance use and self-prescription/misuse separately.

Assessment of psychiatric diagnoses

Participants were also assessed for the presence of psychiatric disorders with an adapted Brazilian version of the MINI International Neuropsychiatric Interview (MINI) (Amorim, 2000), administered by trained clinical psychologists, who did not have access to the DSM-XC responses. The MINI 5.0.0 is a structured and standardized diagnostic interview based on the 4th edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) (American Psychiatric Association, 1994), and was used to assess the following disorders: antisocial personality disorder (APD), attention-deficit/hyperactivity disorder (ADHD), bipolar disorder (BD), generalized anxiety disorder (GAD), major depressive disorder (MDD), post-traumatic stress disorder (PTSD) and social anxiety disorder (SAD). Moreover, to capture the DSM’s impairment criterion, we included an additional question inquiring participants how much impairment the reported symptoms caused in their life, with response options of none, mild, moderate or severe (Manfro et al., 2021; Matte et al., 2015). The presence of clinical impairment was operationally defined as having a score of moderate or severe. The Alcohol Use Disorders Identification Test (AUDIT), a 10-question instrument, was used to identify participants with active alcohol use disorder (AUD). A score of ≥ 15 for men and ≥ 13 for women was considered positive. The AUDIT validity has been previously investigated in Brazil (Meneses-Gaya et al., 2010).

Data analyses

We followed the Standards for Reporting of Diagnostic Accuracy Studies (STARD) 2015 guidelines (Bossuyt et al., 2015), reported in Supplementary table 2.

Screening properties

We estimated the number of true positives, false positives, true negatives, and false negatives for each of the 13 DSM-XC domains to detect individuals with psychiatric disorders. We then estimated the sensitivity, specificity, positive (LR+) and negative (LR-) likelihood ratios (Altman & Bland, 1994; Grimes & Schulz, 2005) for each domain-diagnosis pair.

Sensitivities and specificities higher than 0.75 (Power et al., 2013), LR+ values higher than 2 and LR- values lower than 0.5 were considered clinically useful (Grimes & Schulz, 2005; Jaeschke, Guyatt, & Sackett, 1994). Exact binomial 95% confidence intervals (95% CI) were calculated for sensitivity and specificity (Collett, 2002). Confidence intervals for LR+ and LR- values are based on formulae provided by Simel, Samsa & Matchar (1991).

Analysis of variability in screening properties by framing

We have first described the by-item response differences and calculated the polychoric correlation and Cohen's kappa to measure the reliability between frames for each DSM-XC item. Second, we calculated and compared all screening properties between framings. Overlapping confidence intervals were considered statistically equivalent. Paired Wilcoxon tests with DSM-XC domains as dependent variables were conducted to compare the differences in screening properties between the two framings. Screening properties of the two framings were also compared with the OR rule (reference group) with paired Wilcoxon tests with DSM-XC domains as dependent variables.

Logistic regression models were used to test potential interactions between intensity and frequency that could indicate an advantage of a particular framing. The odds ratios for specific combinations, namely Intensity (-) / Frequency (+), Intensity (+) / Frequency (-) and Intensity (+) / Frequency (+) were calculated with Intensity (-) / Frequency (-) as the reference group for each domain with a corresponding disorder.

We also tested if clinical impairment could be a factor for potential differences in framings instead of the framing effect itself. Because of this, secondary analyses were conducted by removing the clinical impairment criteria. The diagnosis of AUD was based on AUDIT scores and was not included in this analysis.

All analyses were performed using the software R, version 4.1.3. Screening properties were evaluated using the epiR package, version 2.0.44 (Stevenson et al., 2022). Paired Wilcoxon

tests were conducted using the *rstatix* package, version 0.7.0 (Kassambara, 2021). Code is available at osf.io/q54yu.

Results

Sample description

A total of 3,578 subjects with completed DSM-XC and MINI data participated in the 2015 follow-up study. The mean age was 22.57 years. Participants were mainly of the female sex (53.58%). Most individuals had working experience (94.10%), 63.30% had finished high school, and 84.13% were single. At least one psychiatric diagnosis was present in 813 subjects (22.72%). Sample characteristics are provided in Table 1. The prevalence of specific diagnoses ranged from 1.00% (APD) to 10.56% (GAD) (Table 1). A positive screen in the anxiety domain was observed in 50% of participants, followed by substance use at 47% and depression at 43%. Other prevalent domains included somatic symptoms (36%), anger (32%), and personality functioning (30%). Suicidal ideation (9%), psychosis (14%), and dissociation (16%) were the least frequent domains. Full details of the number and proportion of positive screens for each domain is provided in Supplementary table 3. Cross-tabulation of the DSM-XC domains by the results of the reference standards (MINI diagnoses and AUDIT) are provided in Supplementary Tables 4 to 19.

Sensitivity

The anxiety domain demonstrated excellent sensitivity across all internalizing disorders, with sensitivity values ranging from 0.86 to 0.95 (Table 2). In particular, the depression domain exhibited good sensitivity for both GAD (0.80, 95% CI, 0.76-0.84) and MDD (0.95, 95% CI, 0.89-0.98). The personality functioning domain showed adequate sensitivity for MDD (0.86, 95% CI, 0.78-0.92). For externalizing disorders, only the substance use domain displayed good sensitivity for AUD (0.87, 95% CI, 0.81-0.91).

Specificity

Various domains, including suicidal ideation, psychosis, memory, repetitive thoughts and behaviours, and dissociation, exhibited good specificity across all tested disorders (Table 2). Specifically, suicidal ideation had a specificity range of 0.91 to 0.92, psychosis ranged from 0.86 to 0.87, memory from 0.83 to 0.86, repetitive thoughts and behaviours from 0.81 to 0.84,

and dissociation from 0.84 to 0.87. Sleep problems showed good specificity for both GAD (0.79, 95% CI, 0.77-0.80) and PTSD (0.77, 95% CI, 0.76-0.78).

Likelihood ratios

Differences in LR+ values were observed between internalizing and externalizing disorders (Table 3). Several domains, namely suicidal ideation, repetitive thoughts and behaviours, dissociation, and personality functioning, exhibited clinically meaningful LR+ values for all internalizing disorders, but not for externalizing disorders.

The anxiety domain demonstrated meaningful LR- values for all internalizing disorders, ranging from 0.09 to 0.27. The depression domain also displayed good LR- values for most internalizing disorders, specifically GAD, MDD, and PTSD. Other domains, such as anger, personality functioning, and repetitive thoughts and behaviours showed meaningful LR- values for specific disorders. For externalizing disorders, only the substance use domain presented acceptable LR- values for AUD.

Variability in above-threshold domains and screening properties by adverbial framing

Polychoric correlation coefficients between “intensity” and “frequency” framings ranged from 0.77 to 0.90 and Kappa coefficient ranged from 0.50 to 0.66. Detailed results are shown in Supplementary Table 1, and the item response distribution for the DSM-XC is shown in Supplementary Figure 1.

Intensity-framed questions resulted in a similar number of above-threshold domains compared to the original rule (Supplementary Table 2). Frequency-framed questions showed a lower number of above-threshold domains, which was accompanied by significant improvements in LR+ values (Figure 1) mainly for the depression, anger, and anxiety domains. LR- values were not significantly impacted by the framing variation (Figure 2). Wilcoxon tests comparing framings across all DSM-XC domains showed that intensity-framed questions present marginally better sensitivity and LR- values, but worse specificity, and LR+ values when compared to frequency-framed questions (Supplementary Figures 2 and 3, Supplementary tables 20 to 23). The intensity framing showed minor differences in sensitivity, specificity and LR+ when compared to the OR rule, while the frequency framing showed lower sensitivity, but higher specificity and LR+ compared to the OR rule. Differences in LR- were very small for both framings compared to the OR rule (Supplementary tables 24 to 27).

Logistic regressions of specific framing combinations (Supplementary table 28) indicate that for mania-BD, personality functioning-APD, and substance use-AUD pairs, the Intensity (-) / Frequency (+) group had significant higher odds compared to the reference group, while the Intensity (+) / Frequency (-) had not. For the depression domain to detect MDD and anxiety domain to detect GAD and SAD, all operationalizations had a higher likelihood of the corresponding disorder.

Full details of the screening properties by framing are described in Supplementary Tables 29 to 52.

Secondary analyses showed that the findings remained stable even without the MINI clinical impairment criterion (Supplementary figures 4 to 7, Supplementary tables 53 to 78).

Discussion

This is the first study to describe the screening properties of the DSM-XC for a broad spectrum of psychiatric diagnoses in a large community sample. Results suggest that screening for psychiatric conditions using simple and objective operational criteria increases the likelihood of detection or may help to rule out specific diagnoses. The results comprise three main findings: First, the DSM-XC domains are not diagnostic-specific, rather, they function as transdiagnostic screening tools. For example, a negative anxiety domain is useful in decreasing and a positive suicidal ideation domain is useful in increasing the probability of many internalizing disorders. Second, the screening utility of the DSM-XC is lower for externalizing disorders. Third, “intensity” and “frequency” framings as measured by the DSM-XC are only moderately correlated, and the “frequency” framing exhibited a lower number of above-threshold domains and improved screening properties compared to “intensity” framing.

While several screening tools exist for specific diagnosis, few address multiple diagnostic categories simultaneously. Despite limitations in externalizing disorder assessment, for example due to the lack of ADHD domain in the DSM-XC and the lack of a diagnostic interview for AUD in the present cohort, the DSM-XC performance for depression and anxiety is comparable to existing scales, justifying its potential applicability in community settings. The Patient Health Questionnaire-2 (PHQ-2) (Kroenke, Spitzer & Williams, 2003) assesses similar symptoms to the depression domain of the DSM-XC, querying low mood and loss of interest over the past 2 weeks, despite differences in instructions and scoring scales. In

the present study, we found screening properties of the DSM-XC “OR” rule depression domain to detect MDD (LR+ = 2.32, LR- = 0.08, sensitivity = 0.95, specificity = 0.59) to be similar to recent meta-analytic evidence examining the PHQ-2 (LR+ = 2.97, LR- = 0.26, sensitivity = 0.89, specificity = 0.75) (Mitchell et al., 2016). The Generalized Anxiety Disorder two questions scale (GAD-2) with a cut-off of 3 presents more informative properties (LR+ = 4.31, LR- = 0.25, sensitivity = 0.80, specificity = 0.81) (Plummer et al., 2016) compared to the DSM-XC anxiety domain to detect GAD (original “OR” rule LR+ 1.90, LR- = 0.24, sensitivity = 0.87, specificity = 0.54). The differences in performance between legacy measures with similar conceptual content and DSM-XC domains may be attributed to variations in item content, scale and scoring methods (linear vs. algorithm). For instance, the DSM-XC anxiety domain comprises 3 items, includes more symptoms per item and assesses different symptoms, such as panic and anxious avoidance, while the GAD-2 has only 2 items focusing on anxiety and worrying. Additionally, both PHQ-2 and GAD-2 use 4-point scales, while the DSM-XC is rated on a 5-point scale.

A lack of direct correspondence was observed across domains and psychiatric diagnoses. Moreover, some domains have a higher capability than others for screening, and some domains showed utility in screening for multiple disorders (transdiagnostic utility). For example, the depression domain showed a LR+ of 2.32 for detecting MDD, 2.11 for detecting GAD, and 1.94 for PTSD. Conversely, the anxiety domain showed significant LR- values for all internalizing disorders, suggesting its potential as a first screening step. This finding contrasts with the current understanding of the DSM-XC process of further inquiring an individual with disorder-specific questions if they screen positive in a domain. However, the DSM-XC did not show meaningful utility for externalizing disorders, except for a good likelihood ratio of a negative screen in the substance use domain to rule out AUD. Due to identified shortcomings, specific recommended scales for alcohol and substance use should supplement primary care assessments using the DSM-XC. Further modifications in the tool are necessary to address externalizing symptoms, such as impulsivity and disruptive behaviors, aiming at increasing overall screening utility. These targeted improvements could broaden the DSM-XC's scope and solidify its place as a component of community screening strategies.

Strong connections observed in certain domains' screening capabilities likely reflect the interconnectedness of the disorders they captured. The presence of clinical features that are not represented in the diagnostic criteria but are frequent among individuals with certain

mental disorders is a potential explanation for some domains demonstrating screening utility for multiple disorders. For instance, somatic symptoms, while not part of diagnostic criteria, may be more commonly experienced among individuals with GAD and MDD, while anxious distress is a notable feature in both bipolar and major depressive disorders (Bartoli et al., 2023). Another contributing factor is the conceptual similarity and symptom overlap across various diagnostic criteria, with insomnia and irritable mood being among the top non-specific symptoms (Forbes et al., 2023). In our study, sleep problems and anger, for example, demonstrated significant positive likelihood ratios across anxiety, mood, and post-traumatic stress disorder categories. Comorbidity and common temperamental risk factors can contribute to this phenomenon, as individuals with high levels of negative affectivity (neuroticism) are likely to endorse multiple internalizing symptom domains in a non-specific manner due to being oriented towards negative affectivity (Griffith et al., 2010). Additionally, substantial evidence supports the idea that common neural and genetic mechanisms contribute to the association between these disorders (Xie et al., 2023, Jami et al., 2022).

We also examined the contributions of framing questions as intensity or frequency of symptoms, as the questionnaire was revised based on participant feedback regarding the challenge in understanding double-barrelled questions. The DSM-XC showed moderate correlation between framings, with intensity-framed questions leading to a significantly higher proportion of positive screens for most domains compared to frequency-framed questions. For some domains, such as depression, anger, and memory, framing questions as “intensity” approximately doubled the number of individuals that would have been needed to complete Level 2 Measures, compared with “frequency”. Thus, using frequency-framed questions may be a more resource-efficient approach.

Framings also differed in screening properties, with frequency-framed questions presenting generally higher specificity but equivalent sensitivity to detect their correspondent specific disorders. Differences in LR+ favouring the use of the “frequency” framing were found, with minimal compromise in LR- values. The value of likelihood ratios for clinical decision making is significant (Grimes & Schulz, 2005). For example, a positive result in the anxiety domain, with a likelihood ratio of 1.95, translates to a shift from the pre-test probability of GAD from 10.5% to a post-test probability of 18.3%. However, by employing the “frequency” framing, with a LR+ value of 3.11, the post-test probability substantially increases to 26.8% for GAD. As the conversion from pre-test to post-test probability involves

a transformation into odds, we have developed an accessible tool, available at https://pacheco-jpg.shinyapps.io/dsm_xc_shiny_app/ that enables quick estimation of the shift from pre- to post-test probability for the tested disorders within the DSM-XC domains.

Apart from establishing improved screening utility for the DSM-XC, defining standards for adverbial framing of questions can reduce the substantial heterogeneity in mental health assessment (Newson, Hunter, & Thiagarajan, 2020). Previous work suggests higher reliability, criterion validity (Krishnakumar et al., 2021) and stability over time (Krabbe & Forkmann, 2014) for frequency in comparison to intensity, which could explain the higher accuracy of the frequency framing that we found in the present study.

This study has three main strengths. First, we examined a community-based sample, which allows for real-world estimates of the screening properties of the DSM-XC. Second, we investigated a broad spectrum of psychiatric diagnoses, with examples of both internalizing and externalizing disorders. Third, we removed a previous restriction imposed by double-barrelled phrasing, which allowed for examining the contributions of intensity and frequency framings. However, interpreting these results requires considering several limitations. First, results should not be necessarily extrapolated for other age groups as the sample was limited to the age of 22. Second, although the presence of most disorders was determined by a qualified clinical psychologist with a structured interview, we used the AUDIT to evaluate AUD. Third, further research is needed to directly compare the implemented “OR” rule with the original double-barrelled scale. The “OR” rule allowed to characterize the intended meaning of the DSM-XC, which is an important as it addresses the cognitively demanding phrasing of the original scale. A notable concern with the implemented approach is that the order that questions were presented could have affected how individuals rated each item. As the “intensity” framing was always presented first, this piece of information could have influenced how participants answered the following “frequency” question, introducing anchoring bias. However, anchoring bias tends to make results look more similar and we found only moderate correlations between framings. Additionally, the frequency framing, presented after intensity, showed better screening properties, and logistic regression analyses of specific framing subgroups suggests the frequency framing captures relevant information of specific corresponding disorders on more occasions than the intensity framing alone. This suggests that anchoring bias did not occur as frequency framing is not conditioned to intensity framing.

The present results demonstrated that the DSM-XC is useful in screening multiple domains of psychopathology, increasing or decreasing the likelihood of detection of specific diagnoses with simple objective questions. Due to the low direct correspondence between domain and diagnosis, more than one Level 2 Cross-Cutting Symptom Measures could apply after a positive screening in some domains. We also found that adverbial framings have an impact on the DSM-XC's screening performance. Framing questions as "frequency" may be more resource-efficient and could lead to improved detection of positive cases. We recommend that these findings should be replicated in other samples to allow implementation in routine practice and that future examination of scoring thresholds should be done to further optimize the DSM-XC as a screening tool for mental health conditions.

References

- Altman, D. G., & Bland, J. M. (1994). Statistics Notes: Diagnostic tests 1: sensitivity and specificity. *BMJ*, 308(6943), 1552–1552. doi:10.1136/bmj.308.6943.1552
- American Psychiatric Association. (1994). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: American Psychiatric Publishing.
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Arlington, VA: American Psychiatric Publishing.
- Amorim, P. (2000). Mini International Neuropsychiatric Interview (MINI): Mini International Neuropsychiatric Interview (MINI): validation of a short structured diagnostic psychiatric interview. *Brazilian Journal of Psychiatry*, 22(3), 106–115. doi:10.1590/S1516-44462000000300003
- Bartoli, F., Bachi, B., Callovin, T., Palpella, D., Piacenti, S., Morreale, M., ... Carrà, G. (2024). Anxious distress in people with major depressive episodes: a cross-sectional analysis of clinical correlates. *CNS Spectrums*, 29(1), 49–53. doi:10.1017/S1092852923002377
- Bastiaens, L., & Galus, J. (2018). The DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure as a Screening Tool. *Psychiatric Quarterly*, 89(1), 111–115. <https://doi.org/10.1007/s11126-017-9518-7>
- Bjelland, I., Dahl, A. A., Haug, T. T., & Neckelmann, D. (2002). The validity of the Hospital Anxiety and Depression Scale. *Journal of Psychosomatic Research*, 52(2), 69–77. doi:10.1016/S0022-3999(01)00296-3

- Bossuyt, P. M., Reitsma, J. B., Bruns, D. E., Gatsonis, C. A., Glasziou, P. P., Irwig, L., ... Cohen J. F. (2015). STARD 2015: an updated list of essential items for reporting diagnostic accuracy studies. *BMJ*, 351, h5527. doi:10.1136/bmj.h5527
- Collett, D. (2002). *Modelling Binary Data* (2nd ed.). New York, NY: Chapman and Hall/CRC. doi:10.1201/b16654
- Doss, R. A., & Lowmaster, S. E. (2022). Validation of the DSM-5 Level 1 Cross-Cutting Symptom Measure in a Community Sample. *Psychiatry Research*, 318, 114935. doi:10.1016/j.psychres.2022.114935
- Eaton, N. R., Bringmann, L. F., Elmer, T., Fried, E. I., Forbes, M. K., Greene, A. L., ... Waszczuk, M. A. (2023). A review of approaches and models in psychopathology conceptualization research. *Nature Reviews Psychology*, 2, 622–636. doi:10.1038/s44159-023-00218-4
- Forbes, M. K., Neo, B., Nezami, O. M., Fried, E. I., Faure, K., Michelsen, B., Twose, M., & Dras, M. (2023). *Elemental psychopathology: Distilling constituent symptoms and patterns of repetition in the diagnostic criteria of the DSM-5* [Preprint]. PsyArXiv. doi:10.31234/osf.io/u56p2
- Forbes, M. K., Tackett, J. L., Markon, K. E., & Krueger, R. F. (2016). Beyond comorbidity: Toward a dimensional and hierarchical approach to understanding psychopathology across the life span. *Development and Psychopathology*, 28(4pt1), 971–986. doi:10.1017/S0954579416000651
- Gibbons, A. B., Farmer, C., Shaw, J. S., & Chung, J. Y. (2023). Examining the factor structure of the DSM-5 Level 1 cross-cutting symptom measure. *International Journal of Methods in Psychiatric Research*, 32(2), e1953. doi:10.1002/mpr.1953
- Gonçalves, H., Wehrmeister, F. C., Assunção, M. C. F., Tovo-Rodrigues, L., Oliveira, I. O. de, Murray, J., ... Menezes, A. M. B. (2018). Cohort Profile Update: The 1993 Pelotas (Brazil) Birth Cohort follow-up at 22 years. *International Journal of Epidemiology*, 47(5), 1389–1390e. doi:10.1093/ije/dyx249
- Gong, J., Zhang, Y., Yang, Z., Huang, Y., Feng, J., & Zhang, W. (2013). The framing effect in medical decision-making: A review of the literature. *Psychology, Health & Medicine*, 18(6), 645–653. doi:10.1080/13548506.2013.766352
- Griffith, J. W., Zinbarg, R. E., Craske, M. G., Mineka, S., Rose, R. D., Waters, A. M., & Sutton, J. M. (2010). Neuroticism as a common dimension in the internalizing disorders. *Psychological Medicine*, 40(7), 1125–1136. doi:10.1017/S0033291709991449

- Grimes, D. A., & Schulz, K. F. (2005). Refining clinical diagnosis with likelihood ratios. *The Lancet*, 365(9469), 1500–1505. doi:10.1016/S0140-6736(05)66422-7
- Gryczynski, J., Kelly, S. M., Mitchell, S. G., Kirk, A., O'Grady, K. E., & Schwartz, R. P. (2015). Validation and performance of the Alcohol, Smoking and Substance Involvement Screening Test (ASSIST) among adolescent primary care patients. *Addiction*, 110(2), 240–247. doi:10.1111/add.12767
- Jaeschke, R., Guyatt, G. H., & Sackett, D. L. (1994). Users' guides to the medical literature. III. How to use an article about a diagnostic test. B. What are the results and will they help me in caring for my patients? The Evidence-Based Medicine Working Group. *JAMA*, 271(9), 703–707. doi:10.1001/jama.271.9.703
- Jami, E. S., Hammerschlag, A. R., Ip, H. F., Allegrini, A. G., Benyamin, B., Border, R., ... Middeldorp, C. M. (2022). Genome-wide Association Meta-analysis of Childhood and Adolescent Internalizing Symptoms. *Journal of the American Academy of Child & Adolescent Psychiatry*, 61(7), 934–945. doi:10.1016/j.jaac.2021.11.035
- Kassambara, A. (2021). *Package 'rstatix'—Pipe-Friendly Framework for Basic Statistical Tests*. Retrieved from <https://CRAN.R-project.org/package=rstatix>
- Krabbe, J., & Forkmann, T. (2014). Frequency vs. intensity: Framing effects on patients' use of verbal rating scale anchors. *Comprehensive Psychiatry*, 55(8), 1928–1936. doi:10.1016/j.comppsy.2014.06.010
- Krishnakumar, A., Scopel Hoffmann, M., Schoeller, F., Clucas, J. C., Son, J., Müller-Naendrup, L., Salum, G., Milham, M., Lindner, A., & Klein, A. (2021). *Framing mental health questions using context better predicts disability than does frequency* [Preprint]. PsyArXiv. doi:10.31234/osf.io/jm9kc
- Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2003). The Patient Health Questionnaire-2: Validity of a Two-Item Depression Screener. *Medical Care*, 41(11), 1284–1292. doi:10.1097/01.MLR.0000093487.78664.3C
- Levin, I. P., Schneider, S. L., & Gaeth, G. J. (1998). All Frames Are Not Created Equal: A Typology and Critical Analysis of Framing Effects. *Organizational Behavior and Human Decision Processes*, 76(2), 149–188. doi:10.1006/obhd.1998.2804
- Löwe, B., Kroenke, K., & Gräfe, K. (2005). Detecting and monitoring depression with a two-item questionnaire (PHQ-2). *Journal of Psychosomatic Research*, 58(2), 163–171. doi:10.1016/j.jpsychores.2004.09.006

- Löwe, B., Wahl, I., Rose, M., Spitzer, C., Glaesmer, H., Wingenfeld, K., ... Brähler, E. (2010). A 4-item measure of depression and anxiety: Validation and standardization of the Patient Health Questionnaire-4 (PHQ-4) in the general population. *Journal of Affective Disorders*, 122(1–2), 86–95. doi:10.1016/j.jad.2009.06.019
- Mahoney, M. R., Farmer, C., Sinclair, S., Sung, S., Dehaut, K., & Chung, J. Y. (2020). Utilization of the DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure-Adult to Screen Healthy Volunteers for Research Studies. *Psychiatry Research*, 286, 112822. doi:10.1016/j.psychres.2020.112822
- Manfro, P. H., Belem da Silva, C. T., Anselmi, L., Barros, F., Eaton, W. W., Gonçalves, H., Murray, J., Oliveira, I. O., Tovo-Rodrigues, L., Wehrmeister, F. C., Menezes, A. M. B., Rohde, L. A., & Kieling, C. (2021). Depression in a youth population-based sample from Brazil: Prevalence and symptom structure. *Journal of Affective Disorders*, 292, 633–641. doi:10.1016/j.jad.2021.05.073
- Matte, B., Anselmi, L., Salum, G. A., Kieling, C., Gonçalves, H., Menezes, A., ... Rohde, L. A. (2015). ADHD in DSM-5: A field trial in a large, representative sample of 18- to 19-year-old adults. *Psychological Medicine*, 45(2), 361–373. doi:10.1017/S0033291714001470
- McNeely, J., Wu, L.-T., Subramaniam, G., Sharma, G., Cathers, L. A., Svikis, D., ... Schwartz, R. P. (2016). Performance of the Tobacco, Alcohol, Prescription Medication, and Other Substance Use (TAPS) Tool for Substance Use Screening in Primary Care Patients. *Annals of Internal Medicine*, 165(10), 690. doi:10.7326/M16-0317
- Means-Christensen, A. J., Arnau, R. C., Tonidandel, A. M., Bramson, R., & Meagher, M. W. (2005). An Efficient Method of Identifying Major Depression and Panic Disorder in Primary Care. *Journal of Behavioral Medicine*, 28(6), 565–572. doi:10.1007/s10865-005-9023-6
- Meneses-Gaya, C., Zuardi, A. W., Loureiro, S. R., Hallak, J. E., Trzesniak, C., de Azevedo Marques, J. M., ... Crippa, J. A. (2010). Is the full version of the AUDIT really necessary? Study of the validity and internal construct of its abbreviated versions. *Alcoholism, clinical and experimental research*, 34(8), 1417–1424. doi:doi.org/10.1111/j.1530-0277.2010.01225.x
- Mitchell, A. J., Yadegarfar, M., Gill, J., & Stubbs, B. (2016). Case finding and screening clinical utility of the Patient Health Questionnaire (PHQ-9 and PHQ-2) for depression

- in primary care: A diagnostic meta-analysis of 40 studies. *BJPsych Open*, 2(2), 127–138. doi:10.1192/bjpo.bp.115.001685
- Mulvaney-Day, N., Marshall, T., Downey Piscopo, K., Korsen, N., Lynch, S., Karnell, L. H., ... Ghose, S. S. (2018). Screening for Behavioral Health Conditions in Primary Care Settings: A Systematic Review of the Literature. *Journal of General Internal Medicine*, 33(3), 335–346. doi:10.1007/s11606-017-4181-0
- Negeri, Z. F., Levis, B., Sun, Y., He, C., Krishnan, A., Wu, Y., ... Thombs, B. D. (2021). Accuracy of the Patient Health Questionnaire-9 for screening to detect major depression: Updated systematic review and individual participant data meta-analysis. *BMJ*, n2183. doi:10.1136/bmj.n2183
- Newson, J. J., Hunter, D., & Thiagarajan, T. C. (2020). The Heterogeneity of Mental Health Assessment. *Frontiers in Psychiatry*, 11, 76. doi:10.3389/fpsy.2020.00076
- Plana-Ripoll, O., Pedersen, C. B., Holtz, Y., Benros, M. E., Dalsgaard, S., de Jonge, P., ... McGrath, J. J. (2019). Exploring Comorbidity Within Mental Disorders Among a Danish National Population. *JAMA Psychiatry*, 76(3), 259. <https://doi.org/10.1001/jamapsychiatry.2018.3658>
- Plummer, F., Manea, L., Trepel, D., & McMillan, D. (2016). Screening for anxiety disorders with the GAD-7 and GAD-2: A systematic review and diagnostic meta-analysis. *General Hospital Psychiatry*, 39, 24–31. doi:10.1016/j.genhosppsych.2015.11.005
- Power, M., Fell, G., & Wright, M. (2013). Principles for high-quality, high-value testing. *Evidence Based Medicine*, 18(1), 5–10. doi:10.1136/eb-2012-100645
- Simel, D. L., Samsa, G. P., & Matchar, D. B. (1991). Likelihood ratios with confidence: Sample size estimation for diagnostic test studies. *Journal of Clinical Epidemiology*, 44(8), 763–770. doi:10.1016/0895-4356(91)90128-V
- Smith, P. C., Schmidt, S. M., Allensworth-Davies, D., & Saitz, R. (2009). Primary Care Validation of a Single-Question Alcohol Screening Test. *Journal of General Internal Medicine*, 24(7), 783–788. doi:10.1007/s11606-009-0928-6
- Stevenson, M., Sergeant E. Heuer C., Nunes T., Heuer C., Marshall J., ... Rabiee A. (2020). *Package 'epiR'—Tools for the Analysis of Epidemiological Data*. Retrieved from <https://CRAN.R-project.org/package=epiR>
- US Preventive Services Task Force, Barry, M. J., Nicholson, W. K., Silverstein, M., Coker, T. R., Davidson, K. W., ... Wong, J. B. (2023a). Screening for Anxiety Disorders in Adults: US Preventive Services Task Force Recommendation Statement. *JAMA*, 329(24), 2163–2170. doi:10.1001/jama.2023.9301

- US Preventive Services Task Force, Barry, M. J., Nicholson, W. K., Silverstein, M., Chelmow, D., Coker, T. R., ... Wong, J. B. (2023b). Screening for Depression and Suicide Risk in Adults: US Preventive Services Task Force Recommendation Statement. *JAMA*, 329(23), 2057–2067. doi:10.1001/jama.2023.9297
- US Preventive Services Task Force, Curry, S. J., Krist, A. H., Owens, D. K., Barry, M. J., Caughey, A. B., ... Wong, J. B. (2018). Screening and Behavioral Counseling Interventions to Reduce Unhealthy Alcohol Use in Adolescents and Adults: US Preventive Services Task Force Recommendation Statement. *JAMA*, 320(18), 1899–1909. doi:10.1001/jama.2018.16789
- US Preventive Services Task Force, Krist, A. H., Davidson, K. W., Mangione, C. M., Barry, M. J., Cabana, M., ... Wong, J. B. (2020). Screening for Unhealthy Drug Use: US Preventive Services Task Force Recommendation Statement. *JAMA*, 323(22), 2301–2309. doi:10.1001/jama.2020.8020
- Victora, C. G., Araújo, C. L. P., Menezes, A. M. B., Hallal, P. C., Vieira, M. de F., Neutzling, M. B., ... Barros, F. C. (2006). Methodological aspects of the 1993 Pelotas (Brazil) birth cohort study. *Revista de Saúde Pública*, 40(1), 39–46. doi:10.1590/S0034-89102006000100008
- Victora, C. G., Hallal, P. C., Araujo, C. L., Menezes, A. M., Wells, J. C., & Barros, F. C. (2008). Cohort Profile: The 1993 Pelotas (Brazil) Birth Cohort Study. *International Journal of Epidemiology*, 37(4), 704–709. doi:10.1093/ije/dym177
- Vilagut, G., Forero, C. G., Barbaglia, G., & Alonso, J. (2016). Screening for Depression in the General Population with the Center for Epidemiologic Studies Depression (CES-D): A Systematic Review with Meta-Analysis. *PLoS ONE*, 11(5), e0155431. doi:10.1371/journal.pone.0155431
- Waszczuk, M. A., Zimmerman, M., Ruggero, C., Li, K., MacNamara, A., Weinberg, A... Kotov, R. (2017). What do clinicians treat: Diagnoses or symptoms? The incremental validity of a symptom-based, dimensional characterization of emotional disorders in predicting medication prescription patterns. *Comprehensive Psychiatry*, 79, 80–88. doi:10.1016/j.comppsy.2017.04.004
- Xie, C., Xiang, S., Shen, C., Peng, X., Kang, J., Li, Y., ... ZIB Consortium. (2023). A shared neural basis underlying psychiatric comorbidity. *Nature Medicine*, 29(5), 1232–1242. doi:10.1038/s41591-023-02317-4

Tables and figures

Table 1. Sample description

Sample characteristics	Sample (n = 3,578)
Age, mean (SD), y	22.57 (0.33)
Sex	
Female, n (%)	1917 (53.58)
NEET, n (%)	716 (20.85)
Missing, n (%)	66 (1.84)
Attainment	
Incomplete primary or illiterate, n (%)	61 (1.70)
Primary, n (%)	1248 (34.88)
Secondary, n (%)	2065 (57.71)
Tertiary, n (%)	200 (5.59)
Missing, n (%)	4 (0.11)
Family income per capita	
BRL 0 to 1,254, n (%)	2584 (72.22)
BRL 1,255 to 2,004, n (%)	522 (14.59)
BRL 2,005 to 8,640, n (%)	287 (8.02)
BRL > 8,641, n (%)	6 (0.17)
Missing, n (%)	179 (5.00)
Ever worked, n (%)	3367 (94.10)
Currently working, n (%)	2267 (63.36)
Missing, n (%)	211 (5.90)
Marital status	
Married	548 (15.32)
Divorced	19 (0.53)
Single	3010 (84.13)

Widowed	1 (0.00)
Lifetime use of tobacco, n (%)	975 (27.25)
Lifetime use of alcoholic beverages, n (%)	3397 (94.94)
Prevalence of current psychiatric diagnoses	
Alcohol use disorder, n (%)	208 (5.81)
Antisocial personality disorder, n (%)	36 (1.00)
Attention-deficit/hyperactivity disorder, n (%)	162 (4.53)
Bipolar disorder, n (%)	60 (1.68)
Generalized anxiety disorder, n (%)	378 (10.56)
Major depressive disorder, n (%)	105 (2.93)
Post-traumatic stress disorder, n (%)	155 (4.33)
Social anxiety disorder, n (%)	178 (4.97)
Any diagnosis, n (%)	813 (22.72)

Note: NEET: Not in Education, Employment, or Training. BRL: Brazilian *Real* (Currency Unit).

Table 2. Sensitivity and specificity of DSM-XC domains to detect 8 psychiatric disorders.

Accuracy metric, 95% CI,	Domain	Internalizing disorders					Externalizing disorders		
		Generalized anxiety disorder	Social anxiety disorder	Major depressive disorder	Post-traumatic stress disorder	Bipolar disorder	Attention-deficit/hyperactivity disorder	Alcohol use disorder	Antisocial personality disorder
Sensitivity	Depression	0.80	0.77	0.95	0.79	0.77	0.69	0.59	0.58
		(0.76 - 0.84)	(0.70 - 0.83)	(0.89 - 0.98)	(0.72 - 0.85)	(0.64 - 0.87)	(0.61 - 0.76)	(0.52 - 0.66)	(0.41 - 0.74)
	Anger	0.70	0.61	0.76	0.72	0.75	0.59	0.50	0.44
		(0.65 - 0.74)	(0.53 - 0.68)	(0.67 - 0.84)	(0.64 - 0.79)	(0.62 - 0.85)	(0.51 - 0.67)	(0.43 - 0.57)	(0.28 - 0.62)
	Mania	0.43	0.46	0.46	0.55	0.65	0.43	0.40	0.42
		(0.38 - 0.48)	(0.38 - 0.53)	(0.36 - 0.56)	(0.47 - 0.63)	(0.52 - 0.77)	(0.35 - 0.51)	(0.34 - 0.47)	(0.26 - 0.59)
	Anxiety	0.87	0.86	0.95	0.94	0.95	0.75	0.63	0.69
		(0.83 - 0.90)	(0.80 - 0.91)	(0.89 - 0.98)	(0.88 - 0.97)	(0.86 - 0.99)	(0.67 - 0.81)	(0.57 - 0.70)	(0.52 - 0.84)
	Somatic Symptoms	0.67	0.60	0.75	0.72	0.78	0.54	0.49	0.50
		(0.62 - 0.71)	(0.53 - 0.67)	(0.66 - 0.83)	(0.65 - 0.79)	(0.66 - 0.88)	(0.46 - 0.62)	(0.42 - 0.56)	(0.33 - 0.67)
	Suicidal Ideation	0.20	0.25	0.38	0.27	0.33	0.19	0.15	0.11
		(0.16 - 0.25)	(0.19 - 0.32)	(0.29 - 0.48)	(0.20 - 0.35)	(0.22 - 0.47)	(0.13 - 0.25)	(0.10 - 0.20)	(0.03 - 0.26)
	Psychosis	0.28	0.30	0.34	0.39	0.53	0.29	0.29	0.39
		(0.23 - 0.32)	(0.24 - 0.38)	(0.25 - 0.44)	(0.32 - 0.48)	(0.40 - 0.66)	(0.22 - 0.37)	(0.23 - 0.36)	(0.23 - 0.57)
	Sleep Problems	0.54	0.56	0.69	0.65	0.57	0.47	0.48	0.39
		(0.49 - 0.59)	(0.48 - 0.63)	(0.59 - 0.77)	(0.57 - 0.73)	(0.43 - 0.69)	(0.39 - 0.55)	(0.41 - 0.55)	(0.23 - 0.57)
	Memory	0.38	0.42	0.39	0.46	0.48	0.37	0.28	0.25
		(0.33 - 0.43)	(0.34 - 0.49)	(0.30 - 0.49)	(0.38 - 0.55)	(0.35 - 0.62)	(0.30 - 0.45)	(0.22 - 0.35)	(0.12 - 0.42)
	Repetitive Thoughts and Behaviors	0.49	0.46	0.63	0.68	0.68	0.42	0.38	0.42
		(0.44 - 0.54)	(0.38 - 0.53)	(0.53 - 0.72)	(0.60 - 0.76)	(0.55 - 0.80)	(0.34 - 0.50)	(0.31 - 0.44)	(0.26 - 0.59)
	Dissociation	0.40	0.39	0.58	0.49	0.60	0.36	0.30	0.42
		(0.35 - 0.45)	(0.32 - 0.47)	(0.48 - 0.68)	(0.41 - 0.57)	(0.47 - 0.72)	(0.29 - 0.44)	(0.24 - 0.37)	(0.26 - 0.59)
	Personality Functioning	0.63	0.62	0.86	0.75	0.72	0.52	0.53	0.61
		(0.58 - 0.68)	(0.54 - 0.69)	(0.78 - 0.92)	(0.68 - 0.82)	(0.59 - 0.83)	(0.44 - 0.60)	(0.46 - 0.60)	(0.43 - 0.77)
	Substance Use	0.59	0.60	0.63	0.65	0.67	0.58	0.87	0.81
		(0.54 - 0.64)	(0.52 - 0.67)	(0.53 - 0.72)	(0.56 - 0.72)	(0.53 - 0.78)	(0.50 - 0.66)	(0.81 - 0.91)	(0.64 - 0.92)

Specificity	Depression	0.62	0.59	0.59	0.59	0.58	0.59	0.58	0.58
		(0.60 - 0.64)	(0.58 - 0.61)	(0.57 - 0.61)	(0.57 - 0.61)	(0.56 - 0.60)	(0.57 - 0.60)	(0.57 - 0.60)	(0.56 - 0.59)
	Anger	0.73	0.70	0.69	0.70	0.69	0.69	0.69	0.68
		(0.71 - 0.74)	(0.68 - 0.71)	(0.68 - 0.71)	(0.68 - 0.71)	(0.67 - 0.70)	(0.68 - 0.71)	(0.68 - 0.71)	(0.67 - 0.70)
	Mania	0.74	0.73	0.72	0.73	0.72	0.73	0.73	0.72
		(0.72 - 0.75)	(0.71 - 0.74)	(0.71 - 0.74)	(0.72 - 0.75)	(0.71 - 0.74)	(0.71 - 0.74)	(0.71 - 0.74)	(0.70 - 0.73)
	Anxiety	0.54	0.52	0.51	0.52	0.51	0.51	0.51	0.50
		(0.52 - 0.56)	(0.50 - 0.53)	(0.49 - 0.53)	(0.50 - 0.53)	(0.49 - 0.52)	(0.49 - 0.53)	(0.49 - 0.52)	(0.48 - 0.52)
	Somatic	0.68	0.65	0.65	0.66	0.65	0.65	0.65	0.64
	Symptoms	(0.66 - 0.69)	(0.64 - 0.67)	(0.64 - 0.67)	(0.64 - 0.67)	(0.63 - 0.66)	(0.63 - 0.66)	(0.63 - 0.66)	(0.62 - 0.66)
	Suicidal	0.92	0.92	0.92	0.92	0.92	0.92	0.91	0.91
	Ideation	(0.91 - 0.93)	(0.91 - 0.93)	(0.91 - 0.93)	(0.91 - 0.93)	(0.91 - 0.92)	(0.91 - 0.92)	(0.90 - 0.92)	(0.90 - 0.92)
	Psychosis	0.88	0.87	0.87	0.87	0.87	0.87	0.87	0.86
		(0.86 - 0.89)	(0.86 - 0.88)	(0.85 - 0.88)	(0.86 - 0.88)	(0.86 - 0.88)	(0.86 - 0.88)	(0.86 - 0.88)	(0.85 - 0.87)
	Sleep	0.79	0.77	0.77	0.77	0.76	0.76	0.77	0.75
	Problems	(0.77 - 0.80)	(0.75 - 0.78)	(0.75 - 0.78)	(0.76 - 0.78)	(0.74 - 0.77)	(0.75 - 0.78)	(0.75 - 0.78)	(0.74 - 0.77)
	Memory	0.86	0.85	0.84	0.85	0.84	0.84	0.84	0.83
		(0.85 - 0.87)	(0.83 - 0.86)	(0.83 - 0.85)	(0.84 - 0.86)	(0.83 - 0.85)	(0.83 - 0.86)	(0.83 - 0.85)	(0.82 - 0.85)
	Repetitive	0.84	0.82	0.82	0.83	0.81	0.81	0.82	0.81
	Thoughts and Behaviors	(0.83 - 0.85)	(0.80 - 0.83)	(0.80 - 0.83)	(0.81 - 0.84)	(0.80 - 0.83)	(0.80 - 0.83)	(0.80 - 0.83)	(0.79 - 0.82)
	Dissociation	0.87	0.85	0.85	0.86	0.85	0.85	0.85	0.84
		(0.86 - 0.88)	(0.84 - 0.86)	(0.84 - 0.86)	(0.84 - 0.87)	(0.84 - 0.86)	(0.84 - 0.86)	(0.84 - 0.86)	(0.83 - 0.85)
	Personality	0.74	0.72	0.72	0.72	0.71	0.71	0.72	0.70
	Functioning	(0.72 - 0.76)	(0.70 - 0.73)	(0.70 - 0.73)	(0.71 - 0.74)	(0.69 - 0.72)	(0.70 - 0.73)	(0.70 - 0.73)	(0.69 - 0.72)
	Substance Use	0.54	0.53	0.53	0.53	0.53	0.53	0.55	0.53
		(0.52 - 0.56)	(0.52 - 0.55)	(0.51 - 0.55)	(0.52 - 0.55)	(0.51 - 0.55)	(0.51 - 0.55)	(0.53 - 0.57)	(0.51 - 0.55)

Sensitivity represents the ability of the test to designate an individual with the disorder as positive, and values greater than 0.75 are shown in bold if crossing the confidence interval. Specificity represents the ability of the test to designate an individual without the disorder as negative, and values greater than 0.75 are shown in bold if crossing the confidence interval.

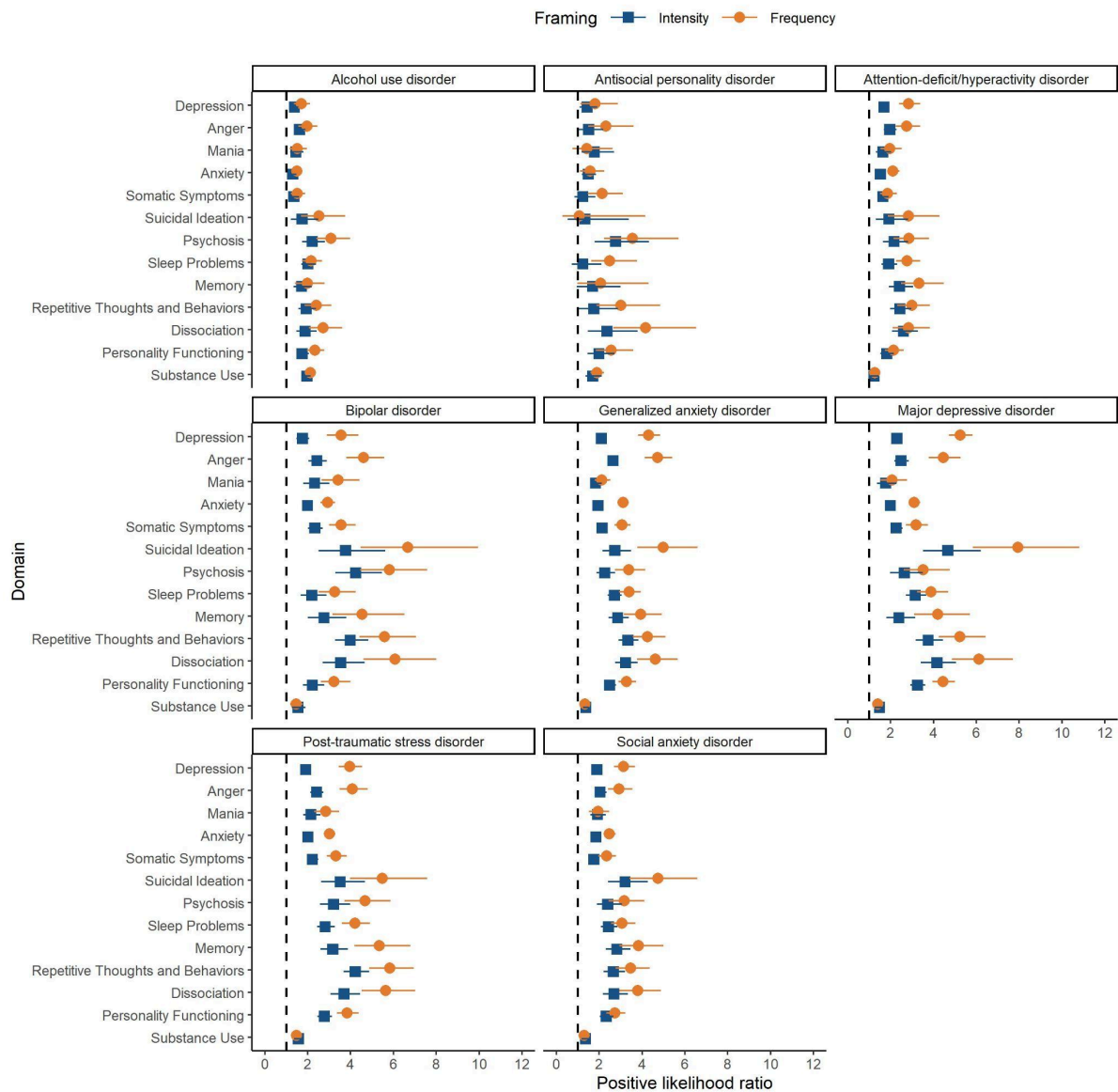
Table 3. Likelihood ratios of DSM-XC domains to detect 8 psychiatric disorders.

Estimate, 95% CI,	Domain	Internalizing disorders				Externalizing disorders			
		Generalized anxiety disorder	Social anxiety disorder	Major depressive disorder	Post-traumati c stress disorder	Bipolar disorder	Attention-defi cit/hyperactivi ty disorder	Alcohol use disorder	Antisocial personality disorder
Positive likelihood ratio	Depression	2.11 (1.97 - 2.25)	1.89 (1.72 - 2.06)	2.32 (2.19 - 2.46)	1.94 (1.77 - 2.12)	1.82 (1.58 - 2.11)	1.67 (1.50 - 1.87)	1.42 (1.26 - 1.60)	1.37 (1.04 - 1.82)
	Anger	2.54 (2.33 - 2.77)	1.99 (1.75 - 2.27)	2.49 (2.21 - 2.80)	2.38 (2.12 - 2.66)	2.40 (2.06 - 2.80)	1.93 (1.69 - 2.22)	1.60 (1.39 - 1.86)	1.40 (0.97 - 2.02)
	Mania	1.62 (1.43 - 1.85)	1.67 (1.41 - 1.98)	1.66 (1.33 - 2.05)	2.04 (1.75 - 2.37)	2.36 (1.95 - 2.87)	1.58 (1.31 - 1.89)	1.47 (1.24 - 1.75)	1.49 (1.01 - 2.20)
	Anxiety	1.90 (1.80 - 2.00)	1.78 (1.66 - 1.90)	1.95 (1.84 - 2.06)	1.94 (1.84 - 2.04)	1.92 (1.80 - 2.05)	1.52 (1.38 - 1.67)	1.28 (1.15 - 1.43)	1.39 (1.11 - 1.73)
	Somatic Symptoms	2.06 (1.88 - 2.24)	1.73 (1.52 - 1.96)	2.16 (1.92 - 2.43)	2.10 (1.88 - 2.34)	2.22 (1.93 - 2.55)	1.54 (1.33 - 1.79)	1.39 (1.20 - 1.61)	1.39 (1.00 - 1.94)
	Suicidal Ideation	2.70 (2.14 - 3.42)	3.15 (2.39 - 4.15)	4.76 (3.64 - 6.23)	3.36 (2.54 - 4.46)	3.94 (2.71 - 5.72)	2.20 (1.56 - 3.09)	1.75 (1.24 - 2.47)	1.25 (0.49 - 3.18)
	Psychosis	2.23 (1.85 - 2.69)	2.32 (1.83 - 2.94)	2.57 (1.95 - 3.40)	3.08 (2.48 - 3.81)	4.02 (3.12 - 5.17)	2.19 (1.70 - 2.83)	2.26 (1.80 - 2.84)	2.84 (1.87 - 4.31)
	Sleep Problems	2.53 (2.26 - 2.84)	2.40 (2.08 - 2.78)	2.93 (2.54 - 3.37)	2.84 (2.49 - 3.24)	2.34 (1.86 - 2.94)	1.98 (1.66 - 2.36)	2.04 (1.75 - 2.38)	1.58 (1.04 - 2.39)
	Memory	2.68 (2.30 - 3.13)	2.72 (2.24 - 3.29)	2.45 (1.91 - 3.15)	3.05 (2.53 - 3.67)	3.01 (2.29 - 3.95)	2.37 (1.91 - 2.94)	1.79 (1.42 - 2.25)	1.51 (0.86 - 2.68)
	Repetitive Thoughts and Behaviors	3.04 (2.67 - 3.46)	2.50 (2.09 - 2.98)	3.44 (2.92 - 4.05)	3.93 (3.46 - 4.48)	3.64 (3.03 - 4.38)	2.27 (1.87 - 2.75)	2.03 (1.68 - 2.45)	2.15 (1.45 - 3.18)
	Dissociation	3.06 (2.63 - 3.57)	2.66 (2.18 - 3.25)	3.95 (3.29 - 4.73)	3.38 (2.83 - 4.05)	3.94 (3.16 - 4.91)	2.43 (1.95 - 3.02)	2.01 (1.61 - 2.50)	2.65 (1.79 - 3.93)
	Personality Functioning	2.41 (2.19 - 2.66)	2.19 (1.93 - 2.49)	3.04 (2.77 - 3.34)	2.71 (2.44 - 3.01)	2.46 (2.08 - 2.90)	1.80 (1.54 - 2.10)	1.86 (1.62 - 2.14)	2.07 (1.59 - 2.70)
	Substance Use	1.28 (1.17 - 1.41)	1.27 (1.12 - 1.44)	1.34 (1.15 - 1.56)	1.38 (1.22 - 1.56)	1.42 (1.18 - 1.70)	1.24 (1.08 - 1.42)	1.92 (1.80 - 2.05)	1.71 (1.45 - 2.02)

Negative likelihood ratio	Depression	0.32	0.39	0.08	0.35	0.40	0.53	0.70	0.72
		(0.26 - 0.39)	(0.30 - 0.51)	(0.03 - 0.19)	(0.26 - 0.48)	(0.25 - 0.64)	(0.42 - 0.66)	(0.59 - 0.83)	(0.49 - 1.07)
	Anger	0.42	0.57	0.34	0.41	0.36	0.59	0.73	0.81
		(0.36 - 0.49)	(0.47 - 0.68)	(0.24 - 0.48)	(0.32 - 0.52)	(0.23 - 0.56)	(0.49 - 0.71)	(0.64 - 0.84)	(0.61 - 1.09)
	Mania	0.78	0.75	0.75	0.62	0.48	0.78	0.82	0.81
		(0.71 - 0.85)	(0.65 - 0.86)	(0.63 - 0.89)	(0.52 - 0.74)	(0.34 - 0.68)	(0.68 - 0.90)	(0.73 - 0.92)	(0.61 - 1.07)
	Anxiety	0.24	0.27	0.09	0.12	0.10	0.50	0.72	0.61
		(0.18 - 0.31)	(0.19 - 0.39)	(0.04 - 0.22)	(0.07 - 0.23)	(0.03 - 0.30)	(0.38 - 0.65)	(0.60 - 0.87)	(0.37 - 1.00)
	Somatic	0.49	0.61	0.38	0.42	0.34	0.70	0.79	0.78
	Symptoms	(0.43 - 0.57)	(0.51 - 0.73)	(0.27 - 0.53)	(0.33 - 0.55)	(0.21 - 0.54)	(0.59 - 0.84)	(0.69 - 0.90)	(0.56 - 1.08)
	Suicidal	0.86	0.81	0.67	0.79	0.73	0.89	0.93	0.98
	Ideation	(0.82 - 0.91)	(0.75 - 0.89)	(0.58 - 0.78)	(0.72 - 0.87)	(0.61 - 0.87)	(0.83 - 0.96)	(0.88 - 0.99)	(0.87 - 1.10)
	Psychosis	0.83	0.80	0.76	0.70	0.54	0.82	0.81	0.71
		(0.78 - 0.88)	(0.73 - 0.88)	(0.66 - 0.87)	(0.61 - 0.79)	(0.41 - 0.71)	(0.74 - 0.90)	(0.74 - 0.89)	(0.55 - 0.92)
	Sleep	0.58	0.58	0.41	0.45	0.57	0.70	0.68	0.81
	Problems	(0.52 - 0.65)	(0.49 - 0.68)	(0.31 - 0.54)	(0.36 - 0.56)	(0.43 - 0.76)	(0.60 - 0.81)	(0.60 - 0.78)	(0.62 - 1.05)
	Memory	0.72	0.69	0.72	0.63	0.62	0.75	0.85	0.90
		(0.67 - 0.78)	(0.61 - 0.78)	(0.62 - 0.85)	(0.55 - 0.73)	(0.48 - 0.79)	(0.66 - 0.84)	(0.78 - 0.93)	(0.74 - 1.09)
	Repetitive	0.61	0.67	0.45	0.38	0.39	0.71	0.77	0.72
	Thoughts and	(0.55 - 0.67)	(0.58 - 0.76)	(0.35 - 0.58)	(0.30 - 0.48)	(0.27 - 0.57)	(0.62 - 0.81)	(0.69 - 0.85)	(0.55 - 0.95)
	Behaviors								
	Dissociation	0.69	0.71	0.49	0.60	0.47	0.75	0.82	0.69
		(0.63 - 0.75)	(0.63 - 0.80)	(0.39 - 0.62)	(0.51 - 0.70)	(0.35 - 0.64)	(0.67 - 0.84)	(0.75 - 0.90)	(0.53 - 0.91)
	Personality	0.50	0.53	0.20	0.34	0.40	0.68	0.66	0.55
	Functioning	(0.44 - 0.58)	(0.44 - 0.64)	(0.12 - 0.32)	(0.26 - 0.45)	(0.27 - 0.60)	(0.58 - 0.79)	(0.57 - 0.76)	(0.37 - 0.83)
	Substance Use	0.76	0.76	0.70	0.66	0.63	0.79	0.24	0.37
		(0.67 - 0.86)	(0.63 - 0.91)	(0.54 - 0.90)	(0.54 - 0.82)	(0.44 - 0.90)	(0.66 - 0.95)	(0.17 - 0.35)	(0.19 - 0.71)

LR values for positive tests greater than 2 (shown in bold if crossing the confidence interval) represent substantial changes in the probability of the target disorder, while LR values for negative tests less than 0.5 (also if crossing the interval) indicate considerable changes in the opposite direction.

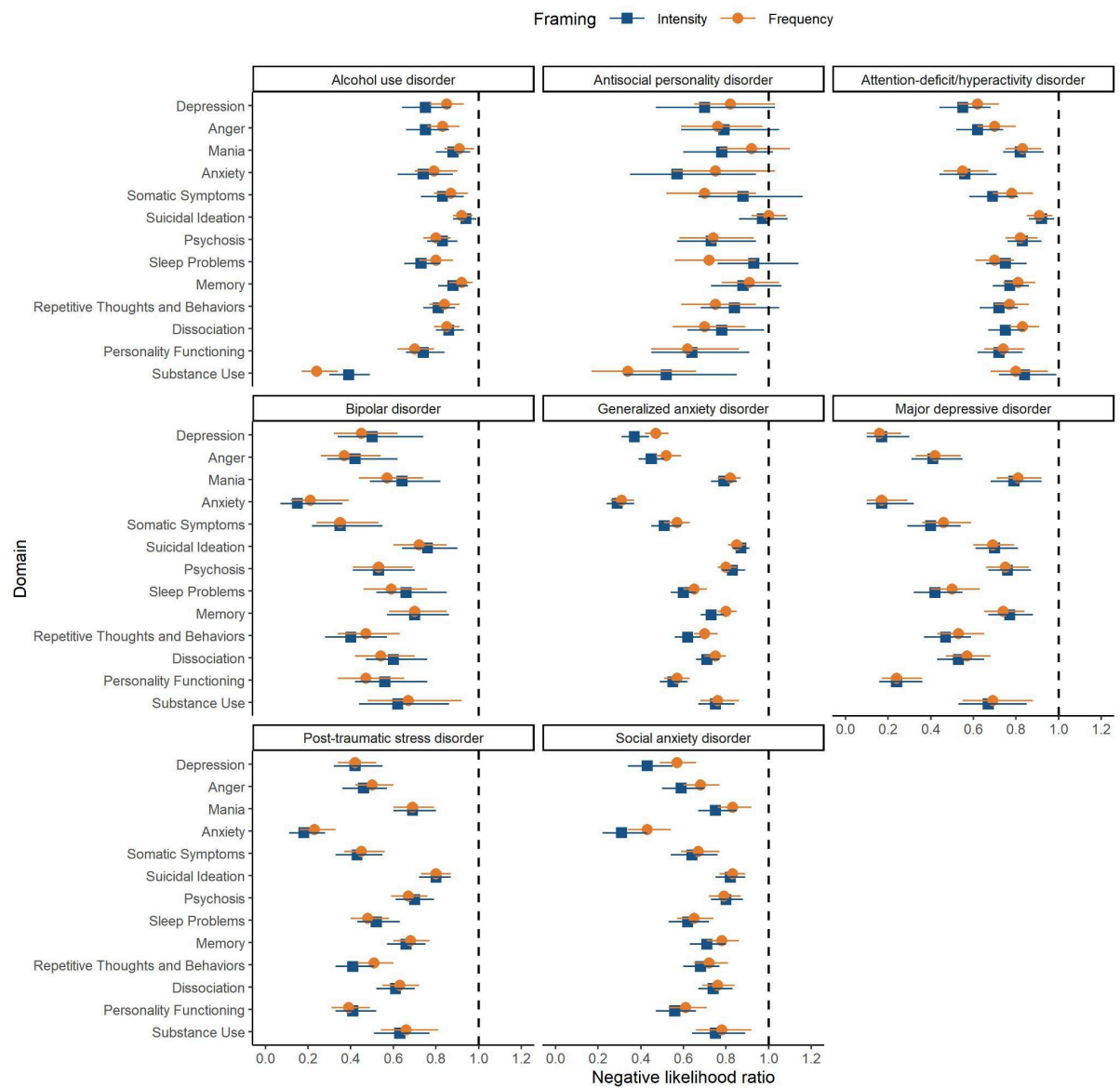
Figure 1. Positive likelihood ratio (LR+) across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings.



Legend:

Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Figure 2. Negative likelihood ratio (LR+) across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings.

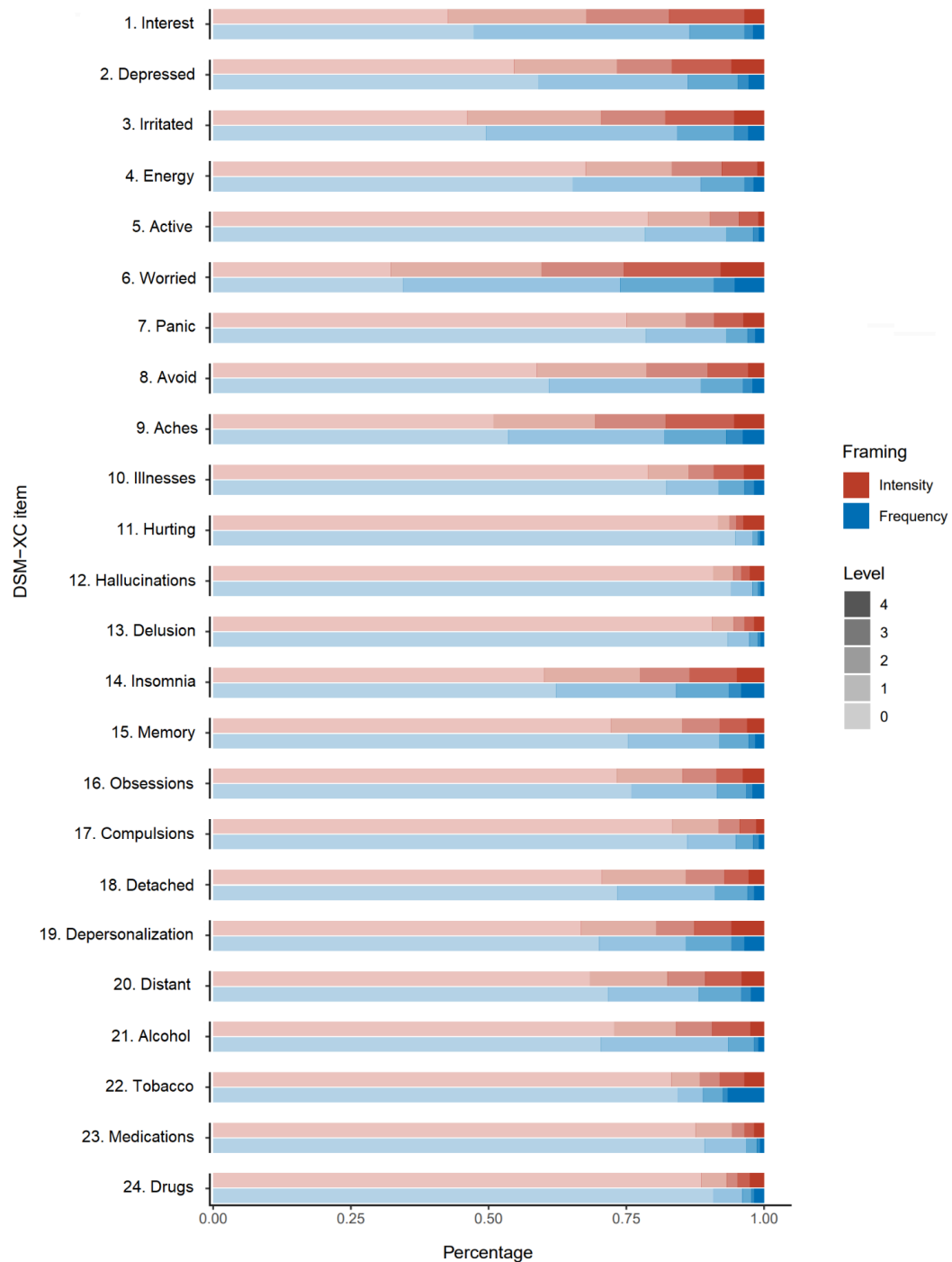


Legend:

Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

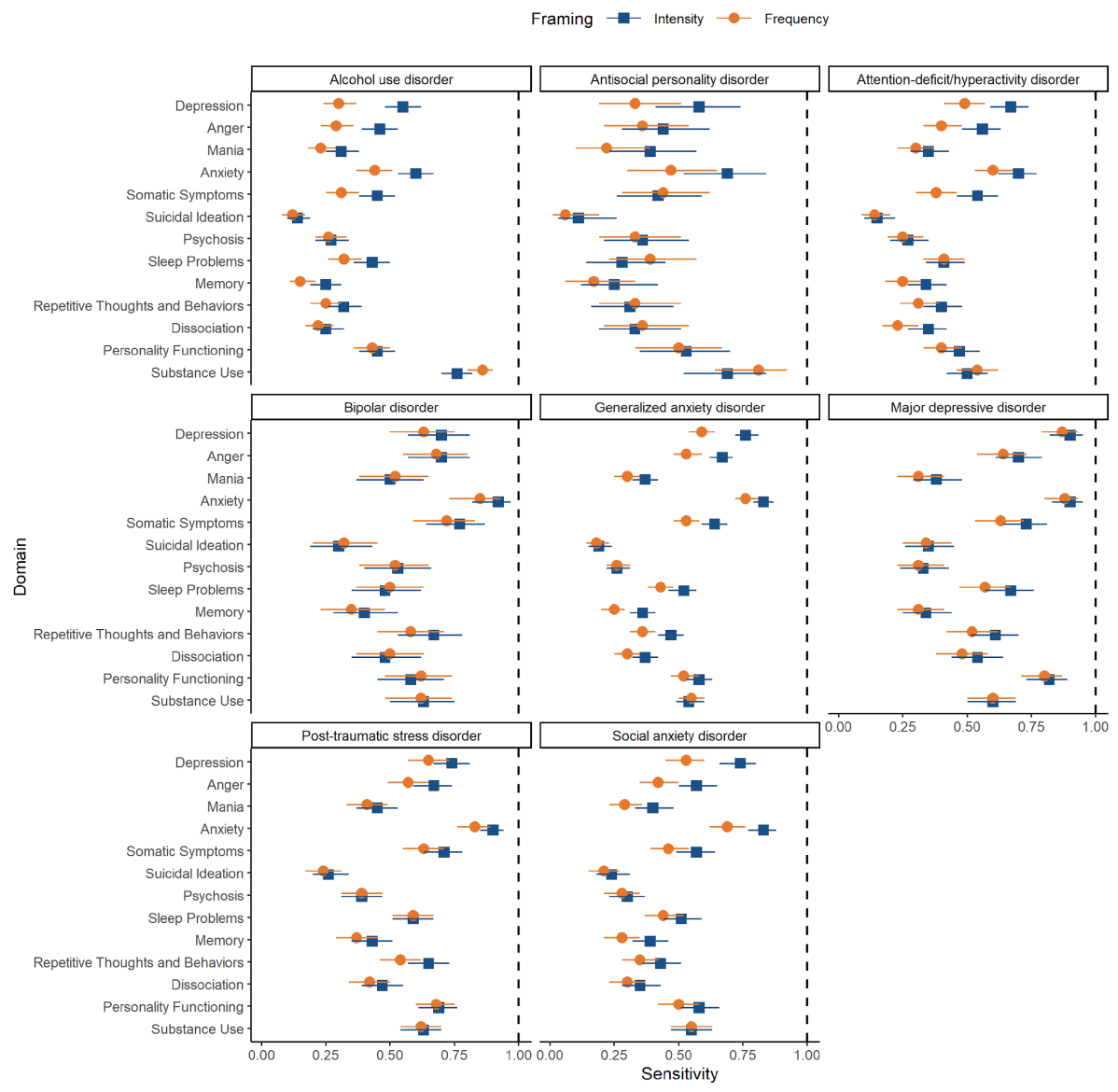
Supplementary material

Supplementary figure 1. DSM-XC item response distribution for the whole sample.



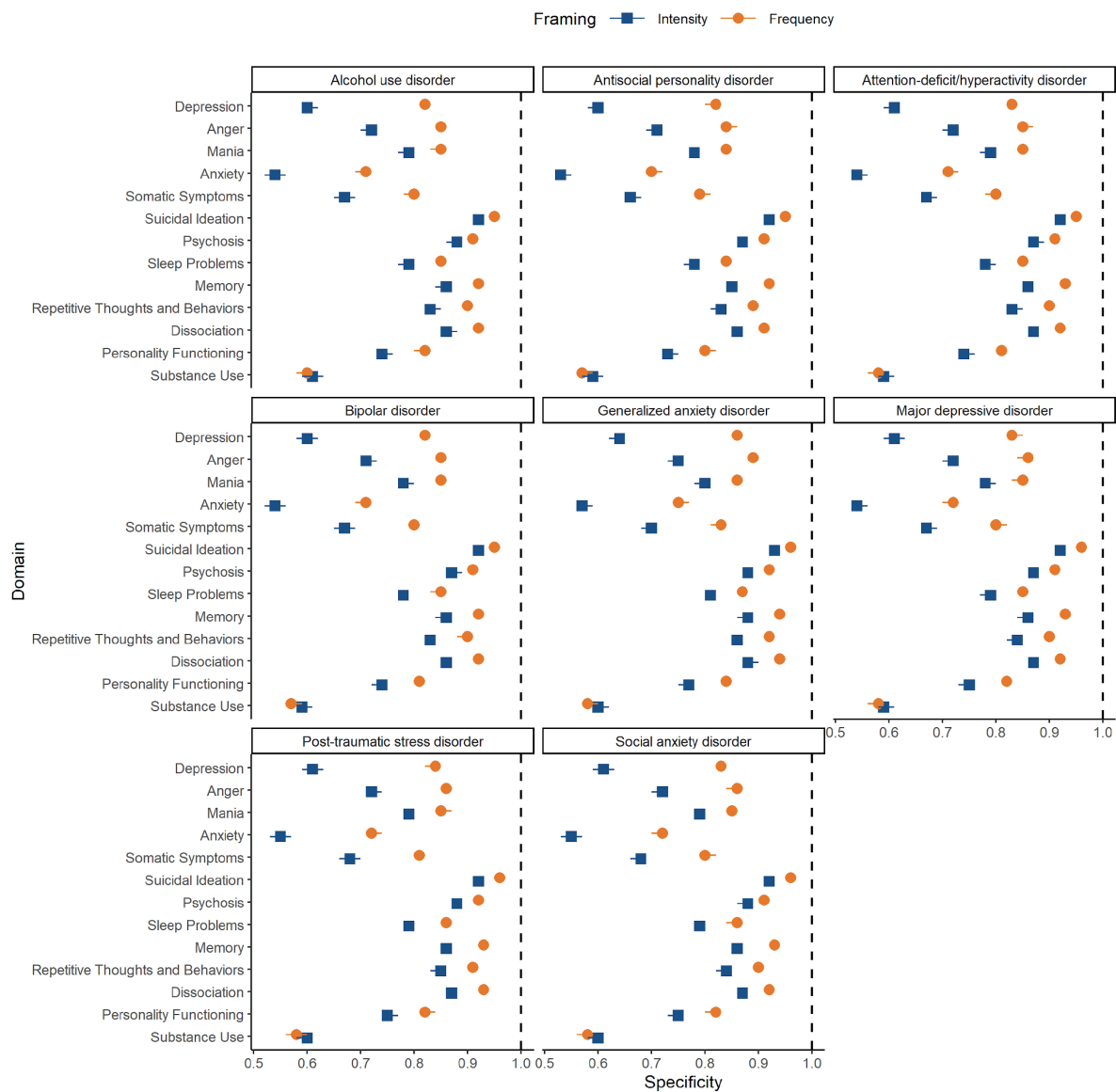
Levels for intensity were 0=none, 1=slight, 2=mild, 3=moderate, and 4=severe. Levels for frequency were 0=not at all, 1=less than a day or two, 2=several days, 3=more than half the days, and 4=nearly every day. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary figure 2. Sensitivity across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings.



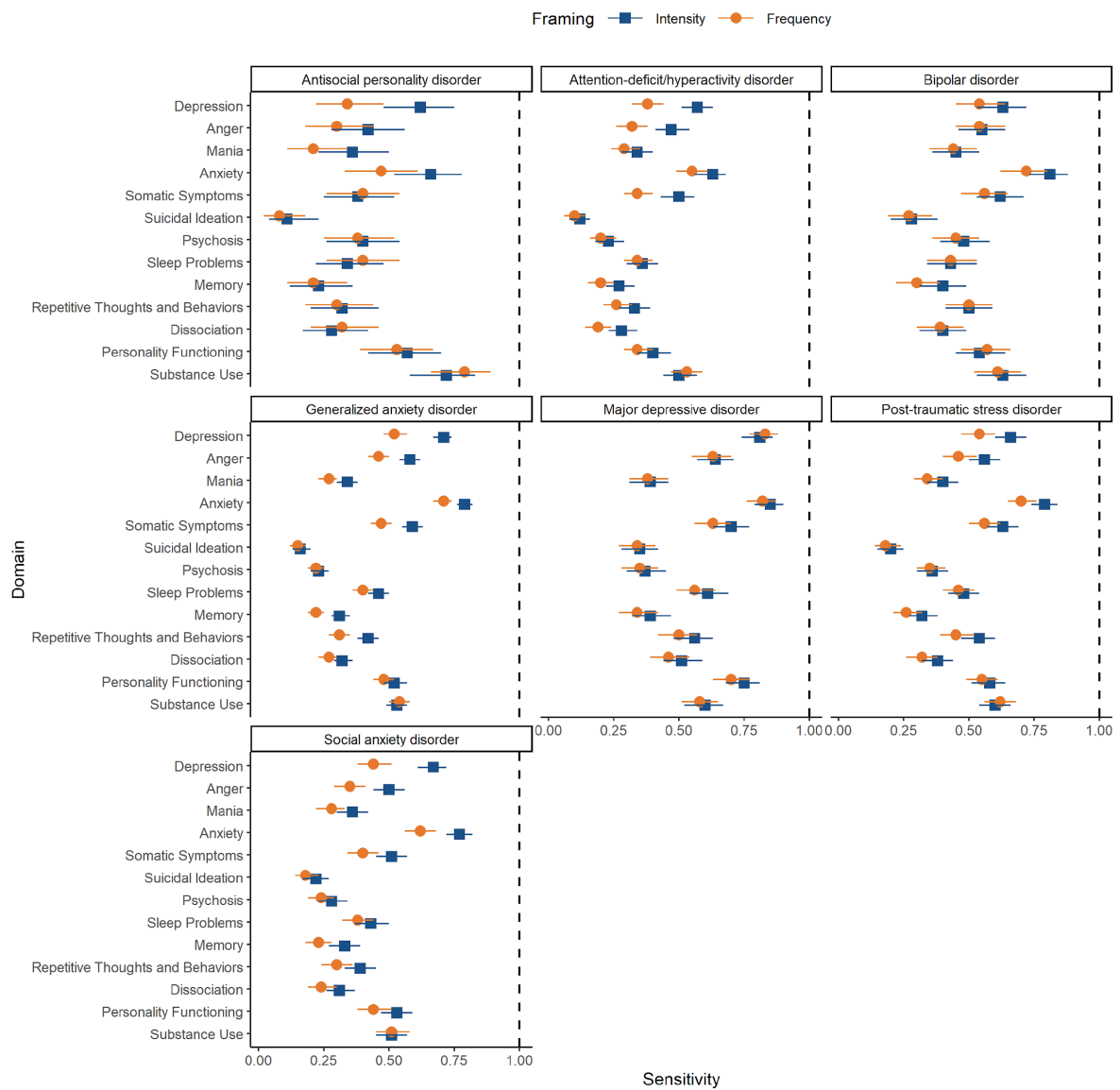
Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary figure 3. Specificity across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings.



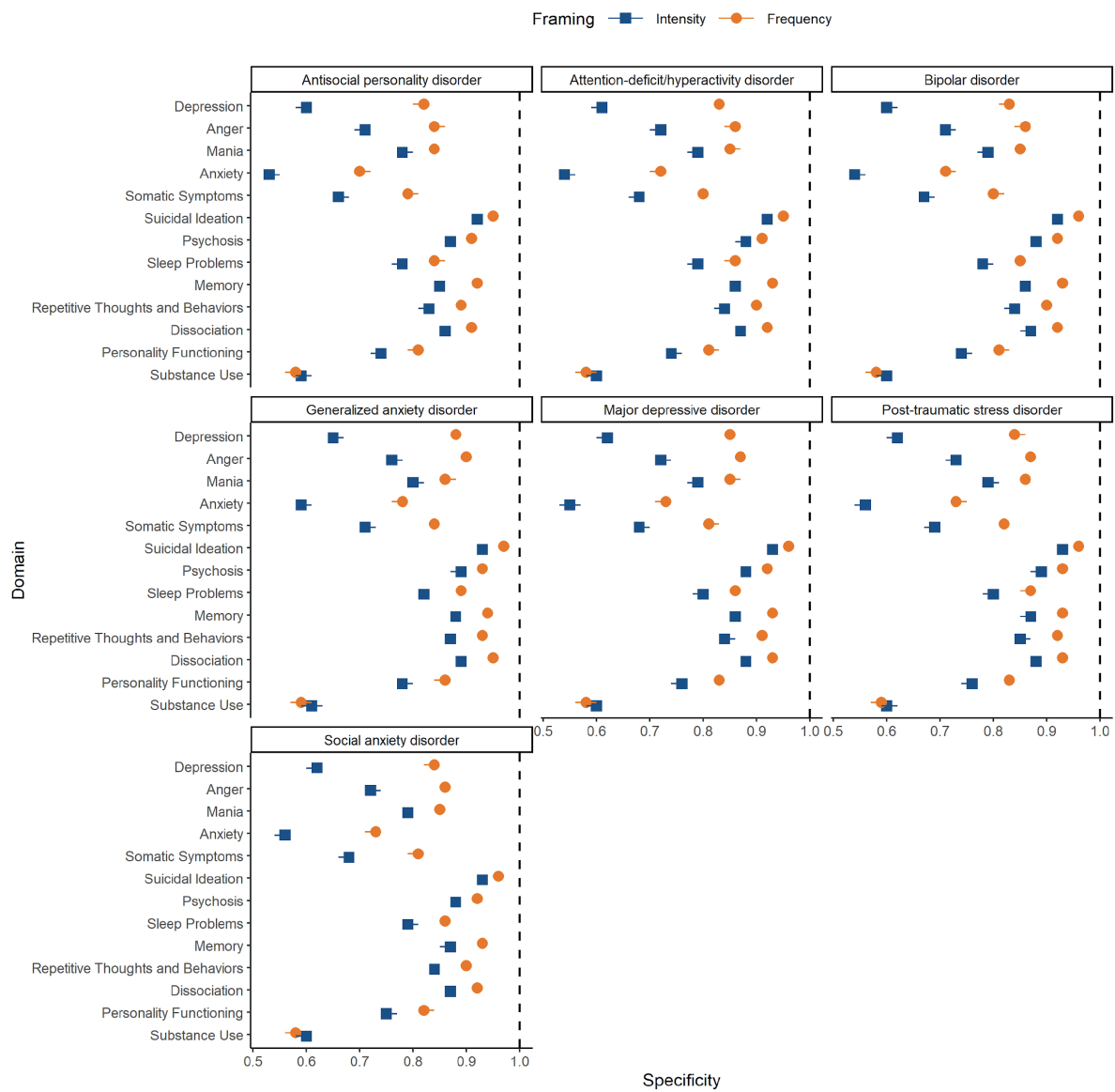
Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary Figure 4. Sensitivity across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings, after removing the clinical impairment criteria.



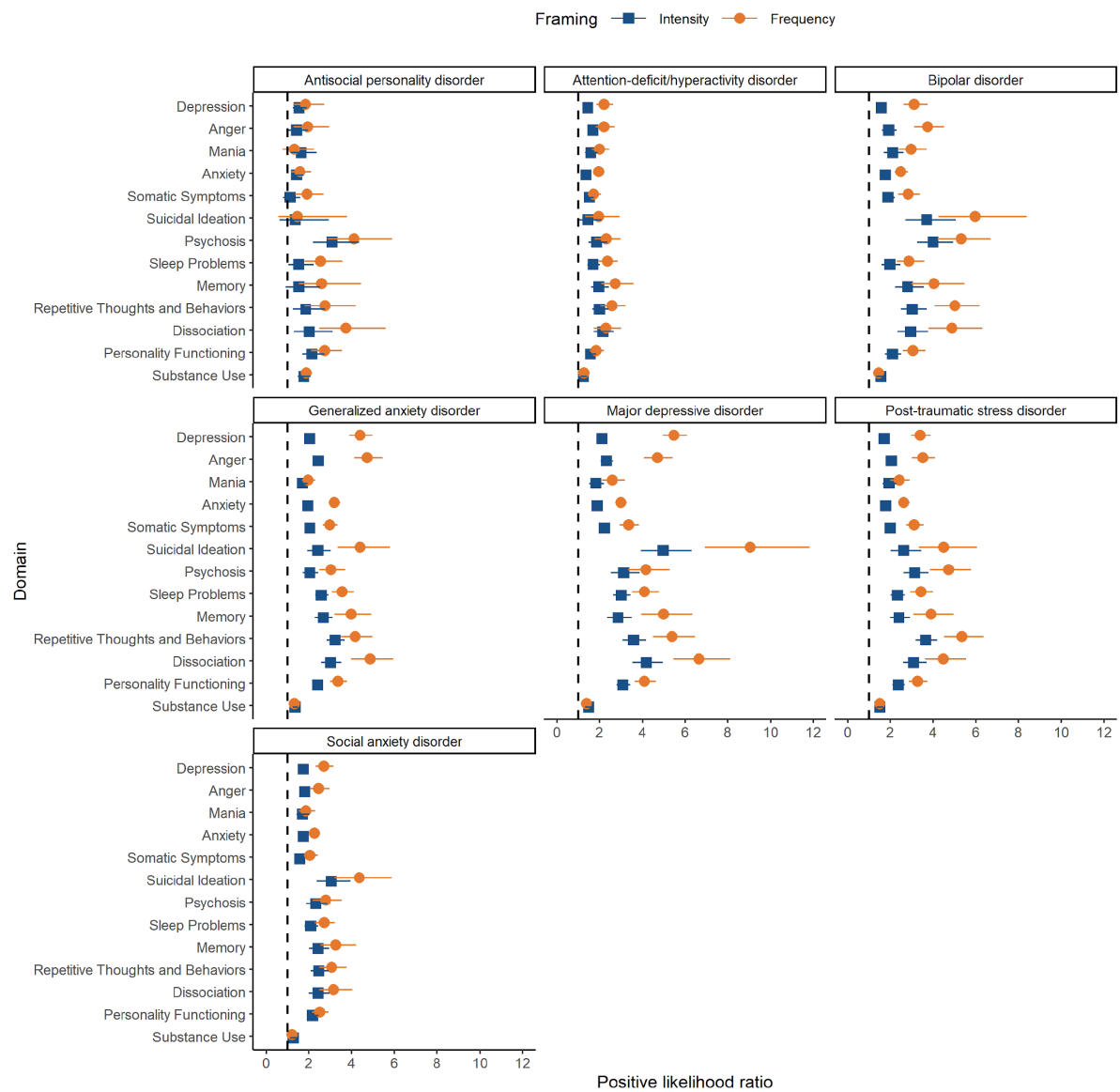
Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary Figure 5. Specificity across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings, after removing the clinical impairment criteria.



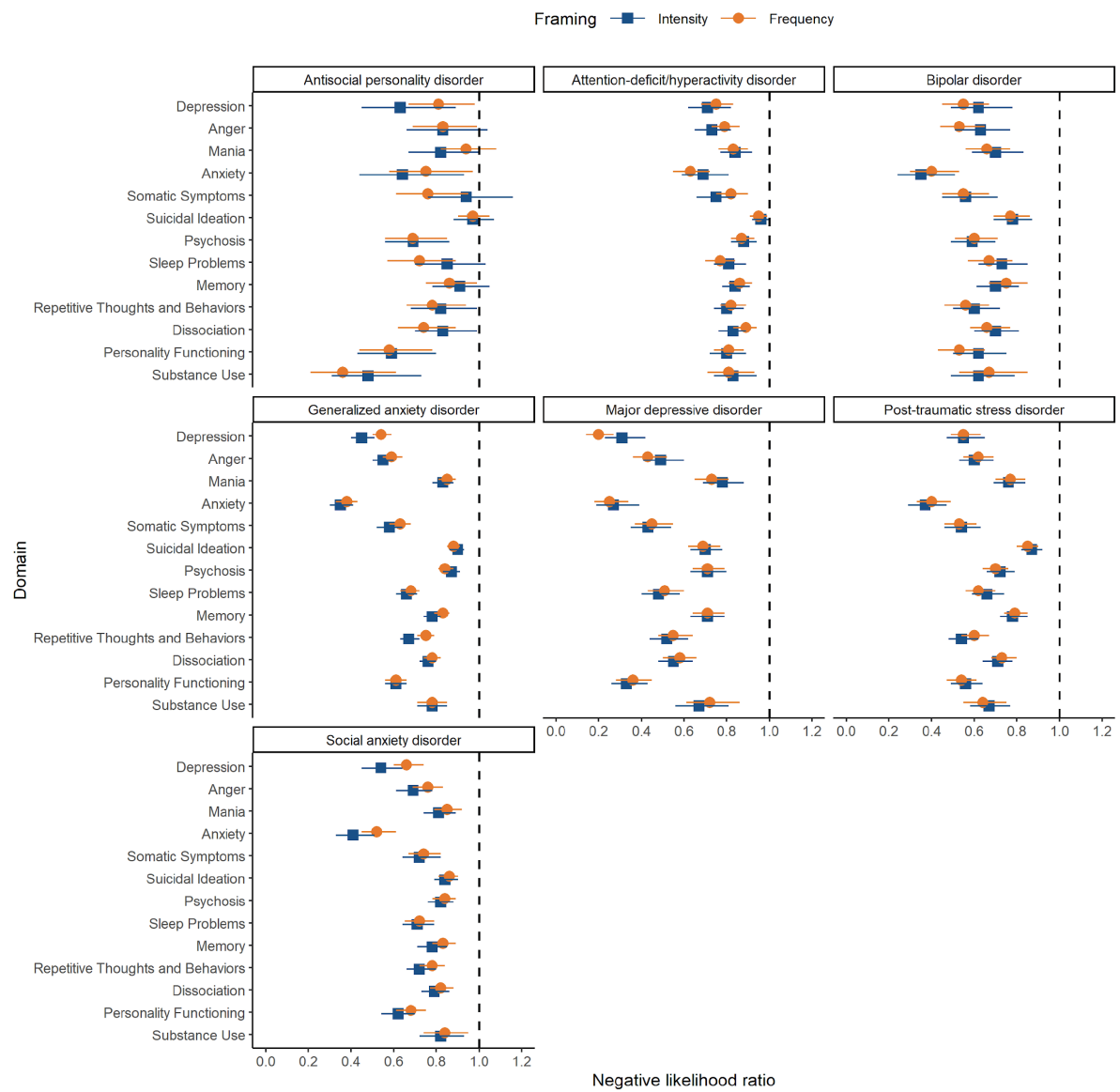
Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary Figure 6. Positive likelihood ratio (LR+) across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings, after removing the clinical impairment criteria.



Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary Figure 7. Negative likelihood ratio (LR+) across DSM-XC domains for psychiatric diagnoses using intensity and frequency framings, after removing the clinical impairment criteria.



Intensity (in blue) and frequency (in yellow) represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 1. Correlation between intensity and frequency framings from the DSM-XC.

Domain	Item	Polychoric correlation ^a	Kappa ^b
I. Depression	1. Little interest or pleasure in doing things?	0.80	0.47
	2. Feeling down, depressed, or hopeless?	0.86	0.56
II. Anger	3. Feeling more irritated, grouchy, or angry than usual?	0.85	0.56
III. Mania	4. Sleeping less than usual, but still have a lot of energy?	0.78	0.50
	5. Starting lots more projects than usual or doing more risky things than usual?	0.84	0.57
IV. Anxiety	6. Feeling nervous, anxious, frightened, worried, or on edge?	0.86	0.51
	7. Feeling panic or being frightened?	0.86	0.61
	8. Avoiding situations that make you anxious?	0.85	0.59
V. Somatic symptoms	9. Unexplained aches and pains (e.g., head, back, joints, abdomen, legs)?	0.88	0.58
	10. Feeling that your illnesses are not being taken seriously enough?	0.89	0.61
VI. Suicidal ideation	11. Thoughts of actually hurting yourself?	0.86	0.49
VII. Psychosis	12. Hearing things other people couldn't hear, such as voices even when no one was around?	0.85	0.54
	13. Feeling that someone could hear your thoughts, or that you could hear what another person was thinking?	0.83	0.52
VIII. Sleep problems	14. Problems with sleep that affected your sleep quality over all?	0.90	0.66
IX. Memory	15. Problems with memory (e.g., learning new information) or with location (e.g., finding your way home)?	0.88	0.64
X. Repetitive thoughts and behaviors	16. Unpleasant thoughts, urges, or images that repeatedly enter your mind?	0.89	0.66
	17. Feeling driven to perform certain behaviors or mental acts over and over again?	0.87	0.62
XI. Dissociation	18. Feeling detached or distant from yourself, your body, your physical surroundings, or your memories?	0.89	0.68
XII. Personality functioning	19. Not knowing who you really are or what you want out of life?	0.89	0.64
	20. Not feeling close to other people or enjoying your relationships with them?	0.87	0.64
XIII. Substance use	21. Drinking at least 4 drinks of any kind of alcohol in a single day?	0.77	0.48
	22. Smoking any cigarettes, a cigar, or pipe, or using snuff or chewing tobacco?	0.86	0.52
	23. Using any of the following medicines ON YOUR OWN, that is, without a doctor's prescription, in greater amounts or longer than prescribed [e.g., painkillers (like Vicodin), stimulants (like Ritalin or Adderall), sedatives or tranquilizers (like sleeping pills or Valium)]?*	0.82	0.60
	24. Using drugs like marijuana, cocaine or crack, club drugs (like ecstasy), hallucinogens (like LSD), heroin, inhalants or solvents (like glue), or methamphetamine (like speed)]?*	0.78	0.50

^aPolychoric correlation measures the strength and direction of the relationship between framings. ^bKappa coefficient measures the beyond chance agreement between two framings. *The last item of the original DSM-XC was split into two (items 23 and 24 in the present instrument), to allow for the exploration of substance use and self-prescription/misuse separately. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 2. Standards for Reporting of Diagnostic Accuracy Studies (STARD) 2015 Checklist.

Section & Topic	No	Item	Reported on page #
TITLE OR ABSTRACT			
	1	Identification as a study of diagnostic accuracy using at least one measure of accuracy (such as sensitivity, specificity, predictive values, or AUC)	1
ABSTRACT			
	2	Structured summary of study design, methods, results, and conclusions (for specific guidance, see STARD for Abstracts)	3
INTRODUCTION			
	3	Scientific and clinical background, including the intended use and clinical role of the index test	4
	4	Study objectives and hypotheses	5
METHODS			
<i>Study design</i>	5	Whether data collection was planned before the index test and reference standard were performed (prospective study) or after (retrospective study)	5
<i>Participants</i>	6	Eligibility criteria	5
	7	On what basis potentially eligible participants were identified (such as symptoms, results from previous tests, inclusion in registry)	5
	8	Where and when potentially eligible participants were identified (setting, location and dates)	5
	9	Whether participants formed a consecutive, random or convenience series	5
<i>Test methods</i>	10a	Index test, in sufficient detail to allow replication	5-6
	10b	Reference standard, in sufficient detail to allow replication	6
	11	Rationale for choosing the reference standard (if alternatives exist)	NA
	12a	Definition of and rationale for test positivity cut-offs or result categories	5

		of the index test, distinguishing pre-specified from exploratory	
	12b	Definition of and rationale for test positivity cut-offs or result categories of the reference standard, distinguishing pre-specified from exploratory	5
	13a	Whether clinical information and reference standard results were available to the performers/readers of the index test	6
	13b	Whether clinical information and index test results were available to the assessors of the reference standard	6
<i>Analysis</i>	14	Methods for estimating or comparing measures of diagnostic accuracy	7
	15	How indeterminate index test or reference standard results were handled	NA
	16	How missing data on the index test and reference standard were handled	5
	17	Any analyses of variability in diagnostic accuracy, distinguishing pre-specified from exploratory	7
	18	Intended sample size and how it was determined	NA
RESULTS			
<i>Participants</i>	19	Flow of participants, using a diagram	NA
	20	Baseline demographic and clinical characteristics of participants	Table 1
	21a	Distribution of severity of disease in those with the target condition	NA
	21b	Distribution of alternative diagnoses in those without the target condition	NA
	22	Time interval and any clinical interventions between index test and reference standard	NA
<i>Test results</i>	23	Cross tabulation of the index test results (or their distribution) by the results of the reference standard	Due to the large number of index and reference tests, cross tabulation was described in tables instead of diagram

			(Supplementary Tables 4 to 19)
	24	Estimates of diagnostic accuracy and their precision (such as 95% confidence intervals)	Tables 2 and 3
	25	Any adverse events from performing the index test or the reference standard	NA
DISCUSSION			
	26	Study limitations, including sources of potential bias, statistical uncertainty, and generalisability	12
	27	Implications for practice, including the intended use and clinical role of the index test	12
OTHER INFORMATION			
	28	Registration number and name of registry	5
	29	Where the full study protocol can be accessed	NA
	30	Sources of funding and other support; role of funders	1

Supplementary table 3. Number and proportion of positive screens and according to DSM-XC scoring rules for each domain.

Domain	OR rule		Intensity		Frequency	
	n	Proportion (95% CI)	n	Proportion (95% CI)	n	Proportion (95% CI)
01. Depression	1524	0.43 (0.41-0.44)	1452	0.41 (0.39-0.42)	665	0.19 (0.17-0.20)
02. Anger	1143	0.32 (0.30-0.34)	1056	0.30 (0.28-0.31)	564	0.16 (0.15-0.17)
03. Mania	1007	0.28 (0.27-0.30)	789	0.22 (0.21-0.23)	564	0.16 (0.15-0.17)
04. Anxiety	1798	0.50 (0.49-0.52)	1677	0.47 (0.45-0.49)	1076	0.30 (0.29-0.32)
05. Somatic symptoms	1290	0.36 (0.34-0.38)	1206	0.34 (0.32-0.35)	751	0.21 (0.20-0.22)
06. Suicidal ideation	318	0.09 (0.08-0.10)	299	0.08 (0.07-0.09)	186	0.05 (0.04-0.06)
07. Psychosis	499	0.14 (0.13-0.15)	474	0.13 (0.12-0.14)	344	0.10 (0.09-0.11)
08. Sleep problems	886	0.25 (0.23-0.26)	805	0.22 (0.21-0.24)	570	0.16 (0.15-0.17)
09. Memory	594	0.17 (0.15-0.18)	534	0.15 (0.14-0.16)	293	0.08 (0.07-0.09)
10. Repetitive thoughts and behaviors	701	0.20 (0.18-0.21)	629	0.18 (0.16-0.19)	403	0.11 (0.10-0.12)
11. Dissociation	572	0.16 (0.15-0.17)	510	0.14 (0.13-0.15)	320	0.09 (0.08-0.10)
12. Personality functioning	1069	0.30 (0.28-0.31)	958	0.27 (0.25-0.28)	709	0.20 (0.19-0.21)
13. Substance use	1696	0.47 (0.46-0.49)	1474	0.41 (0.40-0.43)	1537	0.43 (0.41-0.45)

n = number of positive screens. DSM-XC stipulates a positive screen as rating mild (i.e., 2) or greater on any item within a domain (depression, anger, mania, anxiety, somatic symptoms, sleep problems, memory, repetitive thoughts and behaviors, dissociation, personality functioning, except for substance use, suicidal ideation, and psychosis). For substance use, suicidal ideation, and psychosis, rating slight (i.e., 1) or greater on any item within the domain was considered a positive screen. “OR” rule represents original DSM-XC scoring instructions. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 4. Diagnostic and test results of DSM-XC domains using the original “OR” rule for alcohol use disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	123	85	1401	1969
II. Anger	103	105	1040	2330
III. Mania	84	124	923	2447
IV. Anxiety	132	76	1666	1704
V. Somatic symptoms	102	106	1188	2182
VI. Suicidal ideation	31	177	287	3083
VII. Psychosis	61	147	438	2932
VIII. Sleep problems	99	109	787	2583
IX. Memory	59	149	535	2835
X. Repetitive thoughts and behaviors	78	130	623	2747
XI. Dissociation	63	145	509	2861
XII. Personality functioning	110	98	959	2411
XIII. Substance use	180	28	1516	1854

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 5. Diagnostic and test results of DSM-XC domains using the original “OR” rule for antisocial personality disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	21	15	1503	2039
II. Anger	16	20	1127	2415
III. Mania	15	21	992	2550
IV. Anxiety	25	11	1773	1769
V. Somatic symptoms	18	18	1272	2270
VI. Suicidal ideation	4	32	314	3228
VII. Psychosis	14	22	485	3057
VIII. Sleep problems	14	22	872	2670
IX. Memory	9	27	585	2957
X. Repetitive thoughts and behaviors	15	21	686	2856
XI. Dissociation	15	21	557	2985
XII. Personality functioning	22	14	1047	2495
XIII. Substance use	29	7	1667	1875

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 6. Diagnostic and test results of DSM-XC domains using the original “OR” rule for attention-deficit/hyperactivity disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	112	50	1412	2004
II. Anger	96	66	1047	2369
III. Mania	70	92	937	2479
IV. Anxiety	121	41	1677	1739
V. Somatic symptoms	88	74	1202	2214
VI. Suicidal ideation	30	132	288	3128
VII. Psychosis	47	115	452	2964
VIII. Sleep problems	76	86	810	2606
IX. Memory	60	102	534	2882
X. Repetitive thoughts and behaviors	68	94	633	2783
XI. Dissociation	59	103	513	2903
XII. Personality functioning	84	78	985	2431
XIII. Substance use	94	68	1602	1814

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 7. Diagnostic and test results of DSM-XC domains using the original “OR” rule for bipolar disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	46	14	1478	2040
II. Anger	45	15	1098	2420
III. Mania	39	21	968	2550
IV. Anxiety	57	3	1741	1777
V. Somatic symptoms	47	13	1243	2275
VI. Suicidal ideation	20	40	298	3220
VII. Psychosis	32	28	467	3051
VIII. Sleep problems	34	26	852	2666
IX. Memory	29	31	565	2953
X. Repetitive thoughts and behaviors	41	19	660	2858
XI. Dissociation	36	24	536	2982
XII. Personality functioning	43	17	1026	2492
XIII. Substance use	40	20	1656	1862

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 8. Diagnostic and test results of DSM-XC domains using the original “OR” rule for generalized anxiety disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	304	74	1220	1980
II. Anger	264	114	879	2321
III. Mania	162	216	845	2355
IV. Anxiety	329	49	1469	1731
V. Somatic symptoms	252	126	1038	2162
VI. Suicidal ideation	77	301	241	2959
VII. Psychosis	104	274	395	2805
VIII. Sleep problems	204	174	682	2518
IX. Memory	143	235	451	2749
X. Repetitive thoughts and behaviors	185	193	516	2684
XI. Dissociation	152	226	420	2780
XII. Personality functioning	237	141	832	2368
XIII. Substance use	223	155	1473	1727

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 9. Diagnostic and test results of DSM-XC domains using the original “OR” rule for major depressive disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	100	5	1424	2049
II. Anger	80	25	1063	2410
III. Mania	48	57	959	2514
IV. Anxiety	100	5	1698	1775
V. Somatic symptoms	79	26	1211	2262
VI. Suicidal ideation	40	65	278	3195
VII. Psychosis	36	69	463	3010
VIII. Sleep problems	72	33	814	2659
IX. Memory	41	64	553	2920
X. Repetitive thoughts and behaviors	66	39	635	2838
XI. Dissociation	61	44	511	2962
XII. Personality functioning	90	15	979	2494
XIII. Substance use	66	39	1630	1843

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 10. Diagnostic and test results of DSM-XC domains using the original “OR” rule for post-traumatic stress disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	123	32	1401	2022
II. Anger	111	44	1032	2391
III. Mania	85	70	922	2501
IV. Anxiety	145	10	1653	1770
V. Somatic symptoms	112	43	1178	2245
VI. Suicidal ideation	42	113	276	3147
VII. Psychosis	61	94	438	2985
VIII. Sleep problems	101	54	785	2638
IX. Memory	72	83	522	2901
X. Repetitive thoughts and behaviors	106	49	595	2828
XI. Dissociation	76	79	496	2927
XII. Personality functioning	117	38	952	2471
XIII. Substance use	100	55	1596	1827

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 11. Diagnostic and test results of DSM-XC domains using the original “OR” rule for social anxiety disorder.

Domain	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	137	41	1387	2013
II. Anger	108	70	1035	2365
III. Mania	81	97	926	2474
IV. Anxiety	153	25	1645	1755
V. Somatic symptoms	107	71	1183	2217
VI. Suicidal ideation	45	133	273	3127
VII. Psychosis	54	124	445	2955
VIII. Sleep problems	99	79	787	2613
IX. Memory	74	104	520	2880
X. Repetitive thoughts and behaviors	81	97	620	2780
XI. Dissociation	70	108	502	2898
XII. Personality functioning	110	68	959	2441
XIII. Substance use	106	72	1590	1810

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 12. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for alcohol use disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	114	94	1338	2032
	Frequency	63	145	602	2768
II. Anger	Intensity	96	112	960	2410
	Frequency	61	147	503	2867
III. Mania	Intensity	65	143	724	2646
	Frequency	48	160	516	2854
IV. Anxiety	Intensity	125	83	1552	1818
	Frequency	91	117	985	2385
V. Somatic symptoms	Intensity	93	115	1113	2257
	Frequency	64	144	687	2683
VI. Suicidal ideation	Intensity	29	179	270	3100
	Frequency	25	183	161	3209
VII. Psychosis	Intensity	57	151	417	2953
	Frequency	55	153	289	3081
VIII. Sleep problems	Intensity	89	119	716	2654
	Frequency	67	141	503	2867
IX. Memory	Intensity	51	157	483	2887
	Frequency	32	176	261	3109
X. Repetitive thoughts and behaviors	Intensity	67	141	562	2808
	Frequency	52	156	351	3019
XI. Dissociation	Intensity	53	155	457	2913
	Frequency	46	162	274	3096
XII. Personality functioning	Intensity	93	115	865	2505
	Frequency	89	119	620	2750
XIII. Substance use	Intensity	159	49	1315	2055
	Frequency	178	30	1359	2011

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 13. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for antisocial personality disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	21	15	1431	2111
	Frequency	12	24	653	2889
II. Anger	Intensity	16	20	1040	2502
	Frequency	13	23	551	2991
III. Mania	Intensity	14	22	775	2767
	Frequency	8	28	556	2986
IV. Anxiety	Intensity	25	11	1652	1890
	Frequency	17	19	1059	2483
V. Somatic symptoms	Intensity	15	21	1191	2351
	Frequency	16	20	735	2807
VI. Suicidal ideation	Intensity	4	32	295	3247
	Frequency	2	34	184	3358
VII. Psychosis	Intensity	13	23	461	3081
	Frequency	12	24	332	3210
VIII. Sleep problems	Intensity	10	26	795	2747
	Frequency	14	22	556	2986
IX. Memory	Intensity	9	27	525	3017
	Frequency	6	30	287	3255
X. Repetitive thoughts and behaviors	Intensity	11	25	618	2924
	Frequency	12	24	391	3151
XI. Dissociation	Intensity	12	24	498	3044
	Frequency	13	23	307	3235
XII. Personality functioning	Intensity	19	17	939	2603
	Frequency	18	18	691	2851
XIII. Substance use	Intensity	25	11	1449	2093
	Frequency	29	7	1508	2034

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 14. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for attention-deficit/hyperactivity disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	108	54	1344	2072
	Frequency	79	83	586	2830
II. Anger	Intensity	90	72	966	2450
	Frequency	65	97	499	2917
III. Mania	Intensity	57	105	732	2684
	Frequency	48	114	516	2900
IV. Anxiety	Intensity	113	49	1564	1852
	Frequency	98	64	978	2438
V. Somatic symptoms	Intensity	87	75	1119	2297
	Frequency	61	101	690	2726
VI. Suicidal ideation	Intensity	25	137	274	3142
	Frequency	22	140	164	3252
VII. Psychosis	Intensity	44	118	430	2986
	Frequency	41	121	303	3113
VIII. Sleep problems	Intensity	67	95	738	2678
	Frequency	66	96	504	2912
IX. Memory	Intensity	55	107	479	2937
	Frequency	40	122	253	3163
X. Repetitive thoughts and behaviors	Intensity	65	97	564	2852
	Frequency	50	112	353	3063
XI. Dissociation	Intensity	56	106	454	2962
	Frequency	38	124	282	3134
XII. Personality functioning	Intensity	76	86	882	2534
	Frequency	65	97	644	2772
XIII. Substance use	Intensity	81	81	1393	2023
	Frequency	87	75	1450	1966

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 15. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for bipolar disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	42	18	1410	2108
	Frequency	38	22	627	2891
II. Anger	Intensity	42	18	1014	2504
	Frequency	41	19	523	2995
III. Mania	Intensity	30	30	759	2759
	Frequency	31	29	533	2985
IV. Anxiety	Intensity	55	5	1622	1896
	Frequency	51	9	1025	2493
V. Somatic symptoms	Intensity	46	14	1160	2358
	Frequency	43	17	708	2810
VI. Suicidal ideation	Intensity	18	42	281	3237
	Frequency	19	41	167	3351
VII. Psychosis	Intensity	32	28	442	3076
	Frequency	31	29	313	3205
VIII. Sleep problems	Intensity	29	31	776	2742
	Frequency	30	30	540	2978
IX. Memory	Intensity	24	36	510	3008
	Frequency	21	39	272	3246
X. Repetitive thoughts and behaviors	Intensity	40	20	589	2929
	Frequency	35	25	368	3150
XI. Dissociation	Intensity	29	31	481	3037
	Frequency	30	30	290	3228
XII. Personality functioning	Intensity	35	25	923	2595
	Frequency	37	23	672	2846
XIII. Substance use	Intensity	38	22	1436	2082
	Frequency	37	23	1500	2018

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 16. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for generalized anxiety disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	289	89	1163	2037
	Frequency	224	154	441	2759
II. Anger	Intensity	252	126	804	2396
	Frequency	202	176	362	2838
III. Mania	Intensity	140	238	649	2551
	Frequency	113	265	451	2749
IV. Anxiety	Intensity	314	64	1363	1837
	Frequency	289	89	787	2413
V. Somatic symptoms	Intensity	243	135	963	2237
	Frequency	200	178	551	2649
VI. Suicidal ideation	Intensity	73	305	226	2974
	Frequency	69	309	117	3083
VII. Psychosis	Intensity	100	278	374	2826
	Frequency	98	280	246	2954
VIII. Sleep problems	Intensity	195	183	610	2590
	Frequency	163	215	407	2793
IX. Memory	Intensity	135	243	399	2801
	Frequency	93	285	200	3000
X. Repetitive thoughts and behaviors	Intensity	178	200	451	2749
	Frequency	135	243	268	2932
XI. Dissociation	Intensity	141	237	369	2831
	Frequency	113	265	207	2993
XII. Personality functioning	Intensity	218	160	740	2460
	Frequency	198	180	511	2689
XIII. Substance use	Intensity	206	172	1268	1932
	Frequency	209	169	1328	1872

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 17. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for major depressive disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	94	11	1358	2115
	Frequency	91	14	574	2899
II. Anger	Intensity	74	31	982	2491
	Frequency	67	38	497	2976
III. Mania	Intensity	40	65	749	2724
	Frequency	33	72	531	2942
IV. Anxiety	Intensity	95	10	1582	1891
	Frequency	92	13	984	2489
V. Somatic symptoms	Intensity	77	28	1129	2344
	Frequency	66	39	685	2788
VI. Suicidal ideation	Intensity	37	68	262	3211
	Frequency	36	69	150	3323
VII. Psychosis	Intensity	35	70	439	3034
	Frequency	33	72	311	3162
VIII. Sleep problems	Intensity	70	35	735	2738
	Frequency	60	45	510	2963
IX. Memory	Intensity	36	69	498	2975
	Frequency	33	72	260	3213
X. Repetitive thoughts and behaviors	Intensity	64	41	565	2908
	Frequency	55	50	348	3125
XI. Dissociation	Intensity	57	48	453	3020
	Frequency	50	55	270	3203
XII. Personality functioning	Intensity	86	19	872	2601
	Frequency	84	21	625	2848
XIII. Substance use	Intensity	63	42	1411	2062
	Frequency	63	42	1474	1999

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 18. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for post-traumatic stress disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	115	40	1337	2086
	Frequency	101	54	564	2859
II. Anger	Intensity	104	51	952	2471
	Frequency	88	67	476	2947
III. Mania	Intensity	70	85	719	2704
	Frequency	64	91	500	2923
IV. Anxiety	Intensity	140	15	1537	1886
	Frequency	129	26	947	2476
V. Somatic symptoms	Intensity	110	45	1096	2327
	Frequency	98	57	653	2770
VI. Suicidal ideation	Intensity	41	114	258	3165
	Frequency	37	118	149	3274
VII. Psychosis	Intensity	60	95	414	3009
	Frequency	60	95	284	3139
VIII. Sleep problems	Intensity	91	64	714	2709
	Frequency	91	64	479	2944
IX. Memory	Intensity	67	88	467	2956
	Frequency	57	98	236	3187
X. Repetitive thoughts and behaviors	Intensity	101	54	528	2895
	Frequency	84	71	319	3104
XI. Dissociation	Intensity	73	82	437	2986
	Frequency	65	90	255	3168
XII. Personality functioning	Intensity	107	48	851	2572
	Frequency	105	50	604	2819
XIII. Substance use	Intensity	97	58	1377	2046
	Frequency	96	59	1441	1982

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 19. Diagnostic and test results of DSM-XC domains using the intensity and frequency framings for social anxiety disorder.

Domain	Framing	Test + / Diagnosis +	Test - / Diagnosis +	Test + / Diagnosis -	Test - / Diagnosis -
I. Depression	Intensity	131	47	1321	2079
	Frequency	94	84	571	2829
II. Anger	Intensity	102	76	954	2446
	Frequency	75	103	489	2911
III. Mania	Intensity	72	106	717	2683
	Frequency	52	126	512	2888
IV. Anxiety	Intensity	148	30	1529	1871
	Frequency	123	55	953	2447
V. Somatic symptoms	Intensity	101	77	1105	2295
	Frequency	82	96	669	2731
VI. Suicidal ideation	Intensity	43	135	256	3144
	Frequency	37	141	149	3251
VII. Psychosis	Intensity	53	125	421	2979
	Frequency	49	129	295	3105
VIII. Sleep problems	Intensity	91	87	714	2686
	Frequency	79	99	491	2909
IX. Memory	Intensity	69	109	465	2935
	Frequency	49	129	244	3156
X. Repetitive thoughts and behaviors	Intensity	77	101	552	2848
	Frequency	62	116	341	3059
XI. Dissociation	Intensity	63	115	447	2953
	Frequency	53	125	267	3133
XII. Personality functioning	Intensity	104	74	854	2546
	Frequency	89	89	620	2780
XIII. Substance use	Intensity	98	80	1376	2024
	Frequency	98	80	1439	1961

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 20. Sensitivity differences between DSM-XC intensity and frequency question framings.

Diagnosis	Sensitivity difference (median*)	P-value
Alcohol use disorder	0.08	0.01
Antisocial personality disorder	0.04	0.17
Attention-deficit/hyperactivity disorder	0.09	0.01
Bipolar disorder	0.02	0.29
Generalized anxiety disorder	0.09	<0.01
Major depressive disorder	0.05	<0.01
Post-traumatic stress disorder	0.06	<0.01
Social anxiety disorder	0.09	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 21. Specificity differences between DSM-XC intensity and frequency question framings.

Diagnosis	Specificity difference (median*)	P-value
Alcohol use disorder	-0.07	<0.01
Antisocial personality disorder	-0.07	<0.01
Attention-deficit/hyperactivity disorder	-0.07	<0.01
Bipolar disorder	-0.07	<0.01
Generalized anxiety disorder	-0.07	<0.01
Major depressive disorder	-0.07	<0.01
Post-traumatic stress disorder	-0.07	<0.01
Social anxiety disorder	-0.07	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 22. LR+ differences between DSM-XC intensity and frequency question framings.

Diagnosis	LR+ difference (median*)	P-value
Alcohol use disorder	-0.38	<0.01
Antisocial personality disorder	-0.58	0.01
Attention-deficit/hyperactivity disorder	-0.58	<0.01
Bipolar disorder	-1.45	<0.01
Generalized anxiety disorder	-1.09	<0.01
Major depressive disorder	-1.37	<0.01
Post-traumatic stress disorder	-1.47	<0.01
Social anxiety disorder	-0.76	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 23. LR- differences between DSM-XC intensity and frequency question framings.

Diagnosis	LR- difference (median*)	P-value
Alcohol use disorder	-0.02	0.20
Antisocial personality disorder	0.02	0.60
Attention-deficit/hyperactivity disorder	-0.02	0.11
Bipolar disorder	0.04	0.31
Generalized anxiety disorder	-0.04	0.01
Major depressive disorder	-0.02	0.11
Post-traumatic stress disorder	-0.02	0.18
Social anxiety disorder	-0.05	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 24. Sensitivity differences between DSM-XC OR rule and intensity/frequency question framings.

Diagnosis	Framing being compared to OR rule	Sensitivity difference (median*)	P-value
Alcohol use disorder	Intensity	0.05	<0.01
	Frequency	0.13	<0.01
Antisocial personality disorder	Intensity	0.08	0.01
	Frequency	0.10	<0.01
Attention-deficit/hyperactivity disorder	Intensity	0.04	<0.01
	Frequency	0.12	<0.01
Bipolar disorder	Intensity	0.07	<0.01
	Frequency	0.08	<0.01
Generalized anxiety disorder	Intensity	0.03	<0.01
	Frequency	0.12	<0.01
Major depressive disorder	Intensity	0.04	<0.01
	Frequency	0.09	<0.01
Post-traumatic stress disorder	Intensity	0.04	<0.01
	Frequency	0.09	<0.01
Social anxiety disorder	Intensity	0.04	<0.01
	Frequency	0.12	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. The OR rule was used as the reference group for the comparisons with intensity and frequency. “OR” rule represents original DSM-XC scoring instructions. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 25. Specificity differences between DSM-XC OR rule and intensity/frequency question framings.

Diagnosis	Framing being compared to OR rule	Specificity difference (median*)	P-value
Alcohol use disorder	Intensity	-0.02	<0.01
	Frequency	-0.10	<0.01
Antisocial personality disorder	Intensity	-0.02	<0.01
	Frequency	-0.10	<0.01
Attention-deficit/hyperactivity disorder	Intensity	-0.03	<0.01
	Frequency	-0.10	<0.01
Bipolar disorder	Intensity	-0.02	<0.01
	Frequency	-0.10	<0.01
Generalized anxiety disorder	Intensity	-0.02	<0.01
	Frequency	-0.10	<0.01
Major depressive disorder	Intensity	-0.03	<0.01
	Frequency	-0.10	<0.01
Post-traumatic stress disorder	Intensity	-0.02	<0.01
	Frequency	-0.10	<0.01
Social anxiety disorder	Intensity	-0.02	<0.01
	Frequency	-0.10	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. The OR rule was used as the reference group for the comparisons with intensity and frequency. “OR” rule represents original DSM-XC scoring instructions. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 26. LR+ differences between DSM-XC OR rule and intensity/frequency question framings.

Diagnosis	Framing being compared to OR rule	LR+ difference (median*)	P-value
Alcohol use disorder	Intensity	0.04	0.03
	Frequency	-0.33	<0.01
Antisocial personality disorder	Intensity	0.03	0.75
	Frequency	-0.55	<0.01
Attention-deficit/hyperactivity disorder	Intensity	-0.03	0.29
	Frequency	-0.60	<0.01
Bipolar disorder	Intensity	0.04	0.55
	Frequency	-1.48	<0.01
Generalized anxiety disorder	Intensity	-0.11	<0.01
	Frequency	-1.21	<0.01
Major depressive disorder	Intensity	-0.10	0.02
	Frequency	-1.47	<0.01
Post-traumatic stress disorder	Intensity	-0.12	<0.01
	Frequency	-1.54	<0.01
Social anxiety disorder	Intensity	-0.08	<0.01
	Frequency	-0.83	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. The OR rule was used as the reference group for the comparisons with intensity and frequency. “OR” rule represents original DSM-XC scoring instructions. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 27. LR- differences between DSM-XC OR rule and intensity/frequency question framings.

Diagnosis	Framing being compared to OR rule	LR- difference (median*)	P-value
Alcohol use disorder	Intensity	-0.04	<0.01
	Frequency	-0.07	<0.01
Antisocial personality disorder	Intensity	-0.04	0.15
	Frequency	-0.02	0.29
Attention-deficit/hyperactivity disorder	Intensity	-0.03	<0.01
	Frequency	-0.06	<0.01
Bipolar disorder	Intensity	-0.07	0.01
	Frequency	-0.05	0.01
Generalized anxiety disorder	Intensity	-0.02	<0.01
	Frequency	-0.07	<0.01
Major depressive disorder	Intensity	-0.04	0.01
	Frequency	-0.05	<0.01
Post-traumatic stress disorder	Intensity	-0.04	0.01
	Frequency	-0.05	<0.01
Social anxiety disorder	Intensity	-0.02	0.01
	Frequency	-0.07	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. The OR rule was used as the reference group for the comparisons with intensity and frequency. “OR” rule represents original DSM-XC scoring instructions. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 28. Odds ratios for different framing combinations of specific DSM-XC domains to detect corresponding disorders.

Domain and disorder	Odds ratio (95% CI)			
	Intensity (-) / Frequency (-) (reference group)	Intensity (-) / Frequency (+)	Intensity (+) / Frequency (-)	Intensity (+) / Frequency (+)
Depression domain to detect MDD	0.00 (0.00 - 0.01)	37.25* (11.09 - 125.15)	4.34* (1.45 - 12.98)	68.57* (27.68 - 169.84)
Mania domain to detect BD	0.03 (0.02 - 0.04)	5.23* (2.36 - 11.56)	2.23 (0.98 - 5.07)	8.25* (4.48 - 15.16)
Anxiety domain to detect GAD	0.01 (0.01 - 0.02)	5.00* (2.71 - 9.21)	2.07* (1.35 - 3.18)	14.21* (10.36 - 19.51)
Anxiety domain to detect SAD	0.01 (0.00 - 0.01)	3.03* (1.14 - 8.05)	3.04* (1.78 - 5.21)	9.90* (6.38 - 15.36)
Personality functioning domain to detect APD	0.01 (0.01 - 0.01)	4.95* (1.40 - 17.48)	2.00 (0.66 - 6.12)	4.59* (2.20 - 9.55)
Substance use domain to detect AUD	0.02 (0.01 - 0.02)	6.92* (3.86 - 12.41)	0.84 (0.20 - 3.57)	8.98* (5.97 - 13.51)

Intensity (+) and (-) represents above- and below- threshold domains, respectively for the intensity framing. Frequency (+) and (-) represents above- and below- threshold domains, respectively for the frequency framing. MDD, major depressive disorder. BD, bipolar disorder. GAD, generalized anxiety disorder. SAD, social anxiety disorder. APD, antisocial personality disorder. AUD, alcohol use disorder. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure. *P-value <0.05 for the comparison with reference group, Intensity (-) /Frequency (-).

Supplementary table 29. Sensitivities and specificities of DSM-XC domains for alcohol use disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.55 (0.48-0.62)	0.30 (0.24-0.37)
	Specificity	0.60 (0.59-0.62)	0.82 (0.81-0.83)
II. Anger	Sensitivity	0.46 (0.39-0.53)	0.29 (0.23-0.36)
	Specificity	0.72 (0.70-0.73)	0.85 (0.84-0.86)
III. Mania	Sensitivity	0.31 (0.25-0.38)	0.23 (0.18-0.29)
	Specificity	0.79 (0.77-0.80)	0.85 (0.83-0.86)
IV. Anxiety	Sensitivity	0.60 (0.53-0.67)	0.44 (0.37-0.51)
	Specificity	0.54 (0.52-0.56)	0.71 (0.69-0.72)
V. Somatic symptoms	Sensitivity	0.45 (0.38-0.52)	0.31 (0.25-0.38)
	Specificity	0.67 (0.65-0.69)	0.80 (0.78-0.81)
VI. Suicidal ideation	Sensitivity	0.14 (0.10-0.19)	0.12 (0.08-0.17)
	Specificity	0.92 (0.91-0.93)	0.95 (0.94-0.96)
VII. Psychosis	Sensitivity	0.27 (0.21-0.34)	0.26 (0.21-0.33)
	Specificity	0.88 (0.86-0.89)	0.91 (0.90-0.92)
VIII. Sleep problems	Sensitivity	0.43 (0.36-0.50)	0.32 (0.26-0.39)
	Specificity	0.79 (0.77-0.80)	0.85 (0.84-0.86)
IX. Memory	Sensitivity	0.25 (0.19-0.31)	0.15 (0.11-0.21)
	Specificity	0.86 (0.84-0.87)	0.92 (0.91-0.93)
X. Repetitive thoughts and behaviors	Sensitivity	0.32 (0.26-0.39)	0.25 (0.19-0.31)
	Specificity	0.83 (0.82-0.85)	0.90 (0.89-0.91)
XI. Dissociation	Sensitivity	0.25 (0.20-0.32)	0.22 (0.17-0.28)
	Specificity	0.86 (0.85-0.88)	0.92 (0.91-0.93)
XII. Personality functioning	Sensitivity	0.45 (0.38-0.52)	0.43 (0.36-0.50)
	Specificity	0.74 (0.73-0.76)	0.82 (0.80-0.83)
XIII. Substance use	Sensitivity	0.76 (0.70-0.82)	0.86 (0.80-0.90)
	Specificity	0.61 (0.59-0.63)	0.60 (0.58-0.61)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 30. Sensitivities and specificities of DSM-XC domains for antisocial personality disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.58 (0.41-0.74)	0.33 (0.19-0.51)
	Specificity	0.60 (0.58-0.61)	0.82 (0.80-0.83)
II. Anger	Sensitivity	0.44 (0.28-0.62)	0.36 (0.21-0.54)
	Specificity	0.71 (0.69-0.72)	0.84 (0.83-0.86)
III. Mania	Sensitivity	0.39 (0.23-0.57)	0.22 (0.10-0.39)
	Specificity	0.78 (0.77-0.79)	0.84 (0.83-0.85)
IV. Anxiety	Sensitivity	0.69 (0.52-0.84)	0.47 (0.30-0.65)
	Specificity	0.53 (0.52-0.55)	0.70 (0.69-0.72)
V. Somatic symptoms	Sensitivity	0.42 (0.26-0.59)	0.44 (0.28-0.62)
	Specificity	0.66 (0.65-0.68)	0.79 (0.78-0.81)
VI. Suicidal ideation	Sensitivity	0.11 (0.03-0.26)	0.06 (0.01-0.19)
	Specificity	0.92 (0.91-0.93)	0.95 (0.94-0.96)
VII. Psychosis	Sensitivity	0.36 (0.21-0.54)	0.33 (0.19-0.51)
	Specificity	0.87 (0.86-0.88)	0.91 (0.90-0.92)
VIII. Sleep problems	Sensitivity	0.28 (0.14-0.45)	0.39 (0.23-0.57)
	Specificity	0.78 (0.76-0.79)	0.84 (0.83-0.85)
IX. Memory	Sensitivity	0.25 (0.12-0.42)	0.17 (0.06-0.33)
	Specificity	0.85 (0.84-0.86)	0.92 (0.91-0.93)
X. Repetitive thoughts and behaviors	Sensitivity	0.31 (0.16-0.48)	0.33 (0.19-0.51)
	Specificity	0.83 (0.81-0.84)	0.89 (0.88-0.90)
XI. Dissociation	Sensitivity	0.33 (0.19-0.51)	0.36 (0.21-0.54)
	Specificity	0.86 (0.85-0.87)	0.91 (0.90-0.92)
XII. Personality functioning	Sensitivity	0.53 (0.35-0.70)	0.50 (0.33-0.67)
	Specificity	0.73 (0.72-0.75)	0.80 (0.79-0.82)
XIII. Substance use	Sensitivity	0.69 (0.52-0.84)	0.81 (0.64-0.92)
	Specificity	0.59 (0.57-0.61)	0.57 (0.56-0.59)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 31. Sensitivities and specificities of DSM-XC domains for attention-deficit/hyperactivity disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.67 (0.59-0.74)	0.49 (0.41-0.57)
	Specificity	0.61 (0.59-0.62)	0.83 (0.82-0.84)
II. Anger	Sensitivity	0.56 (0.48-0.63)	0.40 (0.33-0.48)
	Specificity	0.72 (0.70-0.73)	0.85 (0.84-0.87)
III. Mania	Sensitivity	0.35 (0.28-0.43)	0.30 (0.23-0.37)
	Specificity	0.79 (0.77-0.80)	0.85 (0.84-0.86)
IV. Anxiety	Sensitivity	0.70 (0.62-0.77)	0.60 (0.53-0.68)
	Specificity	0.54 (0.53-0.56)	0.71 (0.70-0.73)
V. Somatic symptoms	Sensitivity	0.54 (0.46-0.62)	0.38 (0.30-0.46)
	Specificity	0.67 (0.66-0.69)	0.80 (0.78-0.81)
VI. Suicidal ideation	Sensitivity	0.15 (0.10-0.22)	0.14 (0.09-0.20)
	Specificity	0.92 (0.91-0.93)	0.95 (0.94-0.96)
VII. Psychosis	Sensitivity	0.27 (0.20-0.35)	0.25 (0.19-0.33)
	Specificity	0.87 (0.86-0.89)	0.91 (0.90-0.92)
VIII. Sleep problems	Sensitivity	0.41 (0.34-0.49)	0.41 (0.33-0.49)
	Specificity	0.78 (0.77-0.80)	0.85 (0.84-0.86)
IX. Memory	Sensitivity	0.34 (0.27-0.42)	0.25 (0.18-0.32)
	Specificity	0.86 (0.85-0.87)	0.93 (0.92-0.93)
X. Repetitive thoughts and behaviors	Sensitivity	0.40 (0.33-0.48)	0.31 (0.24-0.39)
	Specificity	0.83 (0.82-0.85)	0.90 (0.89-0.91)
XI. Dissociation	Sensitivity	0.35 (0.27-0.42)	0.23 (0.17-0.31)
	Specificity	0.87 (0.86-0.88)	0.92 (0.91-0.93)
XII. Personality functioning	Sensitivity	0.47 (0.39-0.55)	0.40 (0.33-0.48)
	Specificity	0.74 (0.73-0.76)	0.81 (0.80-0.82)
XIII. Substance use	Sensitivity	0.50 (0.42-0.58)	0.54 (0.46-0.62)
	Specificity	0.59 (0.58-0.61)	0.58 (0.56-0.59)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 32. Sensitivities and specificities of DSM-XC domains for bipolar disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.70 (0.57-0.81)	0.63 (0.50-0.75)
	Specificity	0.60 (0.58-0.62)	0.82 (0.81-0.83)
II. Anger	Sensitivity	0.70 (0.57-0.81)	0.68 (0.55-0.80)
	Specificity	0.71 (0.70-0.73)	0.85 (0.84-0.86)
III. Mania	Sensitivity	0.50 (0.37-0.63)	0.52 (0.38-0.65)
	Specificity	0.78 (0.77-0.80)	0.85 (0.84-0.86)
IV. Anxiety	Sensitivity	0.92 (0.82-0.97)	0.85 (0.73-0.93)
	Specificity	0.54 (0.52-0.56)	0.71 (0.69-0.72)
V. Somatic symptoms	Sensitivity	0.77 (0.64-0.87)	0.72 (0.59-0.83)
	Specificity	0.67 (0.65-0.69)	0.80 (0.79-0.81)
VI. Suicidal ideation	Sensitivity	0.30 (0.19-0.43)	0.32 (0.20-0.45)
	Specificity	0.92 (0.91-0.93)	0.95 (0.94-0.96)
VII. Psychosis	Sensitivity	0.53 (0.40-0.66)	0.52 (0.38-0.65)
	Specificity	0.87 (0.86-0.89)	0.91 (0.90-0.92)
VIII. Sleep problems	Sensitivity	0.48 (0.35-0.62)	0.50 (0.37-0.63)
	Specificity	0.78 (0.77-0.79)	0.85 (0.83-0.86)
IX. Memory	Sensitivity	0.40 (0.28-0.53)	0.35 (0.23-0.48)
	Specificity	0.86 (0.84-0.87)	0.92 (0.91-0.93)
X. Repetitive thoughts and behaviors	Sensitivity	0.67 (0.53-0.78)	0.58 (0.45-0.71)
	Specificity	0.83 (0.82-0.84)	0.90 (0.88-0.91)
XI. Dissociation	Sensitivity	0.48 (0.35-0.62)	0.50 (0.37-0.63)
	Specificity	0.86 (0.85-0.87)	0.92 (0.91-0.93)
XII. Personality functioning	Sensitivity	0.58 (0.45-0.71)	0.62 (0.48-0.74)
	Specificity	0.74 (0.72-0.75)	0.81 (0.80-0.82)
XIII. Substance use	Sensitivity	0.63 (0.50-0.75)	0.62 (0.48-0.74)
	Specificity	0.59 (0.58-0.61)	0.57 (0.56-0.59)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 33. Sensitivities and specificities of DSM-XC domains for generalized anxiety disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.76 (0.72-0.81)	0.59 (0.54-0.64)
	Specificity	0.64 (0.62-0.65)	0.86 (0.85-0.87)
II. Anger	Sensitivity	0.67 (0.62-0.71)	0.53 (0.48-0.59)
	Specificity	0.75 (0.73-0.76)	0.89 (0.88-0.90)
III. Mania	Sensitivity	0.37 (0.32-0.42)	0.30 (0.25-0.35)
	Specificity	0.80 (0.78-0.81)	0.86 (0.85-0.87)
IV. Anxiety	Sensitivity	0.83 (0.79-0.87)	0.76 (0.72-0.81)
	Specificity	0.57 (0.56-0.59)	0.75 (0.74-0.77)
V. Somatic symptoms	Sensitivity	0.64 (0.59-0.69)	0.53 (0.48-0.58)
	Specificity	0.70 (0.68-0.71)	0.83 (0.81-0.84)
VI. Suicidal ideation	Sensitivity	0.19 (0.15-0.24)	0.18 (0.14-0.23)
	Specificity	0.93 (0.92-0.94)	0.96 (0.96-0.97)
VII. Psychosis	Sensitivity	0.26 (0.22-0.31)	0.26 (0.22-0.31)
	Specificity	0.88 (0.87-0.89)	0.92 (0.91-0.93)
VIII. Sleep problems	Sensitivity	0.52 (0.46-0.57)	0.43 (0.38-0.48)
	Specificity	0.81 (0.80-0.82)	0.87 (0.86-0.88)
IX. Memory	Sensitivity	0.36 (0.31-0.41)	0.25 (0.20-0.29)
	Specificity	0.88 (0.86-0.89)	0.94 (0.93-0.95)
X. Repetitive thoughts and behaviors	Sensitivity	0.47 (0.42-0.52)	0.36 (0.31-0.41)
	Specificity	0.86 (0.85-0.87)	0.92 (0.91-0.93)
XI. Dissociation	Sensitivity	0.37 (0.32-0.42)	0.30 (0.25-0.35)
	Specificity	0.88 (0.87-0.90)	0.94 (0.93-0.94)
XII. Personality functioning	Sensitivity	0.58 (0.53-0.63)	0.52 (0.47-0.58)
	Specificity	0.77 (0.75-0.78)	0.84 (0.83-0.85)
XIII. Substance use	Sensitivity	0.54 (0.49-0.60)	0.55 (0.50-0.60)
	Specificity	0.60 (0.59-0.62)	0.58 (0.57-0.60)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 34. Sensitivities and specificities of DSM-XC domains for major depressive disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.90 (0.82-0.95)	0.87 (0.79-0.93)
	Specificity	0.61 (0.59-0.63)	0.83 (0.82-0.85)
II. Anger	Sensitivity	0.70 (0.61-0.79)	0.64 (0.54-0.73)
	Specificity	0.72 (0.70-0.73)	0.86 (0.84-0.87)
III. Mania	Sensitivity	0.38 (0.29-0.48)	0.31 (0.23-0.41)
	Specificity	0.78 (0.77-0.80)	0.85 (0.83-0.86)
IV. Anxiety	Sensitivity	0.90 (0.83-0.95)	0.88 (0.80-0.93)
	Specificity	0.54 (0.53-0.56)	0.72 (0.70-0.73)
V. Somatic symptoms	Sensitivity	0.73 (0.64-0.81)	0.63 (0.53-0.72)
	Specificity	0.67 (0.66-0.69)	0.80 (0.79-0.82)
VI. Suicidal ideation	Sensitivity	0.35 (0.26-0.45)	0.34 (0.25-0.44)
	Specificity	0.92 (0.92-0.93)	0.96 (0.95-0.96)
VII. Psychosis	Sensitivity	0.33 (0.24-0.43)	0.31 (0.23-0.41)
	Specificity	0.87 (0.86-0.88)	0.91 (0.90-0.92)
VIII. Sleep problems	Sensitivity	0.67 (0.57-0.76)	0.57 (0.47-0.67)
	Specificity	0.79 (0.77-0.80)	0.85 (0.84-0.86)
IX. Memory	Sensitivity	0.34 (0.25-0.44)	0.31 (0.23-0.41)
	Specificity	0.86 (0.84-0.87)	0.93 (0.92-0.93)
X. Repetitive thoughts and behaviors	Sensitivity	0.61 (0.51-0.70)	0.52 (0.42-0.62)
	Specificity	0.84 (0.82-0.85)	0.90 (0.89-0.91)
XI. Dissociation	Sensitivity	0.54 (0.44-0.64)	0.48 (0.38-0.58)
	Specificity	0.87 (0.86-0.88)	0.92 (0.91-0.93)
XII. Personality functioning	Sensitivity	0.82 (0.73-0.89)	0.80 (0.71-0.87)
	Specificity	0.75 (0.73-0.76)	0.82 (0.81-0.83)
XIII. Substance use	Sensitivity	0.60 (0.50-0.69)	0.60 (0.05-0.69)
	Specificity	0.59 (0.58-0.61)	0.58 (0.56-0.59)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 35. Sensitivities and specificities of DSM-XC domains for post-traumatic stress disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.74 (0.67-0.81)	0.65 (0.57-0.73)
	Specificity	0.61 (0.59-0.63)	0.84 (0.82-0.85)
II. Anger	Sensitivity	0.67 (0.59-0.74)	0.57 (0.49-0.65)
	Specificity	0.72 (0.71-0.74)	0.86 (0.85-0.87)
III. Mania	Sensitivity	0.45 (0.37-0.53)	0.41 (0.33-0.49)
	Specificity	0.79 (0.78-0.80)	0.85 (0.84-0.87)
IV. Anxiety	Sensitivity	0.90 (0.85-0.94)	0.83 (0.76-0.89)
	Specificity	0.55 (0.53-0.57)	0.72 (0.71-0.74)
V. Somatic symptoms	Sensitivity	0.71 (0.63-0.78)	0.63 (0.55-0.71)
	Specificity	0.68 (0.66-0.70)	0.81 (0.80-0.82)
VI. Suicidal ideation	Sensitivity	0.26 (0.20-0.34)	0.24 (0.17-0.31)
	Specificity	0.92 (0.92-0.93)	0.96 (0.95-0.96)
VII. Psychosis	Sensitivity	0.39 (0.31-0.47)	0.39 (0.31-0.47)
	Specificity	0.88 (0.87-0.89)	0.92 (0.91-0.93)
VIII. Sleep problems	Sensitivity	0.59 (0.51-0.67)	0.59 (0.51-0.67)
	Specificity	0.79 (0.78-0.80)	0.86 (0.85-0.87)
IX. Memory	Sensitivity	0.43 (0.35-0.51)	0.37 (0.29-0.45)
	Specificity	0.86 (0.85-0.87)	0.93 (0.92-0.94)
X. Repetitive thoughts and behaviors	Sensitivity	0.65 (0.57-0.73)	0.54 (0.46-0.62)
	Specificity	0.85 (0.83-0.86)	0.91 (0.90-0.92)
XI. Dissociation	Sensitivity	0.47 (0.39-0.55)	0.42 (0.34-0.50)
	Specificity	0.87 (0.86-0.88)	0.93 (0.92-0.93)
XII. Personality functioning	Sensitivity	0.69 (0.61-0.76)	0.68 (0.60-0.75)
	Specificity	0.75 (0.74-0.77)	0.82 (0.81-0.84)
XIII. Substance use	Sensitivity	0.63 (0.54-0.70)	0.62 (0.54-0.70)
	Specificity	0.60 (0.58-0.61)	0.58 (0.56-0.60)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 36. Sensitivities and specificities of DSM-XC domains for social anxiety disorder.

Domain	Metric	Intensity	Frequency
I. Depression	Sensitivity	0.74 (0.66-0.80)	0.53 (0.45-0.60)
	Specificity	0.61 (0.59-0.63)	0.83 (0.82-0.84)
II. Anger	Sensitivity	0.57 (0.50-0.65)	0.42 (0.35-0.50)
	Specificity	0.72 (0.70-0.73)	0.86 (0.84-0.87)
III. Mania	Sensitivity	0.40 (0.33-0.48)	0.29 (0.23-0.36)
	Specificity	0.79 (0.78-0.80)	0.85 (0.84-0.86)
IV. Anxiety	Sensitivity	0.83 (0.77-0.88)	0.69 (0.62-0.76)
	Specificity	0.55 (0.53-0.57)	0.72 (0.70-0.73)
V. Somatic symptoms	Sensitivity	0.57 (0.49-0.64)	0.46 (0.39-0.54)
	Specificity	0.68 (0.66-0.69)	0.80 (0.79-0.82)
VI. Suicidal ideation	Sensitivity	0.24 (0.18-0.31)	0.21 (0.15-0.27)
	Specificity	0.92 (0.92-0.93)	0.96 (0.95-0.96)
VII. Psychosis	Sensitivity	0.30 (0.23-0.37)	0.28 (0.21-0.35)
	Specificity	0.88 (0.86-0.89)	0.91 (0.90-0.92)
VIII. Sleep problems	Sensitivity	0.51 (0.44-0.59)	0.44 (0.37-0.52)
	Specificity	0.79 (0.78-0.80)	0.86 (0.84-0.87)
IX. Memory	Sensitivity	0.39 (0.32-0.46)	0.28 (0.21-0.35)
	Specificity	0.86 (0.85-0.87)	0.93 (0.92-0.94)
X. Repetitive thoughts and behaviors	Sensitivity	0.43 (0.36-0.51)	0.35 (0.28-0.42)
	Specificity	0.84 (0.82-0.85)	0.90 (0.89-0.91)
XI. Dissociation	Sensitivity	0.35 (0.28-0.43)	0.30 (0.23-0.37)
	Specificity	0.87 (0.86-0.88)	0.92 (0.91-0.93)
XII. Personality functioning	Sensitivity	0.58 (0.51-0.66)	0.50 (0.42-0.58)
	Specificity	0.75 (0.73-0.76)	0.82 (0.80-0.83)
XIII. Substance use	Sensitivity	0.55 (0.47-0.63)	0.55 (0.47-0.63)
	Specificity	0.60 (0.58-0.61)	0.58 (0.56-0.59)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 37. Positive likelihood ratios of DSM-XC domains for alcohol use disorder.

Domain	Intensity	Frequency
I. Depression	1.38 (1.21-1.57)	1.70 (1.36-2.11)
II. Anger	1.62 (1.39-1.89)	1.96 (1.57-2.46)
III. Mania	1.45 (1.18-1.80)	1.51 (1.16-1.96)
IV. Anxiety	1.30 (1.16-1.47)	1.50 (1.27-1.76)
V. Somatic symptoms	1.35 (1.16-1.59)	1.51 (1.22-1.87)
VI. Suicidal ideation	1.74 (1.22-2.49)	2.52 (1.69-3.74)
VII. Psychosis	2.21 (1.74-2.81)	3.08 (2.40-3.97)
VIII. Sleep problems	2.01 (1.70-2.39)	2.16 (1.74-2.67)
IX. Memory	1.71 (1.33-2.20)	1.99 (1.41-2.79)
X. Repetitive thoughts and behaviors	1.93 (1.56-2.39)	2.40 (1.86-3.10)
XI. Dissociation	1.88 (1.47-2.41)	2.72 (2.06-3.60)
XII. Personality functioning	1.74 (1.48-2.05)	2.33 (1.96-2.76)
XIII. Substance use	1.96 (1.80-2.14)	2.12 (1.98-2.27)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 38. Positive likelihood ratios of DSM-XC domains for antisocial personality disorder.

Domain	Intensity	Frequency
I. Depression	1.44 (1.09-1.91)	1.81 (1.13-2.88)
II. Anger	1.51 (1.05-2.19)	2.32 (1.49-3.61)
III. Mania	1.78 (1.17-2.69)	1.42 (0.76-2.62)
IV. Anxiety	1.49 (1.20-1.85)	1.58 (1.11-2.24)
V. Somatic symptoms	1.24 (0.84-1.83)	2.14 (1.48-3.10)
VI. Suicidal ideation	1.33 (0.53-3.38)	1.07 (0.28-4.14)
VII. Psychosis	2.77 (1.78-4.32)	3.56 (2.22-5.71)
VIII. Sleep problems	1.24 (0.73-2.10)	2.48 (1.63-3.76)
IX. Memory	1.69 (0.95-2.99)	2.06 (0.98-4.31)
X. Repetitive thoughts and behaviors	1.75 (1.06-2.88)	3.02 (1.88-4.84)
XI. Dissociation	2.37 (1.48-3.79)	4.17 (2.66-6.52)
XII. Personality functioning	1.99 (1.45-2.72)	2.56 (1.84-3.58)
XIII. Substance use	1.70 (1.36-2.12)	1.89 (1.60-2.23)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 39. Positive likelihood ratios of DSM-XC domains for attention-deficit/hyperactivity disorder.

Domain	Intensity	Frequency
I. Depression	1.69 (1.51-1.90)	2.84 (2.39-3.38)
II. Anger	1.96 (1.69-2.28)	2.75 (2.24-3.37)
III. Mania	1.64 (1.32-2.04)	1.96 (1.53-2.52)
IV. Anxiety	1.52 (1.37-1.70)	2.11 (1.85-2.42)
V. Somatic symptoms	1.64 (1.41-1.91)	1.86 (1.51-2.30)
VI. Suicidal ideation	1.92 (1.32-2.81)	2.83 (1.87-4.29)
VII. Psychosis	2.16 (1.65-2.82)	2.85 (2.14-3.80)
VIII. Sleep problems	1.91 (1.58-2.32)	2.76 (2.26-3.38)
IX. Memory	2.42 (1.92-3.05)	3.33 (2.48-4.47)
X. Repetitive thoughts and behaviors	2.43 (1.98-2.98)	2.99 (2.32-3.84)
XI. Dissociation	2.60 (2.07-3.27)	2.84 (2.11-3.83)
XII. Personality functioning	1.82 (1.53-2.16)	2.13 (1.74-2.60)
XIII. Substance use	1.23 (1.05-1.44)	1.27 (1.09-1.47)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 40. Positive likelihood ratios of DSM-XC domains for bipolar disorder.

Domain	Intensity	Frequency
I. Depression	1.75 (1.47-2.07)	3.55 (2.89-4.36)
II. Anger	2.43 (2.04-2.89)	4.60 (3.80-5.56)
III. Mania	2.32 (1.79-3.01)	3.41 (2.64-4.41)
IV. Anxiety	1.99 (1.83-2.16)	2.92 (2.59-3.28)
V. Somatic symptoms	2.33 (2.01-2.69)	3.56 (3.00-4.23)
VI. Suicidal ideation	3.76 (2.51-5.62)	6.67 (4.47-9.95)
VII. Psychosis	4.24 (3.30-5.46)	5.81 (4.45-7.58)
VIII. Sleep problems	2.19 (1.67-2.87)	3.26 (2.50-4.24)
IX. Memory	2.76 (2.00-3.80)	4.53 (3.15-6.51)
X. Repetitive thoughts and behaviors	3.98 (3.28-4.83)	5.58 (4.41-7.05)
XI. Dissociation	3.54 (2.69-4.65)	6.07 (4.60-7.99)
XII. Personality functioning	2.22 (1.78-2.77)	3.23 (2.61-3.99)
XIII. Substance use	1.55 (1.27-1.89)	1.45 (1.18-1.77)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 41. Positive likelihood ratios of DSM-XC domains for generalized anxiety disorder.

Domain	Intensity	Frequency
I. Depression	2.10 (1.96-2.26)	4.30 (3.81-4.85)
II. Anger	2.65 (2.42-2.91)	4.72 (4.13-5.41)
III. Mania	1.83 (1.57-2.12)	2.12 (1.78-2.53)
IV. Anxiety	1.95 (1.84-2.07)	3.11 (2.86-3.38)
V. Somatic symptoms	2.14 (1.95-2.34)	3.07 (2.72-3.47)
VI. Suicidal ideation	2.73 (2.15-3.48)	4.99 (3.78-6.59)
VII. Psychosis	2.26 (1.87-2.75)	3.37 (2.74-4.15)
VIII. Sleep problems	2.71 (2.40-3.05)	3.39 (2.93-3.93)
IX. Memory	2.86 (2.43-3.37)	3.94 (3.15-4.91)
X. Repetitive thoughts and behaviors	3.34 (2.91-3.83)	4.26 (3.57-5.09)
XI. Dissociation	3.23 (2.75-3.80)	4.62 (3.77-5.66)
XII. Personality functioning	2.49 (2.24-2.78)	3.28 (2.90-3.72)
XIII. Substance use	1.38 (1.24-1.52)	1.33 (1.21-1.47)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 42. Positive likelihood ratios of DSM-XC domains for major depressive disorder.

Domain	Intensity	Frequency
I. Depression	2.29 (2.12-2.47)	5.24 (4.72-5.83)
II. Anger	2.49 (2.18-2.85)	4.46 (3.78-5.26)
III. Mania	1.77 (1.37-2.27)	2.06 (1.53-2.76)
IV. Anxiety	1.99 (1.85-2.13)	3.09 (2.83-3.38)
V. Somatic symptoms	2.26 (1.99-2.56)	3.19 (2.71-3.75)
VI. Suicidal ideation	4.67 (3.52-6.21)	7.94 (5.84-10.8)
VII. Psychosis	2.64 (1.98-3.50)	3.51 (2.60-4.75)
VIII. Sleep problems	3.15 (2.71-3.66)	3.89 (3.24-4.68)
IX. Memory	2.39 (1.81-3.15)	4.20 (3.09-5.70)
X. Repetitive thoughts and behaviors	3.75 (3.16-4.44)	5.23 (4.25-6.44)
XI. Dissociation	4.16 (3.42-5.06)	6.13 (4.86-7.72)
XII. Personality functioning	3.26 (2.93-3.63)	4.45 (3.95-5.01)
XIII. Substance use	1.48 (1.26-1.74)	1.41 (1.20-1.66)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 43. Positive likelihood ratios of DSM-XC domains for post-traumatic stress disorder.

Domain	Intensity	Frequency
I. Depression	1.90 (1.72-2.10)	3.95 (3.45-4.54)
II. Anger	2.41 (2.13-2.73)	4.08 (3.48-4.79)
III. Mania	2.15 (1.79-2.59)	2.83 (2.30-3.47)
IV. Anxiety	2.01 (1.89-2.14)	3.01 (2.75-3.29)
V. Somatic symptoms	2.22 (1.98-2.48)	3.31 (2.89-3.81)
VI. Suicidal ideation	3.51 (2.63-4.68)	5.48 (3.97-7.57)
VII. Psychosis	3.20 (2.57-3.98)	4.67 (3.72-5.86)
VIII. Sleep problems	2.81 (2.43-3.26)	4.20 (3.59-4.90)
IX. Memory	3.17 (2.60-3.87)	5.33 (4.19-6.78)
X. Repetitive thoughts and behaviors	4.22 (3.68-4.86)	5.82 (4.86-6.95)
XI. Dissociation	3.69 (3.06-4.45)	5.63 (4.52-7.01)
XII. Personality functioning	2.78 (2.46-3.13)	3.84 (3.37-4.37)
XIII. Substance use	1.56 (1.37-1.77)	1.47 (1.29-1.67)
DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.		

Supplementary table 44. Positive likelihood ratios of DSM-XC domains for social anxiety disorder.

Domain	Intensity	Frequency
I. Depression	1.89 (1.72-2.09)	3.14 (2.69-3.68)
II. Anger	2.04 (1.78-2.34)	2.93 (2.42-3.55)
III. Mania	1.92 (1.59-2.32)	1.94 (1.52-2.47)
IV. Anxiety	1.85 (1.71-1.99)	2.47 (2.20-2.76)
V. Somatic symptoms	1.75 (1.52-2.00)	2.34 (1.97-2.78)
VI. Suicidal ideation	3.21 (2.41-4.27)	4.74 (3.42-6.58)
VII. Psychosis	2.40 (1.89-3.07)	3.17 (2.44-4.12)
VIII. Sleep problems	2.43 (2.08-2.85)	3.07 (2.56-3.69)
IX. Memory	2.83 (2.31-3.47)	3.84 (2.94-5.01)
X. Repetitive thoughts and behaviors	2.66 (2.21-3.21)	3.47 (2.77-4.35)
XI. Dissociation	2.69 (2.17-3.34)	3.79 (2.94-4.88)
XII. Personality functioning	2.33 (2.03-2.67)	2.74 (2.33-3.23)
XIII. Substance use	1.36 (1.18-1.56)	1.30 (1.13-1.49)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 45. Negative likelihood ratios of DSM-XC domains for alcohol use disorder.

Domain	Intensity	Frequency
I. Depression	0.75 (0.64-0.87)	0.85 (0.77-0.93)
II. Anger	0.75 (0.66-0.86)	0.83 (0.76-0.91)
III. Mania	0.88 (0.80-0.96)	0.91 (0.84-0.98)
IV. Anxiety	0.74 (0.62-0.88)	0.79 (0.70-0.90)
V. Somatic symptoms	0.83 (0.73-0.93)	0.87 (0.79-0.95)
VI. Suicidal ideation	0.94 (0.88-0.99)	0.92 (0.88-0.97)
VII. Psychosis	0.83 (0.76-0.90)	0.80 (0.74-0.87)
VIII. Sleep problems	0.73 (0.65-0.82)	0.80 (0.72-0.88)
IX. Memory	0.88 (0.81-0.95)	0.92 (0.86-0.97)
X. Repetitive thoughts and behaviors	0.81 (0.74-0.89)	0.84 (0.77-0.91)
XI. Dissociation	0.86 (0.80-0.93)	0.85 (0.79-0.91)
XII. Personality functioning	0.74 (0.66-0.84)	0.70 (0.62-0.79)
XIII. Substance use	0.39 (0.30-0.49)	0.24 (0.17-0.34)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 46. Negative likelihood ratios of DSM-XC domains for antisocial personality disorder.

Domain	Intensity	Frequency
I. Depression	0.70 (0.47-1.03)	0.82 (0.65-1.03)
II. Anger	0.79 (0.59-1.05)	0.76 (0.59-0.97)
III. Mania	0.78 (0.60-1.02)	0.92 (0.77-1.10)
IV. Anxiety	0.57 (0.35-0.94)	0.75 (0.55-1.03)
V. Somatic symptoms	0.88 (0.67-1.16)	0.70 (0.52-0.94)
VI. Suicidal ideation	0.97 (0.86-1.09)	1.00 (0.92-1.08)
VII. Psychosis	0.73 (0.57-0.94)	0.74 (0.58-0.93)
VIII. Sleep problems	0.93 (0.76-1.14)	0.72 (0.56-0.94)
IX. Memory	0.88 (0.73-1.06)	0.91 (0.78-1.05)
X. Repetitive thoughts and behaviors	0.84 (0.68-1.05)	0.75 (0.59-0.94)
XI. Dissociation	0.78 (0.62-0.98)	0.70 (0.55-0.89)
XII. Personality functioning	0.64 (0.45-0.91)	0.62 (0.45-0.86)
XIII. Substance use	0.52 (0.32-0.85)	0.34 (0.17-0.66)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 47. Negative likelihood ratios of DSM-XC domains for attention-deficit/hyperactivity disorder.

Domain	Intensity	Frequency
I. Depression	0.55 (0.44-0.68)	0.62 (0.53-0.72)
II. Anger	0.62 (0.52-0.74)	0.70 (0.62-0.80)
III. Mania	0.82 (0.74-0.93)	0.83 (0.75-0.92)
IV. Anxiety	0.56 (0.44-0.71)	0.55 (0.46-0.67)
V. Somatic symptoms	0.69 (0.58-0.81)	0.78 (0.69-0.88)
VI. Suicidal ideation	0.92 (0.86-0.98)	0.91 (0.85-0.97)
VII. Psychosis	0.83 (0.76-0.92)	0.82 (0.75-0.90)
VIII. Sleep problems	0.75 (0.66-0.85)	0.70 (0.61-0.79)
IX. Memory	0.77 (0.69-0.86)	0.81 (0.74-0.89)
X. Repetitive thoughts and behaviors	0.72 (0.63-0.81)	0.77 (0.70-0.86)
XI. Dissociation	0.75 (0.67-0.84)	0.83 (0.77-0.91)
XII. Personality functioning	0.72 (0.62-0.83)	0.74 (0.65-0.84)
XIII. Substance use	0.84 (0.72-0.99)	0.80 (0.68-0.95)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 48. Negative likelihood ratios of DSM-XC domains for bipolar disorder.

Domain	Intensity	Frequency
I. Depression	0.50 (0.34-0.74)	0.45 (0.32-0.62)
II. Anger	0.42 (0.29-0.62)	0.37 (0.26-0.54)
III. Mania	0.64 (0.49-0.82)	0.57 (0.44-0.74)
IV. Anxiety	0.15 (0.07-0.36)	0.21 (0.12-0.39)
V. Somatic symptoms	0.35 (0.22-0.55)	0.35 (0.24-0.53)
VI. Suicidal ideation	0.76 (0.64-0.90)	0.72 (0.60-0.85)
VII. Psychosis	0.53 (0.41-0.70)	0.53 (0.41-0.69)
VIII. Sleep problems	0.66 (0.52-0.85)	0.59 (0.46-0.76)
IX. Memory	0.70 (0.57-0.86)	0.70 (0.58-0.85)
X. Repetitive thoughts and behaviors	0.40 (0.28-0.57)	0.47 (0.34-0.63)
XI. Dissociation	0.60 (0.47-0.76)	0.54 (0.42-0.70)
XII. Personality functioning	0.56 (0.42-0.76)	0.47 (0.34-0.65)
XIII. Substance use	0.62 (0.44-0.86)	0.67 (0.48-0.92)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 49. Negative likelihood ratios of DSM-XC domains for generalized anxiety disorder.

Domain	Intensity	Frequency
I. Depression	0.37 (0.31-0.44)	0.47 (0.42-0.53)
II. Anger	0.45 (0.39-0.51)	0.52 (0.47-0.59)
III. Mania	0.79 (0.73-0.85)	0.82 (0.76-0.87)
IV. Anxiety	0.29 (0.24-0.37)	0.31 (0.26-0.37)
V. Somatic symptoms	0.51 (0.45-0.59)	0.57 (0.51-0.63)
VI. Suicidal ideation	0.87 (0.83-0.91)	0.85 (0.81-0.89)
VII. Psychosis	0.83 (0.78-0.89)	0.80 (0.76-0.85)
VIII. Sleep problems	0.60 (0.54-0.66)	0.65 (0.60-0.71)
IX. Memory	0.73 (0.68-0.79)	0.80 (0.76-0.85)
X. Repetitive thoughts and behaviors	0.62 (0.56-0.68)	0.70 (0.65-0.76)
XI. Dissociation	0.71 (0.66-0.77)	0.75 (0.70-0.80)
XII. Personality functioning	0.55 (0.49-0.62)	0.57 (0.51-0.63)
XIII. Substance use	0.75 (0.67-0.84)	0.76 (0.68-0.86)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 50. Negative likelihood ratios of DSM-XC domains for major depressive disorder.

Domain	Intensity	Frequency
I. Depression	0.17 (0.10-0.30)	0.16 (0.10-0.26)
II. Anger	0.41 (0.31-0.55)	0.42 (0.33-0.54)
III. Mania	0.79 (0.68-0.92)	0.81 (0.71-0.92)
IV. Anxiety	0.17 (0.10-0.32)	0.17 (0.10-0.29)
V. Somatic symptoms	0.40 (0.29-0.54)	0.46 (0.36-0.59)
VI. Suicidal ideation	0.70 (0.61-0.81)	0.69 (0.60-0.79)
VII. Psychosis	0.76 (0.67-0.87)	0.75 (0.66-0.86)
VIII. Sleep problems	0.42 (0.32-0.55)	0.50 (0.40-0.63)
IX. Memory	0.77 (0.67-0.88)	0.74 (0.65-0.84)
X. Repetitive thoughts and behaviors	0.47 (0.37-0.59)	0.53 (0.43-0.65)
XI. Dissociation	0.53 (0.43-0.65)	0.57 (0.47-0.68)
XII. Personality functioning	0.24 (0.16-0.36)	0.24 (0.17-0.36)
XIII. Substance use	0.67 (0.53-0.85)	0.69 (0.55-0.88)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 51. Negative likelihood ratios of DSM-XC domains for post-traumatic stress disorder.

Domain	Intensity	Frequency
I. Depression	0.42 (0.32-0.55)	0.42 (0.34-0.52)
II. Anger	0.46 (0.36-0.57)	0.50 (0.42-0.60)
III. Mania	0.69 (0.60-0.80)	0.69 (0.60-0.79)
IV. Anxiety	0.18 (0.11-0.28)	0.23 (0.16-0.33)
V. Somatic symptoms	0.43 (0.33-0.55)	0.45 (0.37-0.56)
VI. Suicidal ideation	0.80 (0.72-0.87)	0.80 (0.73-0.87)
VII. Psychosis	0.70 (0.61-0.79)	0.67 (0.59-0.76)
VIII. Sleep problems	0.52 (0.43-0.63)	0.48 (0.40-0.58)
IX. Memory	0.66 (0.57-0.75)	0.68 (0.60-0.77)
X. Repetitive thoughts and behaviors	0.41 (0.33-0.51)	0.51 (0.43-0.60)
XI. Dissociation	0.61 (0.52-0.70)	0.63 (0.55-0.72)
XII. Personality functioning	0.41 (0.33-0.52)	0.39 (0.31-0.49)
XIII. Substance use	0.63 (0.51-0.77)	0.66 (0.54-0.81)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 52. Negative likelihood ratios of DSM-XC domains for social anxiety disorder.

Domain	Intensity	Frequency
I. Depression	0.43 (0.34-0.55)	0.57 (0.49-0.66)
II. Anger	0.59 (0.50-0.70)	0.68 (0.60-0.77)
III. Mania	0.75 (0.67-0.85)	0.83 (0.76-0.92)
IV. Anxiety	0.31 (0.22-0.43)	0.43 (0.34-0.54)
V. Somatic symptoms	0.64 (0.54-0.76)	0.67 (0.59-0.77)
VI. Suicidal ideation	0.82 (0.75-0.89)	0.83 (0.77-0.89)
VII. Psychosis	0.80 (0.73-0.88)	0.79 (0.72-0.87)
VIII. Sleep problems	0.62 (0.53-0.72)	0.65 (0.57-0.74)
IX. Memory	0.71 (0.63-0.80)	0.78 (0.71-0.86)
X. Repetitive thoughts and behaviors	0.68 (0.60-0.77)	0.72 (0.65-0.81)
XI. Dissociation	0.74 (0.67-0.83)	0.76 (0.69-0.84)
XII. Personality functioning	0.56 (0.47-0.66)	0.61 (0.53-0.71)
XIII. Substance use	0.75 (0.64-0.89)	0.78 (0.66-0.92)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 53. Sensitivities and specificities of DSM-XC domains for antisocial personality disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.62 (0.48-0.75)	0.34 (0.22-0.48)	0.64 (0.50-0.77)
	Specificity	0.60 (0.58-0.61)	0.82 (0.80-0.83)	0.58 (0.56-0.59)
II. Anger	Sensitivity	0.42 (0.28-0.56)	0.30 (0.18-0.44)	0.42 (0.28-0.56)
	Specificity	0.71 (0.69-0.72)	0.84 (0.83-0.86)	0.68 (0.67-0.70)
III. Mania	Sensitivity	0.36 (0.23-0.50)	0.21 (0.11-0.34)	0.40 (0.26-0.54)
	Specificity	0.78 (0.77-0.80)	0.84 (0.83-0.85)	0.72 (0.71-0.74)
IV. Anxiety	Sensitivity	0.66 (0.52-0.78)	0.47 (0.33-0.61)	0.68 (0.54-0.80)
	Specificity	0.53 (0.52-0.55)	0.70 (0.69-0.72)	0.50 (0.48-0.52)
V. Somatic symptoms	Sensitivity	0.38 (0.25-0.52)	0.40 (0.26-0.54)	0.45 (0.32-0.60)
	Specificity	0.66 (0.65-0.68)	0.79 (0.78-0.81)	0.64 (0.62-0.66)
VI. Suicidal ideation	Sensitivity	0.11 (0.04-0.23)	0.08 (0.02-0.18)	0.11 (0.04-0.23)
	Specificity	0.92 (0.91-0.93)	0.95 (0.94-0.96)	0.91 (0.90-0.92)
VII. Psychosis	Sensitivity	0.40 (0.26-0.54)	0.38 (0.25-0.52)	0.42 (0.28-0.56)
	Specificity	0.87 (0.86-0.88)	0.91 (0.90-0.92)	0.86 (0.85-0.88)
VIII. Sleep problems	Sensitivity	0.34 (0.22-0.48)	0.40 (0.26-0.54)	0.42 (0.28-0.56)
	Specificity	0.78 (0.76-0.79)	0.84 (0.83-0.86)	0.75 (0.74-0.77)
IX. Memory	Sensitivity	0.23 (0.12-0.36)	0.21 (0.11-0.34)	0.28 (0.17-0.42)
	Specificity	0.85 (0.84-0.86)	0.92 (0.91-0.93)	0.84 (0.82-0.85)
X. Repetitive thoughts and behaviors	Sensitivity	0.32 (0.20-0.46)	0.30 (0.18-0.44)	0.40 (0.26-0.54)
	Specificity	0.83 (0.81-0.84)	0.89 (0.88-0.90)	0.81 (0.79-0.82)
XI. Dissociation	Sensitivity	0.28 (0.17-0.42)	0.32 (0.20-0.46)	0.36 (0.23-0.50)
	Specificity	0.86 (0.85-0.87)	0.91 (0.90-0.92)	0.84 (0.83-0.85)
XII. Personality functioning	Sensitivity	0.57 (0.42-0.70)	0.53 (0.39-0.67)	0.64 (0.50-0.77)
	Specificity	0.74 (0.72-0.75)	0.81 (0.79-0.82)	0.71 (0.69-0.72)
XIII. Substance use	Sensitivity	0.72 (0.58-0.83)	0.79 (0.66-0.89)	0.79 (0.66-0.89)
	Specificity	0.59 (0.58-0.61)	0.58 (0.56-0.59)	0.53 (0.51-0.55)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 54. Sensitivities and specificities of DSM-XC domains for attention-deficit/hyperactivity disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.57 (0.51-0.63)	0.38 (0.32-0.44)	0.59 (0.53-0.65)
	Specificity	0.61 (0.59-0.62)	0.83 (0.82-0.84)	0.59 (0.57-0.60)
II. Anger	Sensitivity	0.47 (0.41-0.54)	0.32 (0.26-0.38)	0.50 (0.44-0.57)
	Specificity	0.72 (0.70-0.73)	0.86 (0.84-0.87)	0.70 (0.68-0.71)
III. Mania	Sensitivity	0.34 (0.28-0.40)	0.29 (0.24-0.35)	0.43 (0.37-0.49)
	Specificity	0.79 (0.77-0.80)	0.85 (0.84-0.87)	0.73 (0.71-0.75)
IV. Anxiety	Sensitivity	0.63 (0.56-0.68)	0.55 (0.49-0.61)	0.67 (0.61-0.73)
	Specificity	0.54 (0.53-0.56)	0.72 (0.70-0.73)	0.51 (0.49-0.53)
V. Somatic symptoms	Sensitivity	0.50 (0.43-0.56)	0.34 (0.29-0.40)	0.51 (0.45-0.57)
	Specificity	0.68 (0.66-0.69)	0.80 (0.79-0.81)	0.65 (0.63-0.67)
VI. Suicidal ideation	Sensitivity	0.12 (0.08-0.16)	0.10 (0.06-0.14)	0.14 (0.10-0.19)
	Specificity	0.92 (0.91-0.93)	0.95 (0.94-0.96)	0.92 (0.91-0.92)
VII. Psychosis	Sensitivity	0.23 (0.18-0.29)	0.20 (0.16-0.26)	0.24 (0.19-0.30)
	Specificity	0.88 (0.86-0.89)	0.91 (0.90-0.92)	0.87 (0.86-0.88)
VIII. Sleep problems	Sensitivity	0.36 (0.30-0.42)	0.34 (0.29-0.40)	0.41 (0.35-0.47)
	Specificity	0.79 (0.77-0.80)	0.86 (0.84-0.87)	0.77 (0.75-0.78)
IX. Memory	Sensitivity	0.27 (0.22-0.33)	0.20 (0.15-0.25)	0.31 (0.25-0.36)
	Specificity	0.86 (0.85-0.87)	0.93 (0.92-0.94)	0.84 (0.83-0.86)
X. Repetitive thoughts and behaviors	Sensitivity	0.33 (0.27-0.39)	0.26 (0.21-0.32)	0.35 (0.30-0.42)
	Specificity	0.84 (0.82-0.85)	0.90 (0.89-0.91)	0.82 (0.80-0.83)
XI. Dissociation	Sensitivity	0.28 (0.23-0.34)	0.19 (0.14-0.24)	0.31 (0.25-0.36)
	Specificity	0.87 (0.86-0.88)	0.92 (0.91-0.93)	0.85 (0.84-0.86)
XII. Personality functioning	Sensitivity	0.40 (0.34-0.47)	0.34 (0.29-0.40)	0.45 (0.39-0.52)
	Specificity	0.74 (0.73-0.76)	0.81 (0.80-0.83)	0.71 (0.70-0.73)
XIII. Substance use	Sensitivity	0.50 (0.44-0.57)	0.53 (0.47-0.59)	0.57 (0.51-0.63)
	Specificity	0.60 (0.58-0.61)	0.58 (0.56-0.60)	0.53 (0.52-0.55)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 55. Sensitivities and specificities of DSM-XC domains for bipolar disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.63 (0.53-0.72)	0.54 (0.45-0.64)	0.72 (0.63-0.80)
	Specificity	0.60 (0.59-0.62)	0.83 (0.81-0.84)	0.58 (0.57-0.60)
II. Anger	Sensitivity	0.55 (0.46-0.64)	0.54 (0.45-0.64)	0.63 (0.53-0.72)
	Specificity	0.71 (0.70-0.73)	0.86 (0.84-0.87)	0.69 (0.68-0.71)
III. Mania	Sensitivity	0.45 (0.36-0.54)	0.44 (0.35-0.53)	0.59 (0.49-0.68)
	Specificity	0.79 (0.77-0.80)	0.85 (0.84-0.86)	0.73 (0.71-0.74)
IV. Anxiety	Sensitivity	0.81 (0.73-0.88)	0.72 (0.62-0.80)	0.85 (0.78-0.91)
	Specificity	0.54 (0.53-0.56)	0.71 (0.70-0.73)	0.51 (0.49-0.53)
V. Somatic symptoms	Sensitivity	0.62 (0.53-0.71)	0.56 (0.47-0.65)	0.66 (0.57-0.75)
	Specificity	0.67 (0.66-0.69)	0.80 (0.79-0.82)	0.65 (0.63-0.67)
VI. Suicidal ideation	Sensitivity	0.28 (0.20-0.38)	0.27 (0.19-0.36)	0.30 (0.22-0.39)
	Specificity	0.92 (0.91-0.93)	0.96 (0.95-0.96)	0.92 (0.91-0.93)
VII. Psychosis	Sensitivity	0.48 (0.39-0.58)	0.45 (0.36-0.54)	0.48 (0.39-0.58)
	Specificity	0.88 (0.87-0.89)	0.92 (0.91-0.92)	0.87 (0.86-0.88)
VIII. Sleep problems	Sensitivity	0.43 (0.34-0.53)	0.43 (0.34-0.53)	0.52 (0.42-0.61)
	Specificity	0.78 (0.77-0.80)	0.85 (0.84-0.86)	0.76 (0.75-0.78)
IX. Memory	Sensitivity	0.40 (0.31-0.49)	0.30 (0.22-0.39)	0.46 (0.36-0.55)
	Specificity	0.86 (0.85-0.87)	0.93 (0.92-0.93)	0.84 (0.83-0.86)
X. Repetitive thoughts and behaviors	Sensitivity	0.50 (0.41-0.59)	0.50 (0.41-0.59)	0.57 (0.47-0.66)
	Specificity	0.84 (0.82-0.85)	0.90 (0.89-0.91)	0.82 (0.80-0.83)
XI. Dissociation	Sensitivity	0.40 (0.31-0.49)	0.39 (0.30-0.48)	0.47 (0.38-0.57)
	Specificity	0.87 (0.85-0.88)	0.92 (0.91-0.93)	0.85 (0.84-0.86)
XII. Personality functioning	Sensitivity	0.54 (0.45-0.64)	0.57 (0.47-0.66)	0.67 (0.58-0.76)
	Specificity	0.74 (0.73-0.76)	0.81 (0.80-0.83)	0.71 (0.70-0.73)
XIII. Substance use	Sensitivity	0.63 (0.53-0.72)	0.61 (0.52-0.70)	0.67 (0.58-0.76)
	Specificity	0.60 (0.58-0.61)	0.58 (0.56-0.59)	0.53 (0.52-0.55)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 56. Sensitivities and specificities of DSM-XC domains for generalized anxiety disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.71 (0.67-0.74)	0.52 (0.48-0.57)	0.75 (0.71-0.78)
	Specificity	0.65 (0.64-0.67)	0.88 (0.87-0.89)	0.64 (0.62-0.65)
II. Anger	Sensitivity	0.58 (0.54-0.62)	0.46 (0.42-0.50)	0.63 (0.59-0.66)
	Specificity	0.76 (0.75-0.78)	0.90 (0.89-0.91)	0.74 (0.72-0.76)
III. Mania	Sensitivity	0.34 (0.30-0.38)	0.27 (0.23-0.30)	0.41 (0.37-0.45)
	Specificity	0.80 (0.79-0.82)	0.86 (0.85-0.88)	0.74 (0.73-0.76)
IV. Anxiety	Sensitivity	0.79 (0.76-0.82)	0.71 (0.67-0.74)	0.83 (0.80-0.86)
	Specificity	0.59 (0.58-0.61)	0.78 (0.76-0.79)	0.56 (0.54-0.58)
V. Somatic symptoms	Sensitivity	0.59 (0.55-0.63)	0.47 (0.43-0.51)	0.61 (0.57-0.65)
	Specificity	0.71 (0.70-0.73)	0.84 (0.83-0.85)	0.69 (0.67-0.71)
VI. Suicidal ideation	Sensitivity	0.16 (0.13-0.20)	0.15 (0.12-0.18)	0.17 (0.14-0.21)
	Specificity	0.93 (0.92-0.94)	0.97 (0.96-0.97)	0.93 (0.92-0.94)
VII. Psychosis	Sensitivity	0.23 (0.20-0.27)	0.22 (0.19-0.25)	0.24 (0.21-0.28)
	Specificity	0.89 (0.87-0.90)	0.93 (0.92-0.94)	0.88 (0.87-0.89)
VIII. Sleep problems	Sensitivity	0.46 (0.42-0.50)	0.40 (0.36-0.44)	0.50 (0.46-0.54)
	Specificity	0.82 (0.81-0.83)	0.89 (0.88-0.90)	0.80 (0.79-0.82)
IX. Memory	Sensitivity	0.31 (0.28-0.35)	0.22 (0.19-0.25)	0.34 (0.30-0.38)
	Specificity	0.88 (0.87-0.89)	0.94 (0.94-0.95)	0.87 (0.86-0.88)
X. Repetitive thoughts and behaviors	Sensitivity	0.42 (0.38-0.46)	0.31 (0.27-0.35)	0.44 (0.40-0.48)
	Specificity	0.87 (0.86-0.88)	0.93 (0.92-0.93)	0.85 (0.84-0.86)
XI. Dissociation	Sensitivity	0.32 (0.29-0.36)	0.27 (0.23-0.30)	0.36 (0.32-0.40)
	Specificity	0.89 (0.88-0.90)	0.95 (0.94-0.95)	0.88 (0.87-0.89)
XII. Personality functioning	Sensitivity	0.52 (0.48-0.57)	0.48 (0.44-0.52)	0.58 (0.54-0.62)
	Specificity	0.78 (0.77-0.80)	0.86 (0.84-0.87)	0.76 (0.74-0.77)
XIII. Substance use	Sensitivity	0.53 (0.49-0.57)	0.54 (0.50-0.58)	0.57 (0.53-0.61)
	Specificity	0.61 (0.59-0.63)	0.59 (0.57-0.61)	0.55 (0.53-0.56)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 57. Sensitivities and specificities of DSM-XC domains for major depressive disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.81 (0.74-0.86)	0.83 (0.77-0.88)	0.90 (0.85-0.94)
	Specificity	0.62 (0.60-0.63)	0.85 (0.84-0.86)	0.60 (0.58-0.62)
II. Anger	Sensitivity	0.64 (0.57-0.71)	0.63 (0.55-0.70)	0.73 (0.65-0.79)
	Specificity	0.72 (0.71-0.74)	0.87 (0.86-0.88)	0.70 (0.69-0.72)
III. Mania	Sensitivity	0.39 (0.31-0.46)	0.38 (0.31-0.46)	0.49 (0.42-0.57)
	Specificity	0.79 (0.77-0.80)	0.85 (0.84-0.87)	0.73 (0.71-0.74)
IV. Anxiety	Sensitivity	0.85 (0.79-0.90)	0.82 (0.76-0.87)	0.92 (0.87-0.95)
	Specificity	0.55 (0.53-0.57)	0.73 (0.71-0.74)	0.52 (0.50-0.54)
V. Somatic symptoms	Sensitivity	0.70 (0.63-0.77)	0.63 (0.56-0.70)	0.74 (0.67-0.81)
	Specificity	0.68 (0.67-0.70)	0.81 (0.80-0.83)	0.66 (0.64-0.68)
VI. Suicidal ideation	Sensitivity	0.35 (0.28-0.42)	0.34 (0.27-0.41)	0.37 (0.30-0.44)
	Specificity	0.93 (0.92-0.94)	0.96 (0.96-0.97)	0.93 (0.92-0.93)
VII. Psychosis	Sensitivity	0.37 (0.30-0.45)	0.35 (0.28-0.42)	0.38 (0.31-0.46)
	Specificity	0.88 (0.87-0.89)	0.92 (0.91-0.93)	0.87 (0.86-0.88)
VIII. Sleep problems	Sensitivity	0.61 (0.54-0.69)	0.56 (0.49-0.64)	0.66 (0.58-0.73)
	Specificity	0.80 (0.78-0.81)	0.86 (0.85-0.87)	0.77 (0.76-0.79)
IX. Memory	Sensitivity	0.39 (0.32-0.47)	0.34 (0.27-0.42)	0.44 (0.36-0.51)
	Specificity	0.86 (0.85-0.87)	0.93 (0.92-0.94)	0.85 (0.84-0.86)
X. Repetitive thoughts and behaviors	Sensitivity	0.56 (0.48-0.63)	0.50 (0.42-0.57)	0.59 (0.52-0.66)
	Specificity	0.84 (0.83-0.86)	0.91 (0.90-0.92)	0.82 (0.81-0.84)
XI. Dissociation	Sensitivity	0.51 (0.44-0.59)	0.46 (0.39-0.54)	0.56 (0.48-0.63)
	Specificity	0.88 (0.87-0.89)	0.93 (0.92-0.94)	0.86 (0.85-0.87)
XII. Personality functioning	Sensitivity	0.75 (0.68-0.81)	0.70 (0.63-0.77)	0.79 (0.72-0.85)
	Specificity	0.76 (0.74-0.77)	0.83 (0.82-0.84)	0.73 (0.71-0.74)
XIII. Substance use	Sensitivity	0.60 (0.52-0.67)	0.58 (0.51-0.65)	0.63 (0.55-0.70)
	Specificity	0.60 (0.58-0.61)	0.58 (0.56-0.60)	0.53 (0.52-0.55)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 58. Sensitivities and specificities of DSM-XC domains for post-traumatic stress disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.66 (0.60-0.72)	0.54 (0.47-0.60)	0.72 (0.66-0.77)
	Specificity	0.62 (0.60-0.63)	0.84 (0.83-0.86)	0.60 (0.58-0.61)
II. Anger	Sensitivity	0.56 (0.50-0.62)	0.46 (0.40-0.53)	0.62 (0.56-0.67)
	Specificity	0.73 (0.71-0.74)	0.87 (0.86-0.88)	0.70 (0.69-0.72)
III. Mania	Sensitivity	0.40 (0.34-0.46)	0.34 (0.29-0.40)	0.50 (0.44-0.56)
	Specificity	0.79 (0.78-0.81)	0.86 (0.85-0.87)	0.74 (0.72-0.75)
IV. Anxiety	Sensitivity	0.79 (0.74-0.84)	0.70 (0.65-0.76)	0.85 (0.80-0.89)
	Specificity	0.56 (0.54-0.57)	0.73 (0.72-0.75)	0.53 (0.51-0.54)
V. Somatic symptoms	Sensitivity	0.63 (0.57-0.69)	0.56 (0.50-0.62)	0.66 (0.60-0.72)
	Specificity	0.69 (0.67-0.70)	0.82 (0.81-0.83)	0.66 (0.65-0.68)
VI. Suicidal ideation	Sensitivity	0.20 (0.15-0.25)	0.18 (0.14-0.24)	0.21 (0.16-0.26)
	Specificity	0.93 (0.92-0.93)	0.96 (0.95-0.97)	0.92 (0.91-0.93)
VII. Psychosis	Sensitivity	0.36 (0.30-0.42)	0.35 (0.30-0.41)	0.36 (0.30-0.42)
	Specificity	0.89 (0.87-0.90)	0.93 (0.92-0.93)	0.88 (0.87-0.89)
VIII. Sleep problems	Sensitivity	0.48 (0.42-0.54)	0.46 (0.40-0.52)	0.54 (0.47-0.60)
	Specificity	0.80 (0.78-0.81)	0.87 (0.85-0.88)	0.78 (0.76-0.79)
IX. Memory	Sensitivity	0.32 (0.27-0.38)	0.26 (0.21-0.32)	0.35 (0.30-0.41)
	Specificity	0.87 (0.85-0.88)	0.93 (0.92-0.94)	0.85 (0.84-0.86)
X. Repetitive thoughts and behaviors	Sensitivity	0.54 (0.47-0.60)	0.45 (0.39-0.52)	0.58 (0.52-0.64)
	Specificity	0.85 (0.84-0.87)	0.92 (0.91-0.92)	0.84 (0.82-0.85)
XI. Dissociation	Sensitivity	0.38 (0.32-0.44)	0.32 (0.26-0.38)	0.41 (0.35-0.47)
	Specificity	0.88 (0.87-0.89)	0.93 (0.92-0.94)	0.86 (0.85-0.87)
XII. Personality functioning	Sensitivity	0.58 (0.51-0.64)	0.55 (0.49-0.61)	0.65 (0.59-0.70)
	Specificity	0.76 (0.74-0.77)	0.83 (0.82-0.84)	0.73 (0.71-0.74)
XIII. Substance use	Sensitivity	0.60 (0.54-0.66)	0.62 (0.56-0.68)	0.65 (0.59-0.70)
	Specificity	0.60 (0.59-0.62)	0.59 (0.57-0.60)	0.54 (0.52-0.56)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 59. Sensitivities and specificities of DSM-XC domains for social anxiety disorder, after removing the clinical impairment criteria.

Domain	Metric	Intensity	Frequency	“OR”
I. Depression	Sensitivity	0.67 (0.61-0.72)	0.44 (0.38-0.51)	0.72 (0.66-0.77)
	Specificity	0.62 (0.60-0.63)	0.84 (0.82-0.85)	0.60 (0.58-0.61)
II. Anger	Sensitivity	0.50 (0.44-0.56)	0.35 (0.29-0.41)	0.54 (0.48-0.60)
	Specificity	0.72 (0.71-0.74)	0.86 (0.85-0.87)	0.70 (0.68-0.71)
III. Mania	Sensitivity	0.36 (0.30-0.42)	0.28 (0.22-0.33)	0.42 (0.36-0.48)
	Specificity	0.79 (0.78-0.80)	0.85 (0.84-0.86)	0.73 (0.71-0.74)
IV. Anxiety	Sensitivity	0.77 (0.72-0.82)	0.62 (0.56-0.68)	0.83 (0.78-0.87)
	Specificity	0.56 (0.54-0.57)	0.73 (0.71-0.74)	0.52 (0.51-0.54)
V. Somatic symptoms	Sensitivity	0.51 (0.45-0.57)	0.40 (0.34-0.46)	0.55 (0.49-0.61)
	Specificity	0.68 (0.66-0.69)	0.81 (0.79-0.82)	0.66 (0.64-0.67)
VI. Suicidal ideation	Sensitivity	0.22 (0.17-0.27)	0.18 (0.14-0.23)	0.23 (0.18-0.29)
	Specificity	0.93 (0.92-0.94)	0.96 (0.95-0.97)	0.92 (0.91-0.93)
VII. Psychosis	Sensitivity	0.28 (0.23-0.34)	0.24 (0.19-0.29)	0.28 (0.23-0.34)
	Specificity	0.88 (0.87-0.89)	0.92 (0.91-0.92)	0.87 (0.86-0.88)
VIII. Sleep problems	Sensitivity	0.43 (0.37-0.50)	0.38 (0.32-0.44)	0.49 (0.43-0.55)
	Specificity	0.79 (0.78-0.81)	0.86 (0.85-0.87)	0.77 (0.76-0.79)
IX. Memory	Sensitivity	0.33 (0.27-0.39)	0.23 (0.18-0.28)	0.35 (0.29-0.41)
	Specificity	0.87 (0.85-0.88)	0.93 (0.92-0.94)	0.85 (0.84-0.86)
X. Repetitive thoughts and behaviors	Sensitivity	0.39 (0.33-0.45)	0.30 (0.24-0.36)	0.42 (0.36-0.48)
	Specificity	0.84 (0.83-0.85)	0.90 (0.89-0.91)	0.82 (0.81-0.84)
XI. Dissociation	Sensitivity	0.31 (0.26-0.37)	0.24 (0.19-0.30)	0.35 (0.29-0.41)
	Specificity	0.87 (0.86-0.88)	0.92 (0.91-0.93)	0.86 (0.84-0.87)
XII. Personality functioning	Sensitivity	0.53 (0.47-0.59)	0.44 (0.38-0.51)	0.56 (0.50-0.62)
	Specificity	0.75 (0.74-0.77)	0.82 (0.81-0.84)	0.72 (0.71-0.74)
XIII. Substance use	Sensitivity	0.51 (0.45-0.57)	0.51 (0.45-0.58)	0.57 (0.50-0.63)
	Specificity	0.60 (0.58-0.61)	0.58 (0.56-0.59)	0.53 (0.52-0.55)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 60. Positive likelihood ratios of DSM-XC domains for antisocial personality disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	1.55 (1.25-1.91)	1.85 (1.26-2.71)	1.52 (1.24-1.86)
II. Anger	1.42 (1.02-1.96)	1.94 (1.28-2.95)	1.31 (0.94-1.80)
III. Mania	1.64 (1.14-2.37)	1.32 (0.78-2.25)	1.42 (1.01-1.98)
IV. Anxiety	1.42 (1.17-1.73)	1.58 (1.18-2.11)	1.36 (1.13-1.64)
V. Somatic symptoms	1.12 (0.79-1.59)	1.91 (1.36-2.68)	1.26 (0.93-1.70)
VI. Suicidal ideation	1.36 (0.64-2.92)	1.46 (0.56-3.79)	1.28 (0.60-2.74)
VII. Psychosis	3.08 (2.19-4.35)	4.11 (2.86-5.89)	3.07 (2.20-4.27)
VIII. Sleep problems	1.52 (1.04-2.23)	2.54 (1.81-3.58)	1.69 (1.22-2.34)
IX. Memory	1.53 (0.92-2.53)	2.59 (1.52-4.44)	1.72 (1.12-2.66)
X. Repetitive thoughts and behaviors	1.85 (1.24-2.75)	2.75 (1.81-4.19)	2.05 (1.46-2.88)
XI. Dissociation	2.02 (1.30-3.12)	3.73 (2.49-5.60)	2.29 (1.58-3.30)
XII. Personality functioning	2.15 (1.69-2.74)	2.73 (2.10-3.56)	2.18 (1.78-2.69)
XIII. Substance use	1.76 (1.48-2.09)	1.87 (1.62-2.16)	1.69 (1.47-1.95)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 61. Positive likelihood ratios of DSM-XC domains for attention-deficit/hyperactivity disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	1.45 (1.29-1.62)	2.21 (1.86-2.63)	1.43 (1.29-1.60)
II. Anger	1.68 (1.47-1.93)	2.21 (1.82-2.69)	1.65 (1.45-1.88)
III. Mania	1.59 (1.32-1.91)	2.00 (1.63-2.46)	1.60 (1.38-1.86)
IV. Anxiety	1.37 (1.24-1.52)	1.96 (1.73-2.21)	1.37 (1.25-1.50)
V. Somatic symptoms	1.53 (1.34-1.74)	1.72 (1.44-2.06)	1.47 (1.29-1.67)
VI. Suicidal ideation	1.46 (1.03-2.08)	1.97 (1.31-2.94)	1.67 (1.21-2.29)
VII. Psychosis	1.87 (1.47-2.37)	2.31 (1.77-3.00)	1.86 (1.48-2.34)
VIII. Sleep problems	1.69 (1.42-2.01)	2.37 (1.97-2.86)	1.76 (1.50-2.06)
IX. Memory	1.97 (1.59-2.44)	2.73 (2.08-3.58)	1.97 (1.61-2.40)
X. Repetitive thoughts and behaviors	2.00 (1.66-2.42)	2.57 (2.04-3.23)	1.94 (1.62-2.31)
XI. Dissociation	2.15 (1.74-2.66)	2.29 (1.73-3.02)	2.06 (1.68-2.51)
XII. Personality functioning	1.57 (1.34-1.84)	1.84 (1.53-2.21)	1.59 (1.37-1.83)
XIII. Substance use	1.24 (1.10-1.41)	1.26 (1.12-1.42)	1.23 (1.10-1.37)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 62. Positive likelihood ratios of DSM-XC domains for bipolar disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	1.58 (1.37-1.83)	3.12 (2.60-3.75)	1.74 (1.55-1.96)
II. Anger	1.93 (1.62-2.29)	3.75 (3.12-4.52)	2.04 (1.76-2.36)
III. Mania	2.11 (1.70-2.60)	2.97 (2.38-3.70)	2.16 (1.84-2.54)
IV. Anxiety	1.77 (1.61-1.95)	2.49 (2.20-2.83)	1.74 (1.60-1.89)
V. Somatic symptoms	1.89 (1.63-2.20)	2.83 (2.37-3.37)	1.89 (1.65-2.17)
VI. Suicidal ideation	3.70 (2.71-5.05)	5.97 (4.26-8.37)	3.69 (2.74-4.97)
VII. Psychosis	4.00 (3.25-4.93)	5.31 (4.22-6.69)	3.77 (3.07-4.64)
VIII. Sleep problems	1.98 (1.59-2.46)	2.87 (2.29-3.59)	2.17 (1.80-2.61)
IX. Memory	2.81 (2.22-3.57)	4.05 (3.00-5.47)	2.92 (2.36-3.62)
X. Repetitive thoughts and behaviors	3.03 (2.49-3.69)	5.02 (4.08-6.18)	3.10 (2.61-3.69)
XI. Dissociation	2.96 (2.33-3.76)	4.88 (3.78-6.30)	3.17 (2.58-3.91)
XII. Personality functioning	2.10 (1.76-2.51)	3.06 (2.58-3.64)	2.35 (2.05-2.70)
XIII. Substance use	1.56 (1.34-1.80)	1.45 (1.24-1.68)	1.44 (1.26-1.64)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 63. Positive likelihood ratios of DSM-XC domains for generalized anxiety disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	2.03 (1.89-2.18)	4.39 (3.88-4.97)	2.05 (1.92-2.19)
II. Anger	2.44 (2.22-2.68)	4.73 (4.12-5.44)	2.41 (2.21-2.63)
III. Mania	1.70 (1.49-1.95)	1.96 (1.66-2.30)	1.57 (1.40-1.77)
IV. Anxiety	1.95 (1.84-2.07)	3.19 (2.93-3.47)	1.90 (1.80-2.01)
V. Somatic symptoms	2.04 (1.87-2.23)	2.97 (2.64-3.35)	1.97 (1.81-2.14)
VI. Suicidal ideation	2.42 (1.93-3.03)	4.40 (3.34-5.79)	2.38 (1.91-2.96)
VII. Psychosis	2.04 (1.70-2.44)	3.03 (2.48-3.70)	2.02 (1.69-2.40)
VIII. Sleep problems	2.58 (2.30-2.90)	3.56 (3.09-4.10)	2.52 (2.26-2.80)
IX. Memory	2.67 (2.28-3.12)	3.97 (3.21-4.91)	2.56 (2.21-2.96)
X. Repetitive thoughts and behaviors	3.22 (2.82-3.68)	4.17 (3.50-4.97)	2.96 (2.61-3.36)
XI. Dissociation	3.01 (2.58-3.52)	4.87 (3.98-5.95)	2.95 (2.55-3.41)
XII. Personality functioning	2.41 (2.18-2.67)	3.36 (2.97-3.79)	2.38 (2.16-2.61)
XIII. Substance use	1.35 (1.24-1.48)	1.32 (1.21-1.44)	1.26 (1.16-1.36)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 64. Positive likelihood ratios of DSM-XC domains for major depressive disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	2.11 (1.94-2.29)	5.48 (4.95-6.08)	2.24 (2.10-2.39)
II. Anger	2.32 (2.05-2.62)	4.71 (4.08-5.42)	2.44 (2.20-2.70)
III. Mania	1.82 (1.50-2.21)	2.60 (2.12-3.19)	1.82 (1.55-2.13)
IV. Anxiety	1.89 (1.76-2.03)	3.00 (2.75-3.28)	1.91 (1.80-2.02)
V. Somatic symptoms	2.22 (1.99-2.47)	3.36 (2.95-3.84)	2.18 (1.98-2.41)
VI. Suicidal ideation	4.97 (3.92-6.29)	9.04 (6.92-11.82)	4.97 (3.97-6.23)
VII. Psychosis	3.13 (2.53-3.86)	4.17 (3.32-5.26)	3.00 (2.44-3.68)
VIII. Sleep problems	3.01 (2.63-3.44)	4.09 (3.51-4.77)	2.92 (2.58-3.30)
IX. Memory	2.86 (2.34-3.50)	4.99 (3.93-6.34)	2.87 (2.39-3.45)
X. Repetitive thoughts and behaviors	3.59 (3.08-4.18)	5.38 (4.49-6.45)	3.38 (2.94-3.90)
XI. Dissociation	4.18 (3.53-4.95)	6.65 (5.45-8.12)	4.02 (3.45-4.70)
XII. Personality functioning	3.09 (2.78-3.43)	4.10 (3.64-4.63)	2.89 (2.63-3.17)
XIII. Substance use	1.49 (1.31-1.69)	1.38 (1.21-1.57)	1.34 (1.19-1.51)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 65. Positive likelihood ratios of DSM-XC domains for post-traumatic stress disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	1.72 (1.56-1.89)	3.40 (2.97-3.90)	1.79 (1.64-1.95)
II. Anger	2.05 (1.82-2.31)	3.51 (3.01-4.10)	2.09 (1.87-2.33)
III. Mania	1.94 (1.65-2.27)	2.41 (2.00-2.90)	1.89 (1.65-2.16)
IV. Anxiety	1.79 (1.67-1.93)	2.63 (2.39-2.90)	1.78 (1.67-1.90)
V. Somatic symptoms	2.00 (1.80-2.22)	3.12 (2.75-3.55)	1.97 (1.78-2.17)
VI. Suicidal ideation	2.63 (2.01-3.44)	4.49 (3.33-6.05)	2.61 (2.01-3.38)
VII. Psychosis	3.14 (2.61-3.78)	4.72 (3.87-5.77)	2.98 (2.48-3.58)
VIII. Sleep problems	2.33 (2.02-2.68)	3.43 (2.94-4.00)	2.39 (2.10-2.71)
IX. Memory	2.41 (1.99-2.92)	3.90 (3.08-4.95)	2.35 (1.96-2.82)
X. Repetitive thoughts and behaviors	3.66 (3.18-4.20)	5.36 (4.51-6.37)	3.55 (3.13-4.03)
XI. Dissociation	3.09 (2.59-3.69)	4.48 (3.62-5.55)	2.94 (2.49-3.47)
XII. Personality functioning	2.37 (2.11-2.67)	3.27 (2.87-3.73)	2.39 (2.15-2.65)
XIII. Substance use	1.51 (1.35-1.68)	1.51 (1.36-1.67)	1.40 (1.28-1.54)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 66. Positive likelihood ratios of DSM-XC domains for social anxiety disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	1.74 (1.59-1.91)	2.70 (2.32-3.15)	1.78 (1.64-1.94)
II. Anger	1.81 (1.59-2.06)	2.46 (2.05-2.96)	1.81 (1.60-2.04)
III. Mania	1.70 (1.43-2.03)	1.86 (1.51-2.30)	1.55 (1.33-1.80)
IV. Anxiety	1.74 (1.61-1.88)	2.26 (2.03-2.52)	1.74 (1.63-1.86)
V. Somatic symptoms	1.58 (1.40-1.80)	2.06 (1.76-2.42)	1.60 (1.42-1.80)
VI. Suicidal ideation	3.05 (2.37-3.94)	4.35 (3.21-5.88)	3.00 (2.35-3.84)
VII. Psychosis	2.32 (1.88-2.87)	2.78 (2.18-3.54)	2.22 (1.80-2.73)
VIII. Sleep problems	2.09 (1.79-2.43)	2.71 (2.28-3.22)	2.17 (1.89-2.48)
IX. Memory	2.43 (2.01-2.94)	3.26 (2.54-4.20)	2.31 (1.93-2.77)
X. Repetitive thoughts and behaviors	2.46 (2.08-2.91)	3.06 (2.48-3.77)	2.36 (2.02-2.76)
XI. Dissociation	2.43 (2.00-2.96)	3.16 (2.48-4.02)	2.39 (1.99-2.87)
XII. Personality functioning	2.17 (1.91-2.46)	2.50 (2.15-2.91)	2.01 (1.79-2.27)
XIII. Substance use	1.27 (1.12-1.43)	1.22 (1.08-1.38)	1.21 (1.09-1.36)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 67. Negative likelihood ratios of DSM-XC domains for antisocial personality disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.63 (0.45-0.89)	0.81 (0.67-0.98)	0.62 (0.43-0.89)
II. Anger	0.83 (0.66-1.04)	0.83 (0.69-0.99)	0.86 (0.68-1.08)
III. Mania	0.82 (0.67-1.00)	0.94 (0.82-1.08)	0.84 (0.67-1.04)
IV. Anxiety	0.64 (0.44-0.93)	0.75 (0.58-0.97)	0.64 (0.43-0.95)
V. Somatic symptoms	0.94 (0.76-1.16)	0.76 (0.61-0.95)	0.85 (0.67-1.09)
VI. Suicidal ideation	0.97 (0.88-1.07)	0.97 (0.90-1.05)	0.97 (0.88-1.07)
VII. Psychosis	0.69 (0.56-0.86)	0.69 (0.56-0.85)	0.68 (0.54-0.85)
VIII. Sleep problems	0.85 (0.70-1.03)	0.72 (0.57-0.89)	0.77 (0.62-0.97)
IX. Memory	0.91 (0.78-1.05)	0.86 (0.75-0.99)	0.86 (0.72-1.02)
X. Repetitive thoughts and behaviors	0.82 (0.68-0.99)	0.78 (0.66-0.94)	0.75 (0.60-0.93)
XI. Dissociation	0.83 (0.70-0.99)	0.74 (0.62-0.89)	0.76 (0.62-0.93)
XII. Personality functioning	0.59 (0.43-0.80)	0.58 (0.44-0.78)	0.51 (0.35-0.73)
XIII. Substance use	0.48 (0.31-0.73)	0.36 (0.21-0.61)	0.39 (0.23-0.66)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 68. Negative likelihood ratios of DSM-XC domains for attention-deficit/hyperactivity disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.71 (0.62-0.82)	0.75 (0.68-0.83)	0.70 (0.60-0.81)
II. Anger	0.73 (0.65-0.82)	0.79 (0.73-0.86)	0.71 (0.63-0.81)
III. Mania	0.84 (0.77-0.92)	0.83 (0.76-0.90)	0.78 (0.70-0.87)
IV. Anxiety	0.69 (0.59-0.81)	0.63 (0.55-0.72)	0.64 (0.54-0.77)
V. Somatic symptoms	0.75 (0.66-0.84)	0.82 (0.75-0.90)	0.75 (0.66-0.85)
VI. Suicidal ideation	0.96 (0.92-1.00)	0.95 (0.91-0.99)	0.94 (0.89-0.99)
VII. Psychosis	0.88 (0.82-0.94)	0.87 (0.82-0.93)	0.87 (0.81-0.93)
VIII. Sleep problems	0.81 (0.74-0.89)	0.77 (0.70-0.84)	0.77 (0.69-0.85)
IX. Memory	0.84 (0.78-0.91)	0.86 (0.81-0.92)	0.82 (0.76-0.89)
X. Repetitive thoughts and behaviors	0.80 (0.74-0.88)	0.82 (0.77-0.89)	0.79 (0.72-0.87)
XI. Dissociation	0.83 (0.76-0.89)	0.89 (0.83-0.94)	0.82 (0.75-0.88)
XII. Personality functioning	0.80 (0.72-0.89)	0.81 (0.74-0.88)	0.76 (0.68-0.86)
XIII. Substance use	0.83 (0.74-0.94)	0.81 (0.71-0.93)	0.80 (0.69-0.92)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 69. Negative likelihood ratios of DSM-XC domains for bipolar disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.62 (0.49-0.78)	0.55 (0.45-0.67)	0.47 (0.35-0.64)
II. Anger	0.63 (0.51-0.77)	0.53 (0.44-0.65)	0.54 (0.42-0.68)
III. Mania	0.70 (0.59-0.83)	0.66 (0.56-0.77)	0.57 (0.46-0.71)
IV. Anxiety	0.35 (0.24-0.51)	0.40 (0.30-0.53)	0.29 (0.19-0.45)
V. Somatic symptoms	0.56 (0.45-0.71)	0.55 (0.45-0.67)	0.52 (0.40-0.67)
VI. Suicidal ideation	0.78 (0.69-0.87)	0.77 (0.69-0.86)	0.76 (0.67-0.86)
VII. Psychosis	0.59 (0.49-0.70)	0.60 (0.51-0.71)	0.59 (0.50-0.71)
VIII. Sleep problems	0.73 (0.62-0.85)	0.67 (0.57-0.78)	0.63 (0.52-0.77)
IX. Memory	0.70 (0.61-0.81)	0.75 (0.67-0.85)	0.64 (0.54-0.76)
X. Repetitive thoughts and behaviors	0.60 (0.50-0.72)	0.56 (0.46-0.67)	0.53 (0.43-0.65)
XI. Dissociation	0.70 (0.60-0.81)	0.66 (0.58-0.77)	0.62 (0.52-0.74)
XII. Personality functioning	0.62 (0.50-0.75)	0.53 (0.43-0.65)	0.46 (0.35-0.60)
XIII. Substance use	0.62 (0.49-0.79)	0.67 (0.53-0.85)	0.62 (0.47-0.80)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 70. Negative likelihood ratios of DSM-XC domains for generalized anxiety disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.45 (0.40-0.51)	0.54 (0.50-0.59)	0.40 (0.35-0.46)
II. Anger	0.55 (0.50-0.60)	0.59 (0.55-0.64)	0.51 (0.45-0.56)
III. Mania	0.83 (0.78-0.88)	0.85 (0.81-0.89)	0.80 (0.75-0.86)
IV. Anxiety	0.35 (0.30-0.41)	0.38 (0.33-0.43)	0.30 (0.25-0.36)
V. Somatic symptoms	0.58 (0.52-0.64)	0.63 (0.58-0.68)	0.56 (0.50-0.62)
VI. Suicidal ideation	0.90 (0.86-0.93)	0.88 (0.85-0.91)	0.89 (0.86-0.93)
VII. Psychosis	0.87 (0.83-0.91)	0.84 (0.81-0.88)	0.86 (0.82-0.90)
VIII. Sleep problems	0.66 (0.61-0.71)	0.68 (0.63-0.72)	0.62 (0.58-0.68)
IX. Memory	0.78 (0.74-0.82)	0.83 (0.79-0.86)	0.76 (0.72-0.81)
X. Repetitive thoughts and behaviors	0.67 (0.63-0.72)	0.75 (0.71-0.79)	0.66 (0.61-0.71)
XI. Dissociation	0.76 (0.72-0.80)	0.78 (0.74-0.82)	0.73 (0.69-0.78)
XII. Personality functioning	0.61 (0.56-0.66)	0.61 (0.56-0.66)	0.56 (0.50-0.61)
XIII. Substance use	0.78 (0.71-0.85)	0.78 (0.71-0.85)	0.78 (0.71-0.87)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 71. Negative likelihood ratios of DSM-XC domains for major depressive disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.31 (0.23-0.42)	0.20 (0.14-0.27)	0.17 (0.11-0.26)
II. Anger	0.49 (0.41-0.60)	0.43 (0.36-0.52)	0.39 (0.31-0.50)
III. Mania	0.78 (0.69-0.88)	0.73 (0.65-0.81)	0.70 (0.60-0.81)
IV. Anxiety	0.27 (0.19-0.39)	0.25 (0.18-0.34)	0.16 (0.10-0.26)
V. Somatic symptoms	0.43 (0.35-0.54)	0.45 (0.37-0.55)	0.39 (0.30-0.50)
VI. Suicidal ideation	0.70 (0.63-0.78)	0.69 (0.62-0.77)	0.68 (0.61-0.76)
VII. Psychosis	0.71 (0.63-0.80)	0.71 (0.64-0.79)	0.71 (0.63-0.80)
VIII. Sleep problems	0.48 (0.40-0.58)	0.51 (0.43-0.60)	0.44 (0.36-0.54)
IX. Memory	0.71 (0.63-0.79)	0.71 (0.64-0.79)	0.67 (0.58-0.76)
X. Repetitive thoughts and behaviors	0.52 (0.44-0.62)	0.55 (0.48-0.64)	0.49 (0.41-0.59)
XI. Dissociation	0.55 (0.48-0.64)	0.58 (0.50-0.66)	0.51 (0.43-0.60)
XII. Personality functioning	0.33 (0.26-0.43)	0.36 (0.28-0.45)	0.29 (0.22-0.39)
XIII. Substance use	0.67 (0.56-0.81)	0.72 (0.61-0.86)	0.70 (0.58-0.85)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 72. Negative likelihood ratios of DSM-XC domains for post-traumatic stress disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.55 (0.47-0.65)	0.55 (0.49-0.63)	0.47 (0.39-0.57)
II. Anger	0.60 (0.53-0.69)	0.62 (0.55-0.69)	0.54 (0.47-0.63)
III. Mania	0.76 (0.69-0.84)	0.77 (0.70-0.84)	0.68 (0.60-0.77)
IV. Anxiety	0.37 (0.29-0.47)	0.40 (0.33-0.49)	0.29 (0.22-0.39)
V. Somatic symptoms	0.54 (0.46-0.63)	0.53 (0.46-0.61)	0.51 (0.43-0.60)
VI. Suicidal ideation	0.87 (0.82-0.92)	0.85 (0.80-0.90)	0.86 (0.81-0.92)
VII. Psychosis	0.72 (0.66-0.79)	0.70 (0.64-0.76)	0.73 (0.66-0.80)
VIII. Sleep problems	0.66 (0.59-0.74)	0.62 (0.56-0.70)	0.60 (0.53-0.68)
IX. Memory	0.78 (0.72-0.85)	0.79 (0.74-0.85)	0.76 (0.70-0.83)
X. Repetitive thoughts and behaviors	0.54 (0.48-0.62)	0.60 (0.54-0.67)	0.50 (0.43-0.57)
XI. Dissociation	0.71 (0.64-0.78)	0.73 (0.68-0.80)	0.69 (0.62-0.76)
XII. Personality functioning	0.56 (0.49-0.64)	0.54 (0.47-0.61)	0.49 (0.41-0.57)
XIII. Substance use	0.67 (0.58-0.77)	0.64 (0.55-0.75)	0.66 (0.56-0.77)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 73. Negative likelihood ratios of DSM-XC domains for social anxiety disorder, after removing the clinical impairment criteria.

Domain	Intensity	Frequency	“OR”
I. Depression	0.54 (0.45-0.64)	0.66 (0.60-0.74)	0.47 (0.39-0.57)
II. Anger	0.69 (0.61-0.78)	0.76 (0.69-0.83)	0.65 (0.57-0.74)
III. Mania	0.81 (0.74-0.89)	0.85 (0.79-0.92)	0.80 (0.72-0.88)
IV. Anxiety	0.41 (0.33-0.51)	0.52 (0.45-0.61)	0.33 (0.25-0.43)
V. Somatic symptoms	0.72 (0.64-0.82)	0.74 (0.67-0.82)	0.68 (0.60-0.78)
VI. Suicidal ideation	0.84 (0.79-0.90)	0.86 (0.81-0.90)	0.83 (0.78-0.89)
VII. Psychosis	0.82 (0.76-0.88)	0.84 (0.78-0.89)	0.82 (0.76-0.89)
VIII. Sleep problems	0.71 (0.64-0.79)	0.72 (0.65-0.79)	0.66 (0.58-0.74)
IX. Memory	0.78 (0.71-0.85)	0.83 (0.78-0.89)	0.77 (0.70-0.84)
X. Repetitive thoughts and behaviors	0.72 (0.66-0.80)	0.78 (0.72-0.84)	0.71 (0.64-0.78)
XI. Dissociation	0.79 (0.73-0.86)	0.82 (0.77-0.88)	0.77 (0.70-0.83)
XII. Personality functioning	0.62 (0.54-0.70)	0.68 (0.61-0.75)	0.61 (0.53-0.70)
XIII. Substance use	0.82 (0.72-0.93)	0.84 (0.74-0.95)	0.81 (0.71-0.93)

DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 74. Sensitivity differences between DSM-XC question framings (intensity and frequency), after removing the clinical impairment criteria.

Diagnosis	Sensitivity difference (median*)	P-value
Antisocial personality disorder	0.03	0.20
Attention-deficit/hyperactivity disorder	0.07	<0.01
Bipolar disorder	0.03	0.02
Generalized anxiety disorder	0.07	<0.01
Major depressive disorder	0.03	<0.01
Post-traumatic stress disorder	0.06	<0.01
Social anxiety disorder	0.09	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 75. Specificity differences between DSM-XC question framings (intensity and frequency), after removing the clinical impairment criteria.

Diagnosis	Specificity difference (median*)	P-value
Antisocial personality disorder	-0.07	<0.01
Attention-deficit/hyperactivity disorder	-0.07	<0.01
Bipolar disorder	-0.07	<0.01
Generalized anxiety disorder	-0.08	<0.01
Major depressive disorder	-0.07	<0.01
Post-traumatic stress disorder	-0.07	<0.01
Social anxiety disorder	-0.07	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 76. Diagnostic accuracy differences between DSM-XC question framings (intensity and frequency), after removing the clinical impairment criteria.

Diagnosis	Diagnostic accuracy difference (median*)	P-value
Antisocial personality disorder	-0.07	<0.01
Attention-deficit/hyperactivity disorder	-0.06	<0.01
Bipolar disorder	-0.07	<0.01
Generalized anxiety disorder	-0.05	<0.01
Major depressive disorder	-0.06	<0.01
Post-traumatic stress disorder	-0.07	<0.01
Social anxiety disorder	-0.06	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 77. LR+ differences between DSM-XC question framings (intensity and frequency), after removing the clinical impairment criteria.

Diagnosis	LR+ difference (median*)	P-value
Antisocial personality disorder	-0.58	<0.01
Attention-deficit/hyperactivity disorder	-0.47	<0.01
Bipolar disorder	-1.27	<0.01
Generalized anxiety disorder	-1.14	<0.01
Major depressive disorder	-1.63	<0.01
Post-traumatic stress disorder	-1.30	<0.01
Social anxiety disorder	-0.58	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

Supplementary table 78. LR- differences between DSM-XC question framings (intensity and frequency), after removing the clinical impairment criteria.

Diagnosis	LR- difference (median*)	P-value
Antisocial personality disorder	0.03	0.57
Attention-deficit/hyperactivity disorder	-0.01	0.34
Bipolar disorder	0.02	0.18
Generalized anxiety disorder	-0.03	0.03
Major depressive disorder	-0.00	1.00
Post-traumatic stress disorder	0.00	0.94
Social anxiety disorder	-0.04	<0.01

*Median of the differences between the two groups using paired Wilcoxon tests. Intensity was used as the reference group for the comparisons with frequency. Intensity and frequency represent how the instrument would perform if questions were asked with separate frames. A bold font indicates statistical significance in the comparison between frames or between rules. DSM-XC, DSM-5-TR Level 1 Cross-Cutting Symptom Measure.

4 FINAL REMARKS

The research presented in this thesis evaluated the DSM-XC for the detection, comorbidity assessment, and clinical staging of mental disorders in a community-based sample of youth. By addressing critical gaps in the literature, these three studies provide empirical evidence to support a shift from single-disorder screening toward a transdiagnostic, stage-based framework for youth mental health.

4.1 Synthesis of main findings

Collectively, the findings challenge the traditional disorder-specific paradigm of mental health screening. The first study demonstrated that DSM-XC domains function as transdiagnostic indicators rather than disorder-specific measures, with individual domains consistently influencing the probability of multiple internalizing disorders simultaneously. Furthermore, a comparison revealed that framing questions based on “frequency” (“how often”) yields superior diagnostic accuracy, in particular higher specificity and positive likelihood ratios, compared to “intensity” (“how much”), without compromising sensitivity.

Although the DSM-XC was designed as a multidimensional screening tool, the present results indicate that its structure is heavily oriented toward internalizing symptomatology. These symptoms frequently aggregate into highly comorbid diagnostic groupings (De Vries et al. 2022) that may reflect a common underlying liability (Caspi et al. 2025; Caspi et al. 2020; McGrath et al. 2020; Plana-Ripoll et al. 2019), supported by partially overlapping genetic (Bourque et al. 2024), neurobiological (Xie et al. 2023), and treatment-response profiles (Barlow et al. 2020; Cuijpers et al. 2023; Gosmann et al. 2021; Leichsenring et al. 2022). Consequently, internalizing categories may exhibit limited diagnostic distinctiveness, which constrains their utility for screening, despite capturing phenomenologically distinct experiences (e.g., anhedonia versus acute anxiety). While some degree of shared variance is expected within transdiagnostic frameworks (Simms et al. 2023), multidimensional tools would likely benefit from improved differentiation across domains to reduce excessive interdependence. In this context, broader representation of externalizing and thought-related symptoms may enhance the balance of the DSM-XC. More comprehensive dimensional assessment may also require the inclusion of additional parameters, such as symptom persistence and global functioning (Correll et al. 2025), which are central to clinical staging (Scott et al. 2024) and risk stratification but are not directly captured by the current measure.

The second study further clarified the performance of the DSM-XC in the context of psychiatric comorbidity. Thresholds originally proposed for the DSM-5 (*i.e.*, domain scores ≥ 2 for anxiety, depression, and mania domains, and ≥ 1 for substance use) showed utility for excluding additional diagnoses, as indicated by high sensitivity and negative predictive value. However, they were less effective for confirming these diagnoses due to limited specificity. Increasing domain-level cut-offs resulted in modest improvements in the identification of comorbid conditions, particularly for depressive, anxiety, and substance use disorders, yielding positive likelihood ratios in the range of approximately 2.0. However, this magnitude of likelihood ratio corresponds to only a small shift in post-test diagnostic probability, indicating that domain scores alone remain insufficient for confidently ruling in comorbid diagnoses, even under optimized thresholds. Accordingly, the DSM-XC appears more appropriately positioned as a triage tool to inform the need for more detailed diagnostic evaluation, rather than as a substitute for it. In practical terms, a positive domain score should not be interpreted as confirming the presence of a specific comorbid condition, but rather as an indication that systematic clinical assessment for additional disorders is warranted. For example, in individuals presenting with a high pre-test probability of psychopathology in routine clinical care, elevated DSM-XC scores would support the need to investigate a broader range of potential comorbidities beyond the chief complaint, irrespective of which specific domain thresholds are met. In contrast, aggregate scoring approaches may offer a pragmatic means of identifying individuals with elevated overall symptom burden and increased likelihood of diagnostic complexity. In other words, combining domain-level screening (to support exclusion) with total score-based indicators (to flag clinical complexity) may improve the timely recognition of individuals requiring more comprehensive assessment.

Finally, the third study assessed the application of a transdiagnostic clinical staging framework in the community. The DSM-XC captured coherent cross-cutting symptom profiles across mood, anxiety, and post-traumatic stress disorders, mirroring a potentially unifiable transdiagnostic construct. Stages reflected a continuum of affective and stress-related dysregulation across these conditions and were associated with graded increases in objective burden, including cardiometabolic risk, maladaptive health behaviours, suicidality, and educational and occupational disadvantage. These stages corresponded to distinct gradients of burden: cardiometabolic and behavioural risks emerged in early and subthreshold stages (Stage 1b), whereas suicidality and severe socioeconomic disadvantage were heavily concentrated in persistent or recurrent stages (Stage 3+). Staging trajectories converged along a shared dimension of symptom severity and functional impairment,

indicating comparable clinical burden across diagnostic categories at similar stages. Consistent with life-course evidence (Caspi et al. 2020; Clarke et al. 2025; McGrath et al. 2020; Plana-Ripoll et al. 2019), later stages were marked by greater comorbidity and clinical complexity.

4.2 Implications for clinical assessment and public health

These findings have direct implications for clinical practice and public health planning:

1. **Standardization of assessment:** The superior performance of frequency-based framing indicates that future iterations of the DSM-XC and comparable screening tools should prioritize “how often” phrasing to improve diagnostic precision, possibly by reducing cognitive burden, and enhancing resource efficiency. Standardization of adverbial framing may also reduce heterogeneity across assessment measures and improve comparability across clinical and epidemiological settings.
2. **Implications for multidimensional measurement:** Although the DSM-XC is explicitly conceived as a multidimensional assessment tool, its symptom content appears disproportionately weighted toward internalizing domains. This imbalance likely reflects the strong empirical coherence of internalizing symptoms but may also limit the tool’s capacity to distinguish between partially overlapping conditions that share common latent, genetic, and neurobiological substrates. Consequently, several DSM-XC domains may function more as indicators of shared internalizing liability than as discrete dimensions with independent heuristic value. A more balanced multidimensional measure would benefit from greater representation of externalizing and thought-related symptoms, as well as clearer separation between dimensions, to avoid excessive overlap while preserving the advantages of transdiagnostic assessment. As it stands, any above threshold endorsement of a DSM-XC domain mandates screening for several disorders, not only the one intended by the domain.
3. **Refining comorbidity screening:** Although the DSM-XC is intended to facilitate the detection of comorbidity, its predominantly internalizing structure has implications for how results should be interpreted in clinical practice. In settings where a primary diagnosis has already been established, the use of lower DSM-XC thresholds may be efficient for ruling out additional conditions. However, when the objective is to identify and confirm comorbid disorders, a common requirement in youth mental health care, higher thresholds may be necessary, albeit with limited rule-in capacity. In

this context, the DSM-XC total score may serve as a pragmatic screener for the broader symptom burden associated with psychiatric comorbidity. In time-constrained clinical environments, a total score of 21 or higher could function as an alert to a potentially multimorbid clinical presentation, prompting a comprehensive diagnostic evaluation regardless of the presenting complaint.

4. Stage-informed framework for community mental health assessment: The observed alignment between DSM-XC symptom profiles and transdiagnostic clinical stages raises the possibility that the tool could assist in identifying stage-related gradients of clinical burden in community settings. In this context, DSM-XC–derived symptom patterns may help inform preliminary stratification according to overall symptom severity and functional impact, thereby supporting triage decisions within stepped-care models. However, the extent to which brief multidimensional screeners can be used to operationalize staging-informed allocation of care has not yet been formally evaluated and warrants further investigation. Beyond assessment measure-level considerations, these findings also support the broader applicability of transdiagnostic staging frameworks for organizing mental health needs at the population level. In resource-constrained settings, such as many LMICs, a stage-informed approach may facilitate more proportionate allocation of services across community, primary, and specialized care by targeting intervention intensity according to level of clinical burden rather than discrete diagnostic categories alone (Buchweitz et al. 2026). From a public health perspective, this may represent a scalable hypothesis for service planning in contexts characterized by high prevalence of internalizing symptoms and limited access to specialist assessment, to be tested in future implementation-focused research.

4.3 Methodological considerations and limitations

While this thesis draws on a large, population-based birth cohort and structured diagnostic interviews, several overarching methodological limitations should be acknowledged. First, all analyses were conducted using cross-sectional data collected at age 22, which precludes conclusions regarding the temporal stability of symptom patterns or measurement properties across developmental periods. Accordingly, the performance of the DSM-XC, including domain-level associations and optimal thresholds, may vary over time or across stages of illness.

Second, the sample was restricted to a single birth cohort from Southern Brazil, which may limit the generalizability of findings to other age groups, cultural contexts, or healthcare

systems. Differences in help-seeking behavior, service availability, or symptom expression across settings may influence both tool performance and the applicability of derived thresholds or symptom profiles.

Finally, the set of disorders assessed with structured diagnostic interviews was restricted to selected common mental disorders. The inclusion of additional conditions, such as substance use disorders assessed with dedicated structured modules, psychotic disorders, or neurodevelopmental syndromes, could provide a more comprehensive evaluation of DSM-XC performance across a broader range of psychopathological domains.

4.4 Future research directions

Future research is required to evaluate modifications to the DSM-XC. Additional studies should test the clinical impact of using exclusively frequency-based questions, as well as alternative clinical thresholds for DSM-XC domains to improve the identification of mental disorders in community samples. In parallel, refinements to the DSM-XC, including greater representation of externalizing and thought-related symptoms, the inclusion of metrics capturing chronicity and persistence, and the suppression of redundant or closely related internalizing items, should be evaluated to determine whether they enhance its performance as a multidimensional screening tool. Furthermore, if the measure remains published in its current format, an alternative diagnostic pathway should be formally tested: rather than using DSM-XC domains to trigger disorder-specific Level 2 assessments, research should evaluate whether the presence of any positive item may serve as an indication for comprehensive cross-domain assessment.

Further research is also needed to examine whether DSM-XC–derived symptom profiles can meaningfully contribute to stage-informed allocation of care within youth mental health services. Beyond these considerations, longitudinal studies are required to determine whether DSM-XC–derived symptom patterns prospectively predict stage progression, increasing clinical complexity, or functional decline over time. Tracking individuals across developmental periods will be critical to establish whether specific DSM-XC domains are associated with transitions from nonspecific or subthreshold presentations to persistent and impairing illness. In addition, empirical studies should evaluate whether the DSM-XC total score is capable of differentiating between adjacent early stages (e.g., Stage 0 vs. 1a or Stage 1a vs. 1b), which may provide clinically relevant information for early risk stratification.

Although staging frameworks are often assumed to support proportionate care by matching intervention intensity to illness severity, this premise remains largely untested in community samples. In particular, the assumption that individuals in more advanced stages necessarily require more invasive or resource-intensive interventions requires direct empirical evaluation. Future implementation studies should therefore assess whether stage-informed approaches improve the allocation of limited mental health resources when compared to conventional disorder-based screening strategies in the community.

4.5 Conclusion

This thesis provides empirical evidence regarding the performance of the DSM-XC as a screening tool in youth community settings. Frequency-based response options demonstrated superior diagnostic performance when compared to intensity-based scoring, supporting their use in population-level screening contexts. DSM-XC domains showed transdiagnostic associations across internalizing disorders, challenging assumptions of disorder-specific functioning that underpin many current screening practices, as well as the assumption that the tool functions multidimensionally. These findings suggest that the DSM-XC may be better suited for the identification of internalizing conditions and shared symptom burden than for disorder-specific case detection.

Addressing the detection of co-occurring mental disorders in community settings, the DSM-XC demonstrated utility for excluding additional diagnoses when original domain thresholds were applied in individuals with an established condition, although their specificity for confirming comorbid disorders was limited. Adjusting these thresholds yielded modest gains in diagnostic accuracy across internalizing domains. In this context, the DSM-XC total score emerged as a potential indicator of cumulative symptom burden associated with psychiatric multimorbidity. In practice, elevated total scores may serve as a signal of increased likelihood of co-occurring conditions, warranting a more comprehensive diagnostic assessment beyond the initially identified disorder.

The transdiagnostic clinical staging framework appeared to capture meaningful gradients of burden in this community sample, with successive stages associated with progressive increases in objective indicators such as cardiometabolic risk, maladaptive health behaviours, suicidality, and educational and occupational disadvantage. These findings suggest that staging models may be applicable for structuring population-level mental health assessment based on gradients of real-world risk and impairment.

In parallel, cross-cutting symptom patterns derived from the DSM-XC showed systematic variation across these stages, capturing gradients of affective and stress-related dysregulation across mood, anxiety, and post-traumatic stress conditions. This correspondence indicates that DSM-XC–based symptom profiles may be used to characterize stage-related differences in clinical presentation and could contribute to stage-informed approaches to assessment and service planning.

Taken together, these findings support the use of brief cross-cutting measures such as the DSM-XC as tools for the timely identification of individuals unlikely to meet criteria for additional mental disorders in screening contexts, while also highlighting the broader applicability of transdiagnostic frameworks for structuring mental health assessment and service planning among young people. Moving beyond strictly disorder-specific screening models may represent an important step toward more scalable approaches to youth mental health care in community contexts. Similarly, moving beyond binary present–absent approaches to diagnosis that may obscure meaningful gradients of symptom burden may contribute to more clinically informative assessment strategies.

5 THESIS REFERENCES

American Psychiatric Association. *Diagnostic and statistical manual of mental disorders*. 5th ed. Washington (DC): American Psychiatric Association; 2013.

American Psychiatric Association. *Diagnostic and statistical manual of mental disorders*. 5th ed, text rev. Washington (DC): American Psychiatric Association; 2022.

Arrarás JI, Cuesta MJ, Peralta V, Gil-Berrozpe GJ, Barrado L, Correa O, et al. Psychometric analysis of the psychiatric diagnostic screening questionnaire (PDSQ) and determinants of psychopathology in two outpatient clinics in Navarre (Spain). *An Sist Sanit Navar*. 2023;46(2):e1043.

Barlow DH, Harris BA, Eustis EH, Farchione TJ. The unified protocol for transdiagnostic treatment of emotional disorders. *World Psychiatry*. 2020;19(2):245–6.

Bastiaens L, Galus J. The DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure as a Screening Tool. *Psychiatr. Q.* 2018 Mar;89(1):111–5.

Bourque VR, Poulain C, Proulx C, Moreau CA, Joobor R, Forgeot d’Arc B, et al. Genetic and phenotypic similarity across major psychiatric disorders: a systematic review and quantitative assessment. *Transl Psychiatry*. 2024;14(1):171.

Bravo AJ, Villarosa-Hurlocker MC, Pearson MR, Protective Strategies Study Team. College student mental health: An evaluation of the DSM–5 self-rated Level 1 cross-cutting symptom measure. *Psychol. Assess.* 2018 Oct;30(10):1382–9.

Buchweitz C, Viduani A, Herrman H, Kohrt BA, McGorry P, Salum G, et al. Transdiagnostic stage-based monitoring of public mental health. *Nat Rev Psychol*. 2026 Jan 7 [cited 2026 Jan 20]. Available from: <https://www.nature.com/articles/s44159-025-00527-w>

Caspi A, Houts RM, Ambler A, Danese A, Elliott ML, Hariri A, et al. Longitudinal assessment of mental health disorders and comorbidities across 4 decades among participants in the Dunedin birth cohort study. *JAMA Netw Open*. 2020;3(4):e203221.

Caspi A, Houts RM, Tegner Anker AS, Richmond-Rakerd LS, Andersen SH, Theodore R, et al. Why psychopathology research should avoid studying one mental disorder at a time: an intergenerational and developmental evidence base for understanding “p”. *J Psychopathol Clin Sci*. 2025 Oct 27 [cited 2026 Feb 20]. Available from: <https://doi.apa.org/doi/10.1037/abn0001042>

Chen Y, Aitken Z, Hammond D, Thompson A, Marwaha S, Davey C, et al. Adverse childhood experiences and their differential relationships with transdiagnostic mental health outcomes in young adults. *Psychol. Med.* 2025;55:e147.

Christensen MK, McGrath JJ, Momen N, Weye N, Agerbo E, Pedersen CB, et al. The health care cost of comorbidity in individuals with mental disorders: a Danish register-based study. *Aust N Z J Psychiatry*. 2023;57(6):914–22.

Clarke DE, Kuhl EA. DSM-5 cross-cutting symptom measures: a step towards the future of psychiatric care? *World Psychiatry*. 2014 Oct;13(3):314–6.

Clarke DE, Narrow WE, Regier DA, Kuramoto SJ, Kupfer DJ, Kuhl EA, et al. DSM-5 field trials in the United States and Canada, part I: study design, sampling strategy, implementation, and analytic approaches. *Am J Psychiatry*. 2013;170(1):43–58.

Clarke SL, Soons N, Videler AC, van Alphen SPJ, van H, Dil L, et al. The progress in the field of clinical staging for mental disorders within the last decade: an updated systematic review. *Front Psychiatry*. 2025;15:1473051.

Çökmüş F, Balıkcı K, Aydemir Ö, et al. Reliability and validity of Turkish form of DSM-5 self-rated level 1 cross-cutting symptom scale—adult version. *Anatol J Psychiatry*. 2017;18(2):5.

Correll CU, Cortese S, Solmi M, Boldrini T, Demyttenaere K, Domschke K, et al. Beyond symptom improvement: transdiagnostic and disorder-specific ways to assess functional and quality of life outcomes across mental disorders in adults. *World Psychiatry*. 2025;24(3):296–318.

Cuijpers P, Miguel C, Ciharova M, Ebert D, Harrer M, Karyotaki E. Transdiagnostic treatment of depression and anxiety: a meta-analysis. *Psychol. Med.* 2023 Oct;53(14):6535–46.

Davey CG, McGorry PD. Early intervention for depression in young people: a blind spot in mental health care. *Lancet Psychiatry*. 2019;6(3):267–72.

De Vries YA, Al-Hamzawi A, Alonso J, Andrade LH, Benjet C, Bruffaerts R, et al. Transdiagnostic development of internalizing psychopathology throughout the life course up to age 45: a World Mental Health Surveys report. *Psychol. Med.* 2022 Aug;52(11):2134–43.

Doss RA, Lowmaster SE. Validation of the DSM-5 level 1 cross-cutting symptom measure in a community sample. *Psychiatry Res.* 2022;318:114935.

Dwyer D, Ye RR, Nelson B, McGorry P. Clinical Staging for Psychiatry and Psychology. *Annu. Rev. Clin. Psychol.* 2025 May 7;21(1):497–527.

Eaton NR, Bringmann LF, Elmer T, Fried EI, Forbes MK, Greene AL, et al. A review of approaches and models in psychopathology conceptualization research. *Nat Rev Psychol.* 2023 Aug 7 [cited 2023 Sep 26]. Available from: <https://www.nature.com/articles/s44159-023-00218-4>

Farber GK, Gage S, Kemmer D, White R. Common measures in mental health: a joint initiative by funders and journals. *Lancet Psychiatry*. 2023;10(6):465–70.

Fava GA, Sonino N, Guidi J. Measuring clinical findings: the value of clinimetrics. *Postgrad Med J.* 2025;102(1203):88–94.

Forbes MK, Neo B, Nezami OM, Fried EI, Faure K, Michelsen B, et al. Elemental psychopathology: distilling constituent symptoms and patterns of repetition in the diagnostic criteria of the DSM-5. *Psychol. Med.* 2024 Apr;54(5):886–94.

Forbes MK, Tackett JL, Markon KE, Krueger RF. Beyond comorbidity: Toward a dimensional and hierarchical approach to understanding psychopathology across the life span. *Dev. Psychopathol.* 2016 Nov;28(4pt1):971–86.

Fried EI, Flake JK, Robinaugh DJ. Revisiting the theoretical and methodological foundations of depression measurement. *Nat Rev Psychol*. 2022;1(6):358–68.

Fusar-Poli P, Solmi M, Brondino N, Davies C, Chae C, Politi P, et al. Transdiagnostic psychiatry: a systematic review. *World Psychiatry*. 2019;18(2):192–207.

Gibbons AB, Farmer C, Shaw JS, Chung JY. Examining the factor structure of the DSM-5 level 1 cross-cutting symptom measure. *Int J Methods Psychiatr Res*. 2022 Nov [cited 2022 Nov 2]. Available from: <https://onlinelibrary.wiley.com/doi/10.1002/mpr.1953>

Goel A, Kataria D. Validation of hindi translation of DSM-5 level 1 cross-cutting symptom measure. *Asian J. Psychiatry*. 2018 Apr;34:6–10.

Gong J, Zhang Y, Yang Z, Huang Y, Feng J, Zhang W. The framing effect in medical decision-making: a review of the literature. *Psychol. Health Med*. 2013 Dec;18(6):645–53.

Gosmann NP, Costa MDA, Jaeger MDB, Motta LS, Frozi J, Spanemberg L, et al. Selective serotonin reuptake inhibitors and serotonin and norepinephrine reuptake inhibitors for anxiety, obsessive-compulsive, and stress disorders: a 3-level network meta-analysis. *PLoS Med*. 2021;18(6):e1003664.

Haas LM, Pan PM, Hickie IB, Czepielewski LS. Transdiagnostic clinical staging in youth mental health: a primer for clinical practice. *Braz J Psychiatry*. 2025 [cited 2025 Nov 20]. Available from: <https://www.bjp.org.br/details/3596/en-US/transdiagnostic-clinical-staging-in-youth-mental-health--a-primer-for-clinical-practice>

Haslam N, Holland E, Kuppens P. Categories versus dimensions in personality and psychopathology: a quantitative review of taxometric research. *Psychol Med*. 2012;42(5):903–20.

Hickie IB, Scott EM, Cross SP, et al. Right care, first time: a highly personalised and measurement-based care model to manage youth mental health. *Med J Aust*. 2019;211(Suppl 9):S3–46.

Hyman SE. Psychiatric Disorders: Grounded in Human Biology but Not Natural Kinds. *Perspect. Biol. Med*. 2021;64(1):6–28.

Ishfaq N, Kamal A. Mental health and imprisonment: Measuring cross-cutting symptoms among convicts in Punjab, Pakistan. *Asian J. Psychiatry*. 2019 Aug;44:127–32.

Jami ES, Hammerschlag AR, Ip HF, Allegrini AG, Benyamin B, Border R, et al. Genome-wide association meta-analysis of childhood and adolescent internalizing symptoms. *J Am Acad Child Adolesc Psychiatry*. 2022;61(7):934–45.

Jobim GDS, Amaral JVD, Pacheco JPG, Gadelha A, Miguel EC, Bressan RA, et al. Clinical properties of the short mood and feelings questionnaire: Development of a free calculator based on the Brazilian high-risk cohort study. *J. Psychiatr. Res*. 2025 Oct;190:457–64.

Kieling C, Buchweitz C, Caye A, Silvani J, Ameis SH, Brunoni AR, et al. Worldwide Prevalence and Disability From Mental Disorders Across Childhood and Adolescence:

Evidence From the Global Burden of Disease Study. *JAMA Psychiatry*. 2024 Apr 1;81(4):347.

Kieling C, Salum GA, Pan PM, Bressan RA. Youth mental health services: the right time for a global reach. *World Psychiatry*. 2022 Feb;21(1):86–7.

Kohrt BA, Asher L, Bhardwaj A, Fazel M, Jordans MJD, Mutamba BB, et al. The Role of Communities in Mental Health Care in Low- and Middle-Income Countries: A Meta-Review of Components and Competencies. *Int. J. Environ. Res. Public. Health*. 2018 June 16;15(6):1279.

Kohrt BA, Gurung D, Singh R, Rai S, Neupane M, Turner EL, et al. Is there a mental health diagnostic crisis in primary care? Current research practices in global mental health cannot answer that question. *Epidemiol. Psychiatr. Sci.* 2025;34:e7.

Kotov R, et al. The hierarchical taxonomy of psychopathology (HiTOP): a dimensional alternative to traditional nosologies. *J Abnorm Psychol*. 2017;126(4):454–77.

Krabbe J, Forkmann T. Frequency vs. intensity: Framing effects on patients' use of verbal rating scale anchors. *Compr. Psychiatry*. 2014 Nov;55(8):1928–36.

Krishnakumar A, Scopel Hoffmann M, Schoeller F, Clucas JC, Son J, Müller-Naendrup L, et al. Framing mental health questions using context better predicts disability than does frequency. *PsyArXiv [Preprint]*. 2021 Jan. Available from: <https://osf.io/jm9kc>

Kroenke K, Spitzer RL, Williams JBW. The PHQ-9: Validity of a brief depression severity measure. *J. Gen. Intern. Med*. 2001 Sept;16(9):606–13.

Krueger RF, Eaton NR. Transdiagnostic factors of mental disorders. *World Psychiatry*. 2015 Feb;14(1):27–9.

Leichsenring F, Steinert C, Rabung S, Ioannidis JPA. The efficacy of psychotherapies and pharmacotherapies for mental disorders in adults: an umbrella review and meta-analytic evaluation of recent meta-analyses. *World Psychiatry*. 2022 Feb;21(1):133–45.

Levin IP, Schneider SL, Gaeth GJ. All Frames Are Not Created Equal: A Typology and Critical Analysis of Framing Effects. *Organ. Behav. Hum. Decis. Process*. 1998 Nov;76(2):149–88.

Mahoney MR, Farmer C, Sinclair S, Sung S, Dehaut K, Chung JY. Utilization of the DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure-Adult to Screen Healthy Volunteers for Research Studies. *Psychiatry Res*. 2020 Apr;286:112822.

McGorry PD, Mei C, Dalal N, Alvarez-Jimenez M, Blakemore S-J, Browne V, et al. The Lancet Psychiatry Commission on youth mental health. *Lancet Psychiatry*. 2024 Sept;11(9):731–74.

McGrath JJ, Lim CCW, Plana-Ripoll O, Holtz Y, Agerbo E, Momen NC, et al. Comorbidity within mental disorders: a comprehensive analysis based on 145 990 survey respondents from 27 countries. *Epidemiol. Psychiatr. Sci.* 2020;29:e153.

Mościcki EK, Clarke DE, Kuramoto SJ, Kraemer HC, Narrow WE, Kupfer DJ, et al. Testing DSM-5 in routine clinical practice settings: feasibility and clinical utility. *Psychiatr Serv.* 2013;64(10):952–60.

Narrow WE, Clarke DE, Kuramoto SJ, Kraemer HC, Kupfer DJ, Greiner L, et al. DSM-5 Field Trials in the United States and Canada, Part III: Development and Reliability Testing of a Cross-Cutting Symptom Assessment for DSM-5. *Am. J. Psychiatry.* 2013 Jan;170(1):71–82.

Negeri ZF, Levis B, Sun Y, He C, Krishnan A, Wu Y, et al. Accuracy of the Patient Health Questionnaire-9 for screening to detect major depression: updated systematic review and individual participant data meta-analysis. *BMJ.* 2021 Oct 5;n2183.

Newson JJ, Hunter D, Thiagarajan TC. The Heterogeneity of Mental Health Assessment. *Front. Psychiatry.* 2020 Feb 27;11:76.

Oquendo MA, Abi-Dargham A, Alpert JE, Benton TD, Clarke DE, Compton WM, et al. Initial Strategy for the Future of *DSM*. *Am. J. Psychiatry.* 2026 Jan 28;appi.ajp.20250878.

Patel V, Kieling C, Maulik PK, Divan G. Improving access to care for children with mental disorders: a global perspective. *Arch. Dis. Child.* 2013 May 1;98(5):323–7.

Plana-Ripoll O, Musliner KL, Dalsgaard S, Momen NC, Weye N, Christensen MK, et al. Nature and prevalence of combinations of mental disorders and their association with excess mortality in a population-based cohort study. *World Psychiatry.* 2020 Oct;19(3):339–49.

Plana-Ripoll O, Pedersen CB, Holtz Y, Benros ME, Dalsgaard S, de Jonge P, et al. Exploring Comorbidity Within Mental Disorders Among a Danish National Population. *JAMA Psychiatry.* 2019 Mar 1;76(3):259.

Plummer F, Manea L, Trepel D, McMillan D. Screening for anxiety disorders with the GAD-7 and GAD-2: a systematic review and diagnostic metaanalysis. *Gen. Hosp. Psychiatry.* 2016 Mar;39:24–31.

Ratheesh A, Chen Y, Hammond D, Aitken Z, Shah J, Iorfino F, et al. Progression of Transdiagnostic Stages From Childhood to Young Adulthood. *JAMA Psychiatry* [Internet]. 2025 Sept 24 [cited 2025 Oct 25]; Available from: <https://jamanetwork.com/journals/jamapsychiatry/fullarticle/2839018>

Ratheesh A, Hammond D, Gao C, Marwaha S, Thompson A, Hartmann J, et al. Empirically driven transdiagnostic stages in the development of mood, anxiety and psychotic symptoms in a cohort of youth followed from birth. *Transl Psychiatry.* 2023;13(1):103.

Rose-Clarke K, Bitta M, Evans-Lacko S, Jokinen T, Jordans M, Nyongesa MK, et al. Centring youth mental health discourse on low-income and middle-income countries. *Lancet Psychiatry.* 2024 Sept;11(9):671–2.

Scott J, Iorfino F, Capon W, Crouse J, Nelson B, Chanen AM, et al. Staging 2.0: refining transdiagnostic clinical staging frameworks to enhance reliability and utility for youth mental health. *Lancet Psychiatry.* 2024 June;11(6):461–71.

Scott KM, Lim C, Al-Hamzawi A, Alonso J, Bruffaerts R, Caldas-de-Almeida JM, et al. Association of Mental Disorders With Subsequent Chronic Physical Conditions: World Mental Health Surveys From 17 Countries. *JAMA Psychiatry*. 2016 Feb 1;73(2):150.

Shah JL, Scott J, McGorry PD, Cross SPM, Keshavan MS, Nelson B, et al. Transdiagnostic clinical staging in youth mental health: a first international consensus statement. *World Psychiatry*. 2020;19(2):233–42.

Simms LJ, Wright AGC, Cicero D, Kotov R, Mullins-Sweatt SN, Sellbom M, et al. The hierarchical taxonomy of psychopathology (HiTOP) approach to assessment: summary of the first phase of the HiTOP self-report measure development project. In: Mihura JL, editor. *Oxford handbook of personality and psychopathology assessment*. 2nd ed. Oxford: Oxford University Press; 2023. p. 55–71.

Solmi M, Radua J, Olivola M, Croce E, Soardo L, Salazar De Pablo G, et al. Age at onset of mental disorders worldwide: large-scale meta-analysis of 192 epidemiological studies. *Mol Psychiatry*. 2022;27(1):281–95.

Spitzer RL, Kroenke K, Williams JBW, Löwe B. A Brief Measure for Assessing Generalized Anxiety Disorder: The GAD-7. *Arch. Intern. Med*. 2006 May 22;166(10):1092.

US Preventive Services Task Force, Barry MJ, Nicholson WK, Silverstein M, Chelmos D, Coker TR, et al. Screening for Depression and Suicide Risk in Adults: US Preventive Services Task Force Recommendation Statement. *JAMA*. 2023a June 20;329(23):2057.

US Preventive Services Task Force, Barry MJ, Nicholson WK, Silverstein M, Coker TR, Davidson KW, et al. Screening for Anxiety Disorders in Adults: US Preventive Services Task Force Recommendation Statement. *JAMA*. 2023b June 27;329(24):2163.

Vanes LD, Dolan RJ. Transdiagnostic neuroimaging markers of psychiatric risk: A narrative review. *NeuroImage Clin*. 2021;30:102634.

Xie C, Xiang S, Shen C, Peng X, Kang J, Li Y, et al. A shared neural basis underlying psychiatric comorbidity. *Nat. Med*. 2023 May;29(5):1232–42.

Zimmerman M, Chelminski I. Screening for anxiety disorders in depressed patients. *J. Psychiatr. Res*. 2006 Apr;40(3):267–72.

Zimmerman M, Mattia JJ. Psychiatric diagnosis in clinical practice: Is comorbidity being missed? *Compr. Psychiatry*. 1999 May;40(3):182–91.

Zimmerman M, Mattia JJ. The psychiatric diagnostic screening questionnaire: Development, reliability and validity. *Compr. Psychiatry*. 2001 May;42(3):175–89.

6 APPENDIX

Appendix A: Brazilian Portuguese Adaptation of the DSM-5 Level 1 Cross-Cutting Symptom Measure for the Pelotas Birth Cohort 1993 Study.

The following instrument is the Brazilian Portuguese adaptation of the DSM-5 Level 1 Cross-Cutting Symptom Measure (DSM-XC), as modified and administered during the 22-year follow-up of the 1993 Pelotas Birth Cohort Study. As detailed in the methodology of this thesis, this version was adapted to separate the original double-barreled questions into distinct intensity and frequency scales, thereby reducing cognitive burden and allowing for evaluating the screening capabilities of each framing for this community-based youth sample.

Further details regarding the cohort's overarching methodology, data collection procedures, and complete assessment protocols can be accessed via the study's official repository at: <https://epidemiio-ufpel.org.br/coorte-1993/>. The exact Brazilian Portuguese instrument presented below is also available directly on this website.

Agora as frases são sobre outros sentimentos e comportamentos que podem acontecer com todos. Pensa que eles podem ou não ter ocorrido contigo nas DUAS ÚLTIMAS SEMANAS e assinala as respostas que descrevem melhor quantas vezes e como te incomodou durante este período.		
	O quanto isso te incomodou?	Quantas vezes nas 2 últimas semanas?
Tiveste pouco interesse ou prazer em fazer as coisas:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Te sentiste desanimado(a), deprimido(a) ou sem esperança:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Te sentiste mais irritado(a), mal-humorado(a) ou zangado(a) do que o usual:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste muita energia mesmo dormindo menos do que o normal:	☐ Nada ☐ Muito pouco ☐ Leve	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias

	☐ Moderado ☐ Grave	☐ Mais da metade dos dias ☐ Quase todos os dias
Iniciaste muito mais projetos do que o usual ou fizeste coisas mais arriscadas que o usual:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Te sentiste nervoso(a), ansioso(a), assustado(a), preocupado(a) ou tenso(a):	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste pânico ou te sentiste com medo:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Evitaste situações que o(a) deixam ansioso(a):	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste dores e sofrimentos sem explicação (p. ex., cabeça, costas, articulações, abdome, pernas):	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Sentiste que tuas doenças não estão sendo levadas suficientemente a sério:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste pensamentos de ferir a si mesmo(a):	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Ouviste coisas que outras pessoas não ouviam, como vozes, mesmo quando não havia ninguém por perto:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias

Sentiste que alguém podia ouvir teus pensamentos ou que tu podias ouvir o que outra pessoa estava pensando:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste problemas para dormir que afetaram a qualidade do teu sono em geral:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste problemas com a memória (p. ex., aprender informações novas) ou com localização (p. ex., encontrar o caminho para casa):	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Tiveste pensamentos, impulsos ou imagens desagradáveis que entram repetidamente na tua cabeça:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Te sentiste obrigado(a) a realizar certos comportamentos ou atos mentais repetidamente:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Te sentiste desligado(a) ou distante de ti mesmo(a), do teu corpo, do ambiente físico ao teu redor ou de tuas lembranças:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Ficaste sem saber quem tu realmente é ou o que tu quer da vida:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Não te sentiste próximo(a) a outras pessoas (amigos, familiares, namorado/a...) ou não curtiste tuas relações com elas:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Bebeste no mínimo 4 drinques de qualquer tipo de bebida alcoólica em um único dia:	☐ Nada ☐ Muito pouco ☐ Leve	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias

	☐ Moderado ☐ Grave	☐ Mais da metade dos dias ☐ Quase todos os dias
Fumaste cigarros, charuto ou cachimbo ou usaste rapé ou tabaco de mascar:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Usaste um ou mais dos seguintes medicamentos POR CONTA PRÓPRIA: sem orientação de um médico, ou em quantidades maiores ou por mais tempo do que foi orientado: a) analgésicos como codeína ou Tylex b) estimulantes como metilfenidato ou Ritalina c) sedativos ou tranquilizantes como comprimidos para dormir, diazepam ou Valium, clonazepam ou Rivotril	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias
Usaste alguma destas drogas: maconha, cocaína, crack, ecstasy, alucinógenos (como LSD), heroína, inalantes ou solventes (como cola) ou metanfetamina:	☐ Nada ☐ Muito pouco ☐ Leve ☐ Moderado ☐ Grave	☐ Nenhum dia ☐ Menos de um ou dois dias ☐ Vários dias ☐ Mais da metade dos dias ☐ Quase todos os dias